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(54) **METHODS OF TREATING GASTROINTESTINAL INFLAMMATORY DISEASE**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

Provided herein are variant $\alpha 4\beta 7$ integrin antibodies and methods of use.

20 Claims, 17 Drawing Sheets

Specification includes a Sequence Listing.

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WO	2024123793	A1	6/2024
WO	2024129960	A2	6/2024
WO	2024151852	A2	7/2024
WO	2024182425	A1	9/2024
WO	2024188443	A1	9/2024
WO	2024205498	A1	10/2024
WO	2024206308	A2	10/2024
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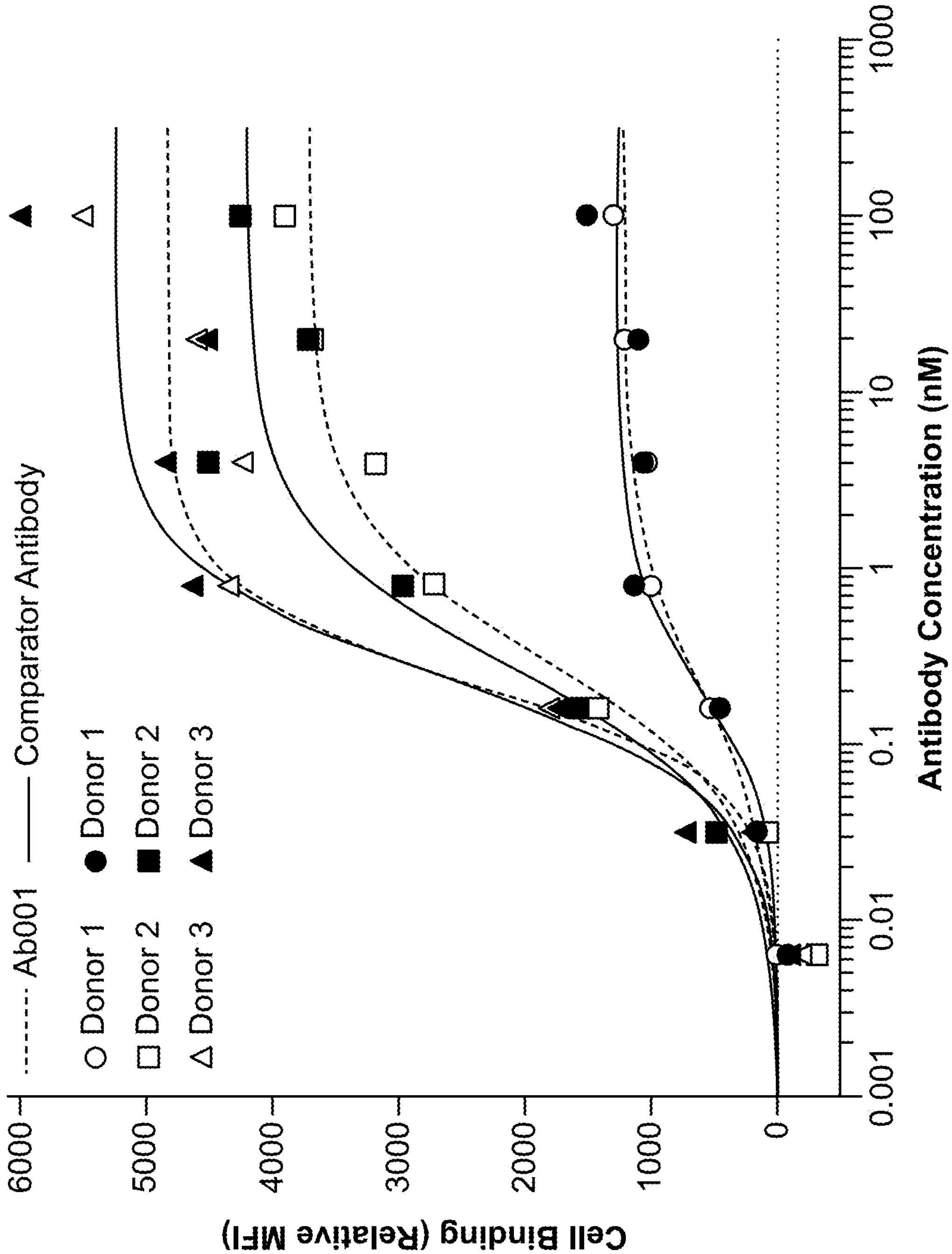

Ab001
Donor 1
Donor 2
Donor 3
Comparator Antibody
Donor 1
Donor 2
Donor 3
Cell Binding (Relative MFI)
6000
5000
4000
3000
2000
1000
0
0.001
0.01
0.1
1
10
100
1000
Antibody Concentration (nM)

FIG. 1A

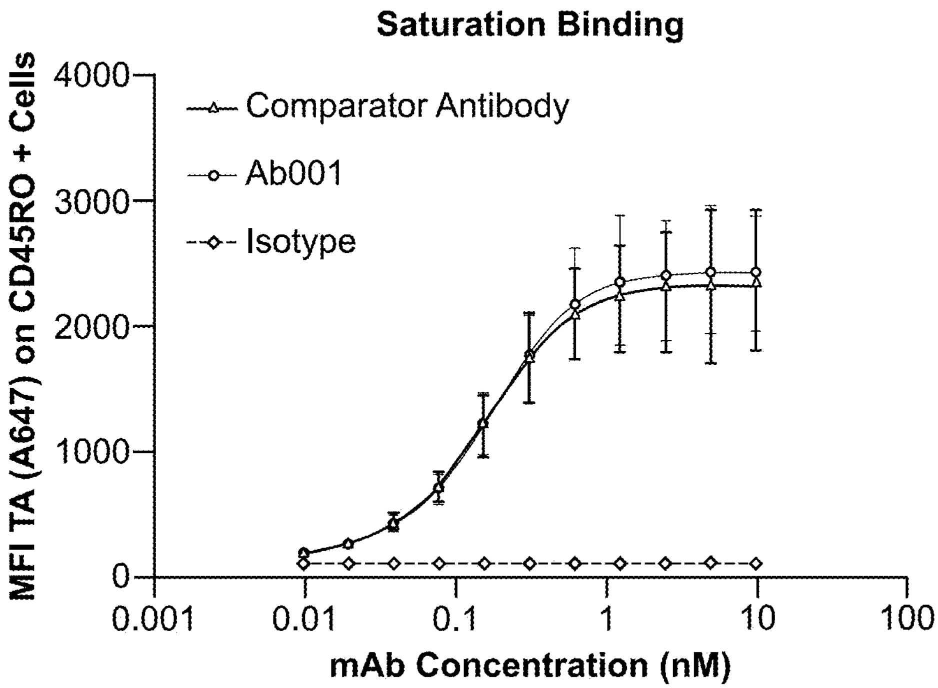

Saturation Binding
MFI TA (A647) on CD45RO + Cells
4000
3000
2000
1000
0
Comparator Antibody
Ab001
Isotype
0.001
0.01
0.1
1
10
100
mAb Concentration (nM)

FIG. 1B

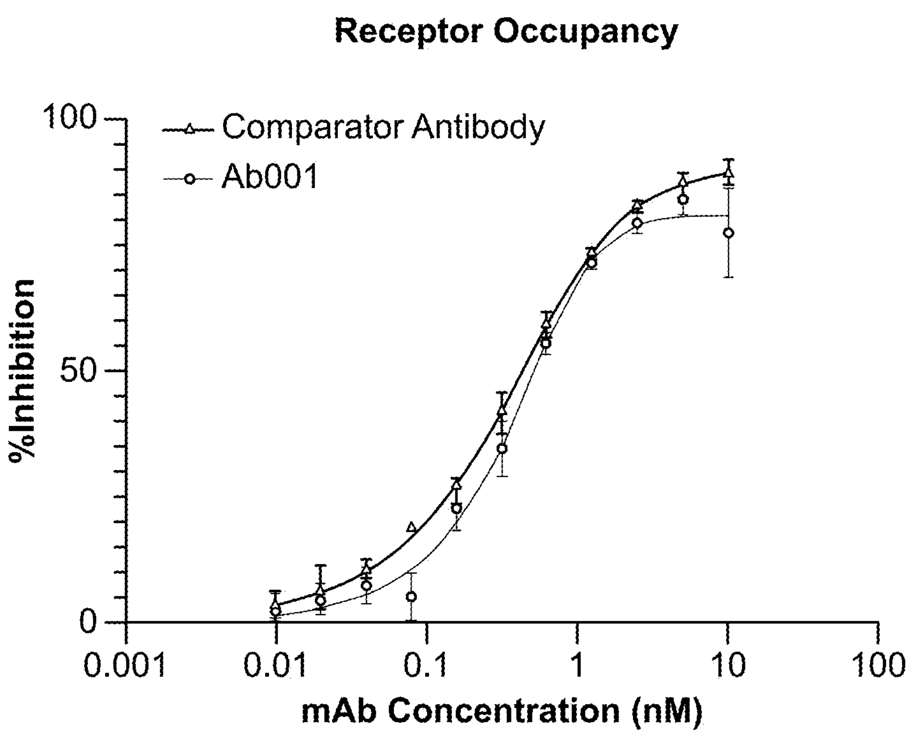
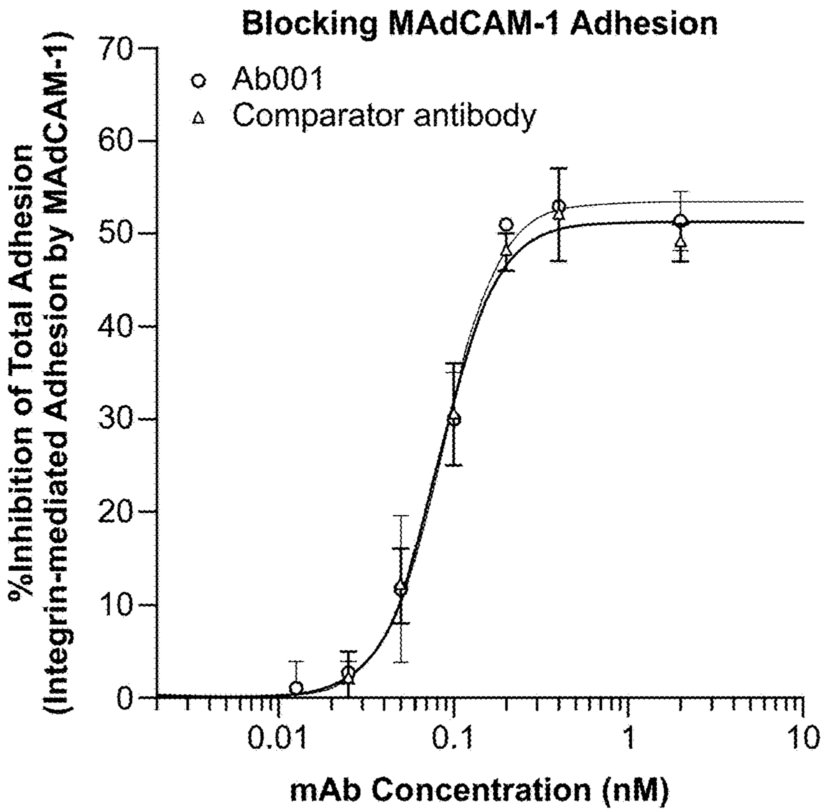
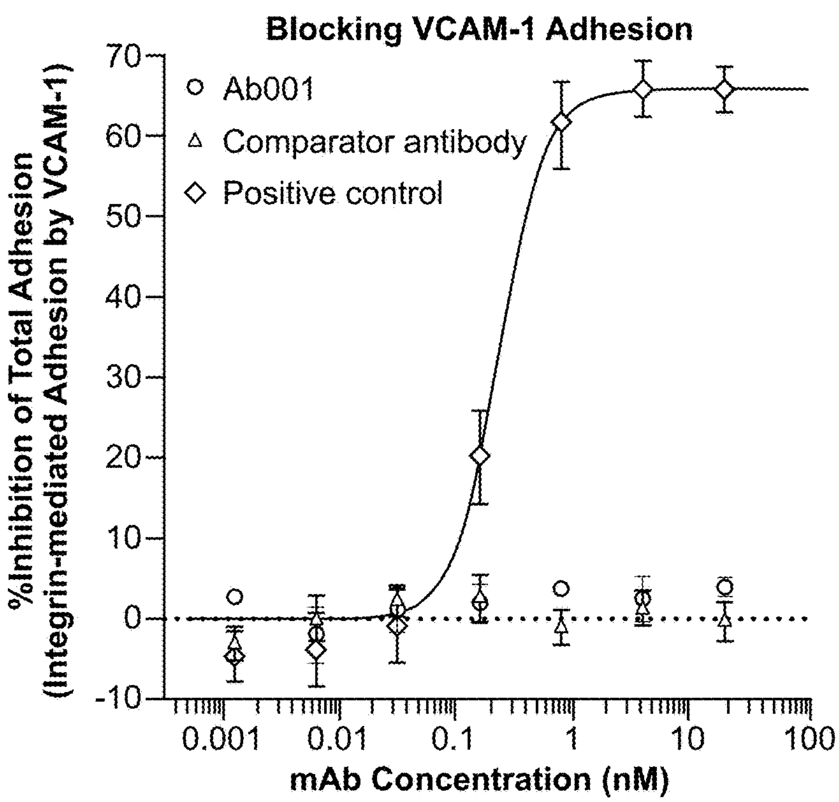

Receptor Occupancy
%Inhibition
100
50
0
Comparator Antibody
Ab001
0.001
0.01
0.1
1
10
100
mAb Concentration (nM)

FIG. 1C



Blocking MAdCAM-1 Adhesion
%Inhibition of Total Adhesion
(Integrin-mediated Adhesion by MAdCAM-1)
Ab001
Comparator antibody
70
60
50
40
30
20
10
0
0.01
0.1
1
10
mAb Concentration (nM)

FIG. 2A



Blocking VCAM-1 Adhesion
%Inhibition of Total Adhesion
(Integrin-mediated Adhesion by VCAM-1)
Ab001
Comparator antibody
Positive control
70
60
50
40
30
20
10
0
-10
0.001
0.01
0.1
1
10
100
mAb Concentration (nM)

FIG. 2B

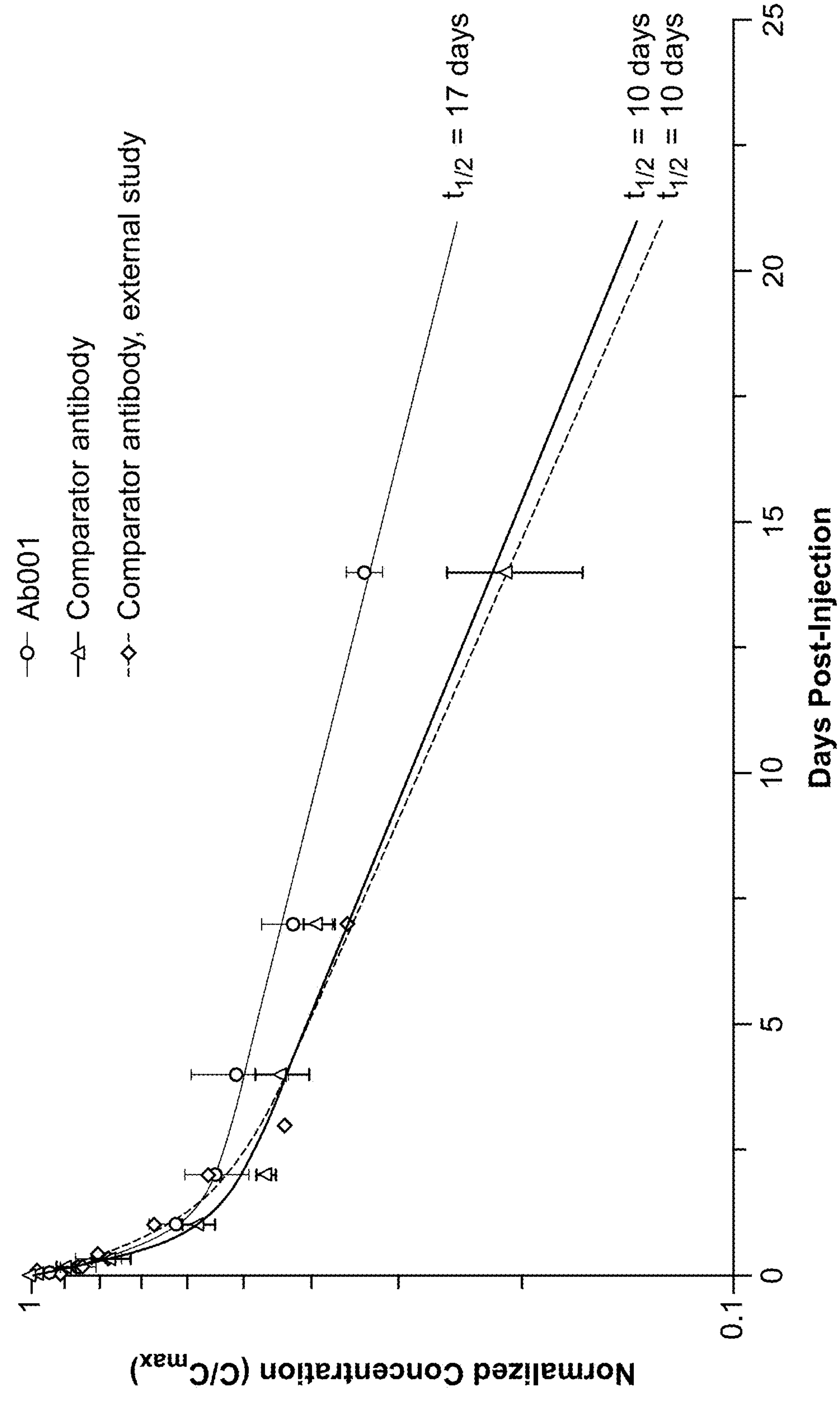
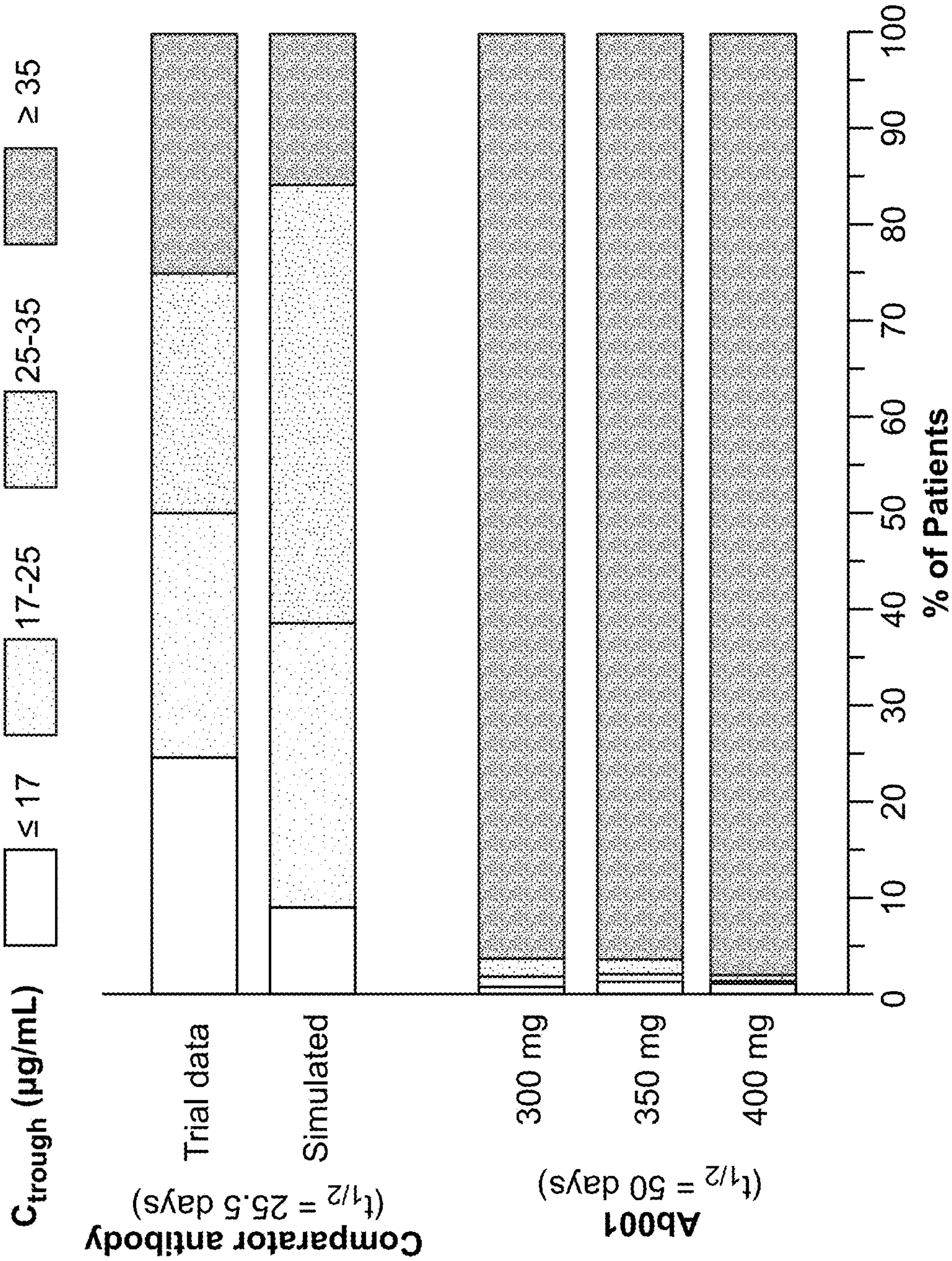

Ab001
Comparator antibody
Comparator antibody, external study
$t_{1/2}$ = 17 days
$t_{1/2}$ = 10 days
$t_{1/2}$ = 10 days
Normalized Concentration (C/C_{max})
1
0.1
0
5
10
15
20
25
Days Post-Injection

FIG. 3



C_{trough} (μg/mL)
≤ 17
17-25
25-35
≥ 35
Comparator antibody
($t_{1/2}$ = 25.5 days)
Trial data
Simulated
Ab001
($t_{1/2}$ = 50 days)
300 mg
350 mg
400 mg
0
10
20
30
40
50
60
70
80
90
100
% of Patients

FIG. 4

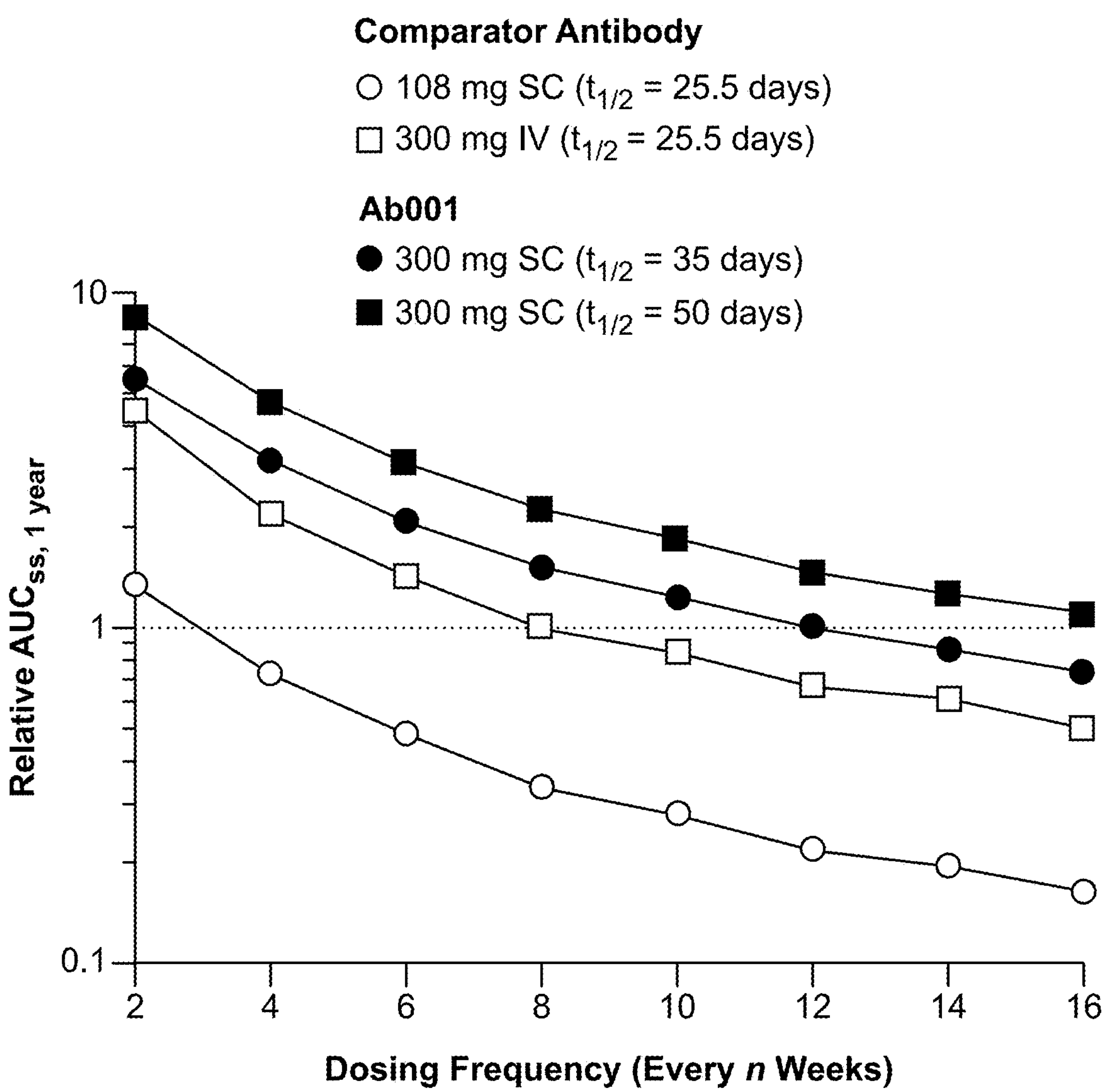

Comparator Antibody
108 mg SC ($t_{1/2}$ = 25.5 days)
300 mg IV ($t_{1/2}$ = 25.5 days)
Ab001
300 mg SC ($t_{1/2}$ = 35 days)
300 mg SC ($t_{1/2}$ = 50 days)
Relative AUC_{ss}, 1 year
10
1
0.1
2
4
6
8
10
12
14
16
Dosing Frequency (Every *n* Weeks)

FIG. 5

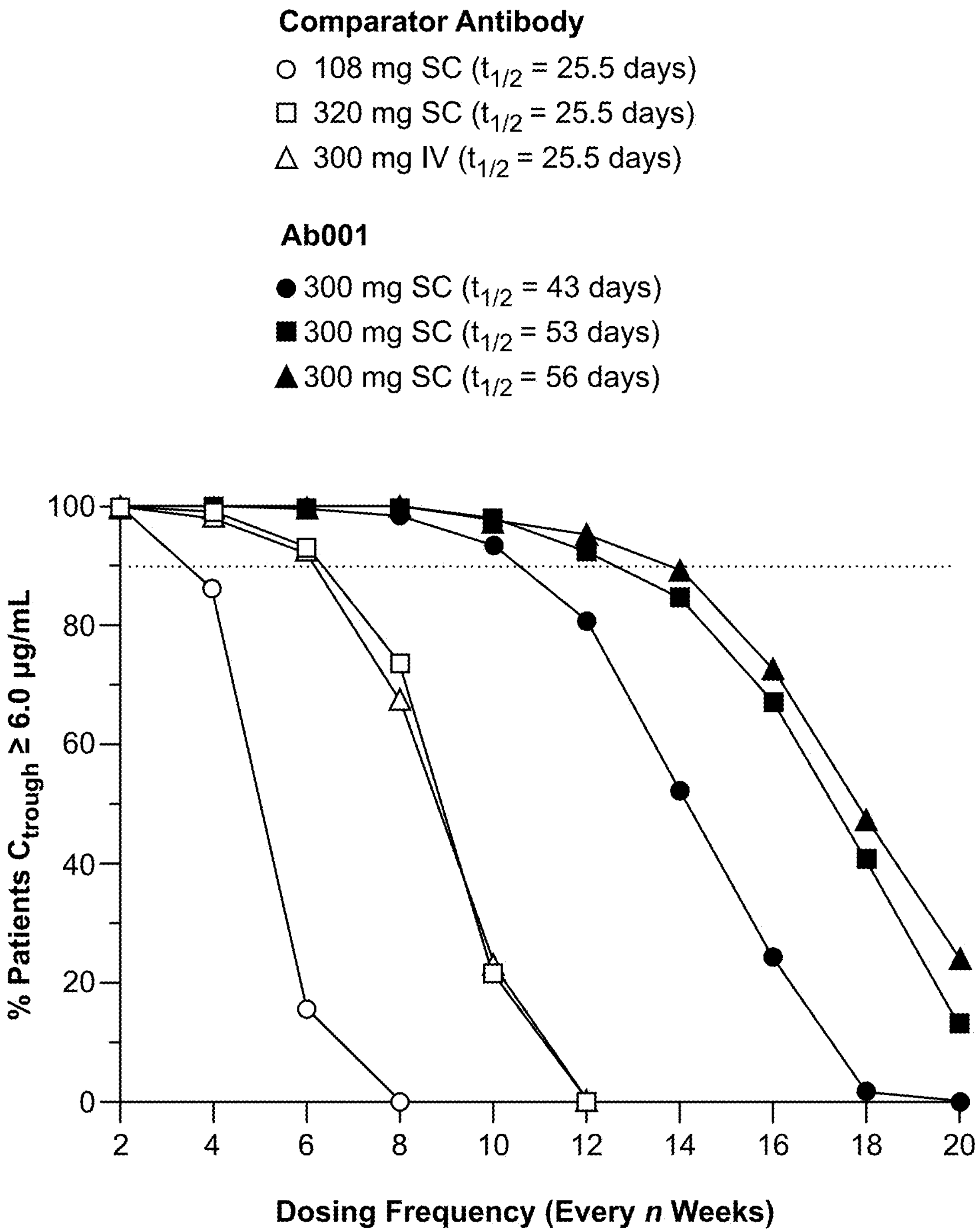

Comparator Antibody
○ 108 mg SC ($t_{1/2}$ = 25.5 days)
□ 320 mg SC ($t_{1/2}$ = 25.5 days)
△ 300 mg IV ($t_{1/2}$ = 25.5 days)
Ab001
● 300 mg SC ($t_{1/2}$ = 43 days)
■ 300 mg SC ($t_{1/2}$ = 53 days)
▲ 300 mg SC ($t_{1/2}$ = 56 days)
% Patients C_{trough} ≥ 6.0 μg/mL
0
20
40
60
80
100
2
4
6
8
10
12
14
16
18
20
Dosing Frequency (Every *n* Weeks)

FIG. 6

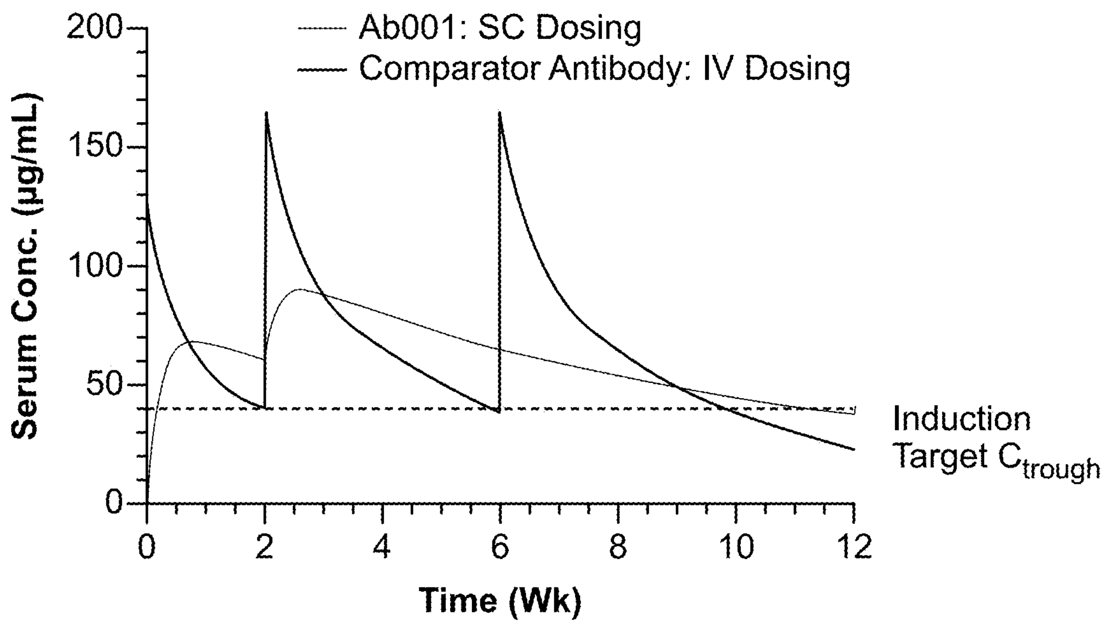

Ab001: SC Dosing
Comparator Antibody: IV Dosing
Serum Conc. (µg/mL)
200
150
100
50
0
0
2
4
6
8
10
12
Time (Wk)
Induction Target C_{trough}

FIG. 7A

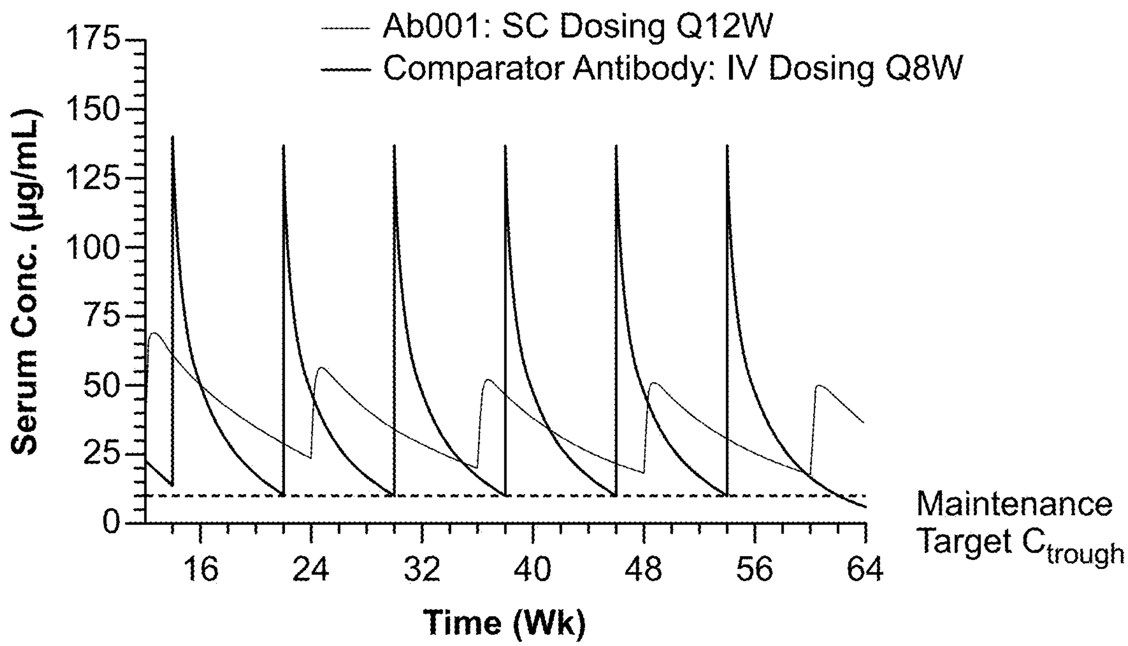

Ab001: SC Dosing Q12W
Comparator Antibody: IV Dosing Q8W
Serum Conc. (µg/mL)
175
150
125
100
75
50
25
0
16
24
32
40
48
56
64
Time (Wk)
Maintenance Target C_{trough}

FIG. 7B

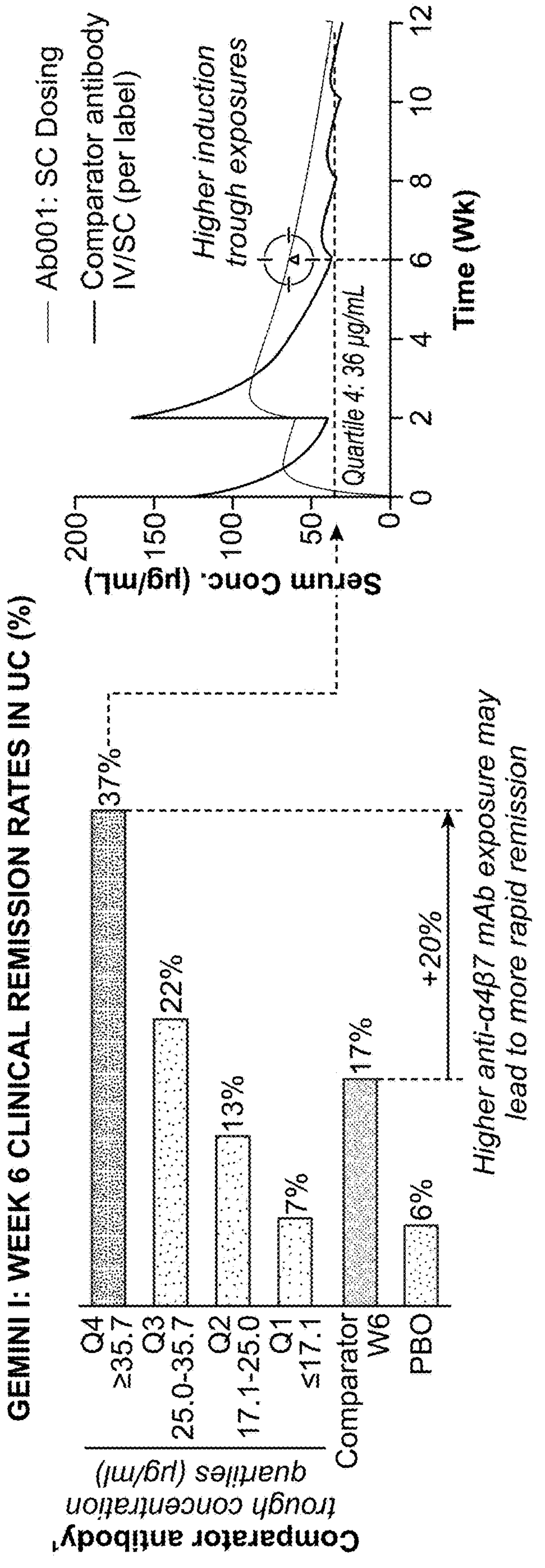

GEMINI I: WEEK 6 CLINICAL REMISSION RATES IN UC (%)
Comparator antibody¹ trough concentration quantiles (µg/ml)
Q4 ≥35.7
Q3 25.0-35.7
Q2 17.1-25.0
Q1 ≤17.1
Comparator W6
PBO
37%
22%
13%
7%
17%
6%
+20%
Higher anti-α4β7 mAb exposure may lead to more rapid remission
Ab001: SC Dosing
Comparator antibody IV/SC (per label)
Higher induction trough exposures
Quartile 4: 36 µg/mL
Serum Conc. (µg/mL)
Time (Wk)
200
150
100
50
0
0
2
4
6
8
10
12

FIG. 7C

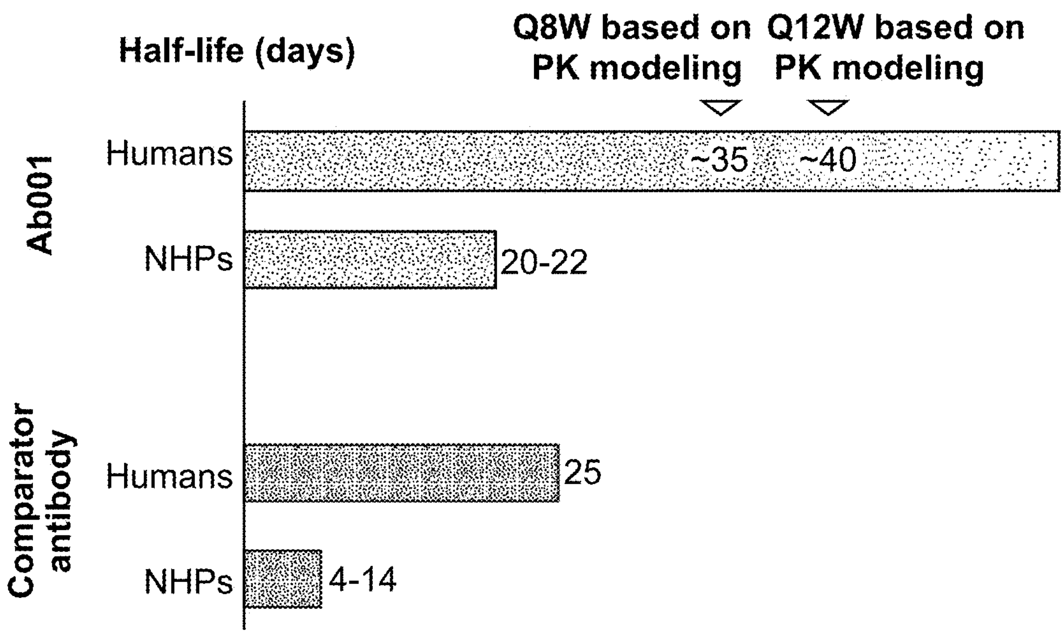

Half-life (days)
Q8W based on PK modeling
Q12W based on PK modeling
Ab001
Humans
~35
~40
NHPs
20-22
Comparator antibody
Humans
25
NHPs
4-14

FIG. 7D

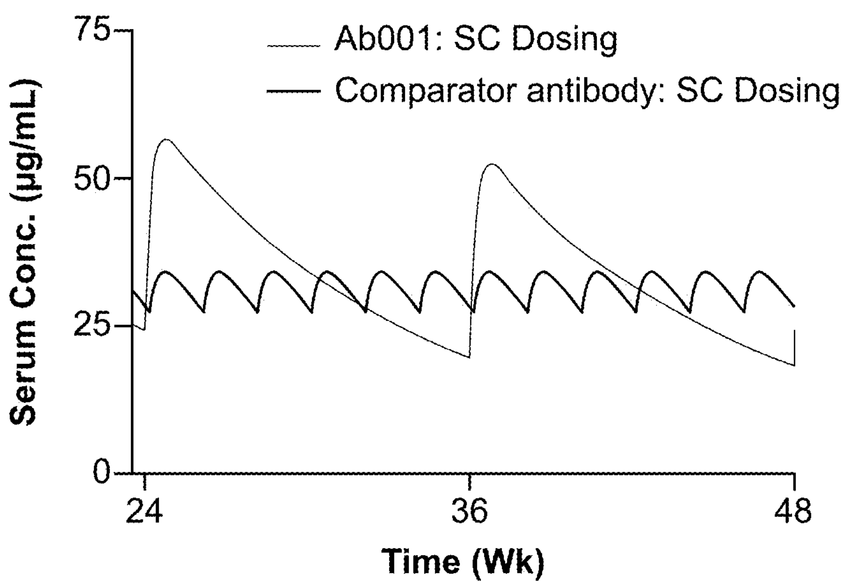

Ab001: SC Dosing
Comparator antibody: SC Dosing
Serum Conc. (µg/mL)
75
50
25
0
24
36
48
Time (Wk)

FIG. 7E

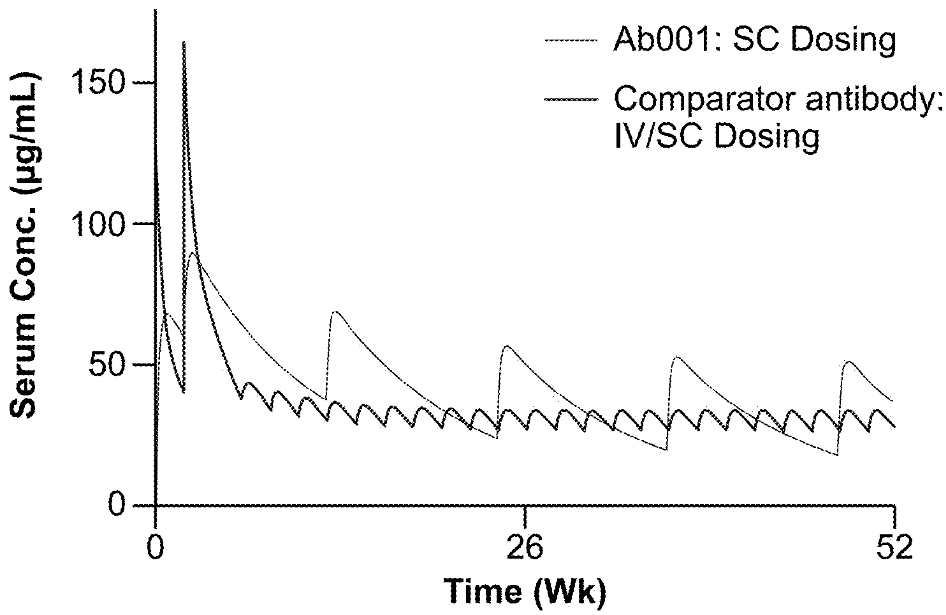

Ab001: SC Dosing
Comparator antibody: IV/SC Dosing
Serum Conc. (µg/mL)
150
100
50
0
0
26
52
Time (Wk)

FIG. 7F

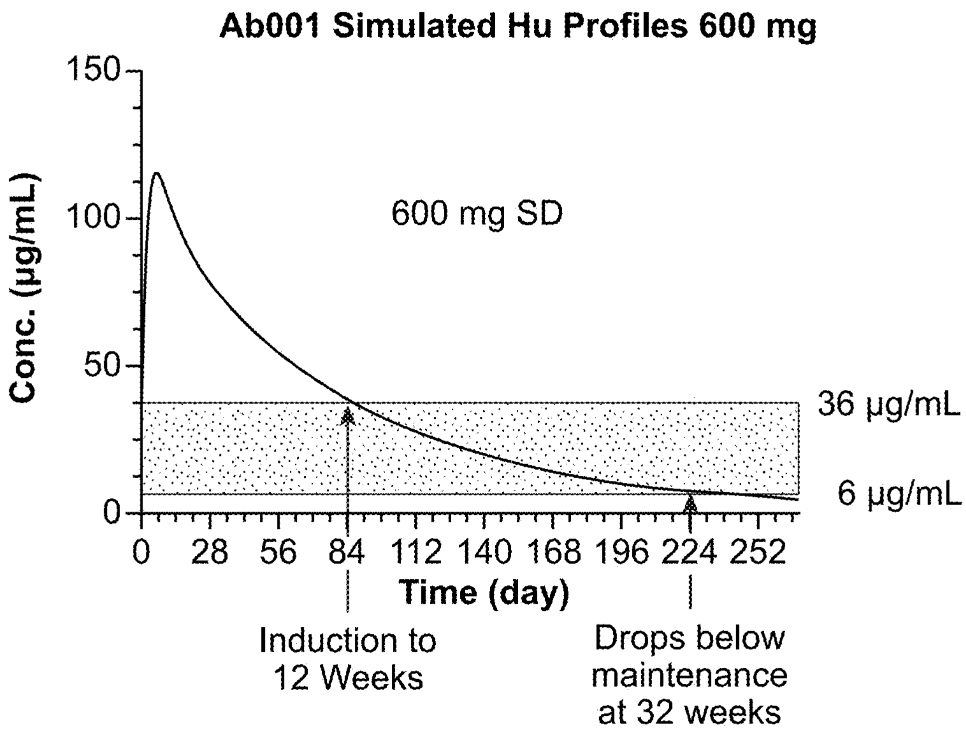

Ab001 Simulated Hu Profiles 600 mg
600 mg SD
Conc. (µg/mL)
150
100
50
0
36 µg/mL
6 µg/mL
0 28 56 84 112 140 168 196 224 252
Time (day)
Induction to 12 Weeks
Drops below maintenance at 32 weeks

FIG. 8A

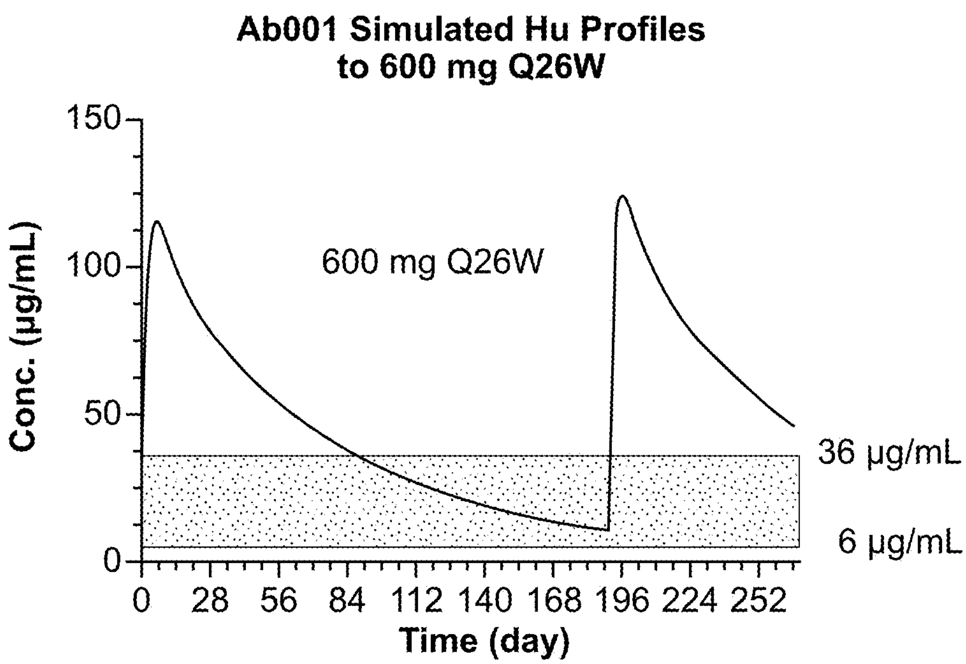

Ab001 Simulated Hu Profiles to 600 mg Q26W
600 mg Q26W
Conc. (µg/mL)
150
100
50
0
36 µg/mL
6 µg/mL
0 28 56 84 112 140 168 196 224 252
Time (day)

FIG. 8B

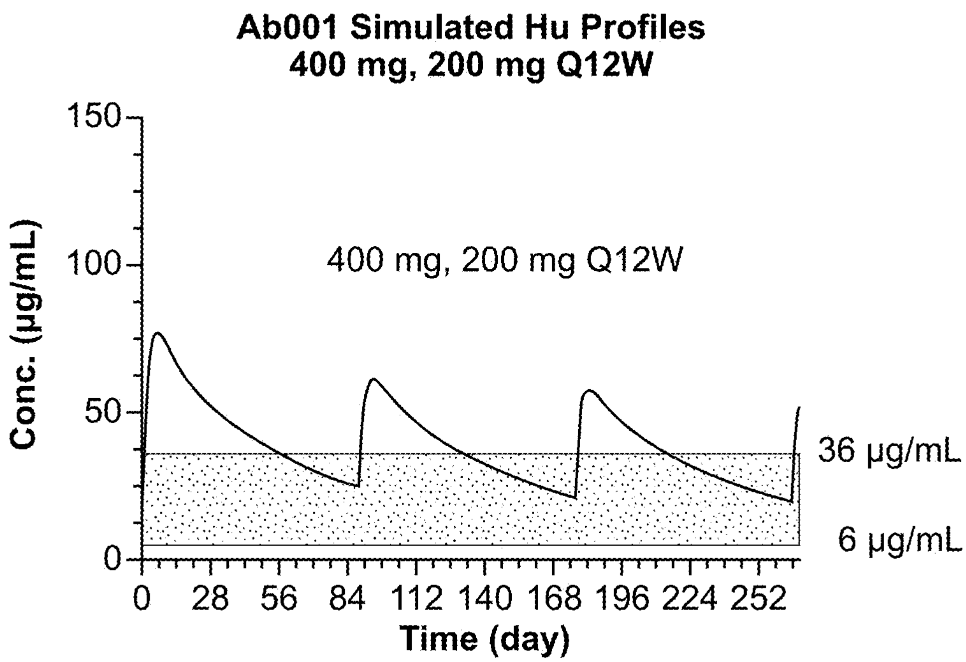

Ab001 Simulated Hu Profiles
400 mg, 200 mg Q12W
Conc. (µg/mL)
150
100
50
0
400 mg, 200 mg Q12W
36 µg/mL
6 µg/mL
0
28
56
84
112
140
168
196
224
252
Time (day)

FIG. 8C

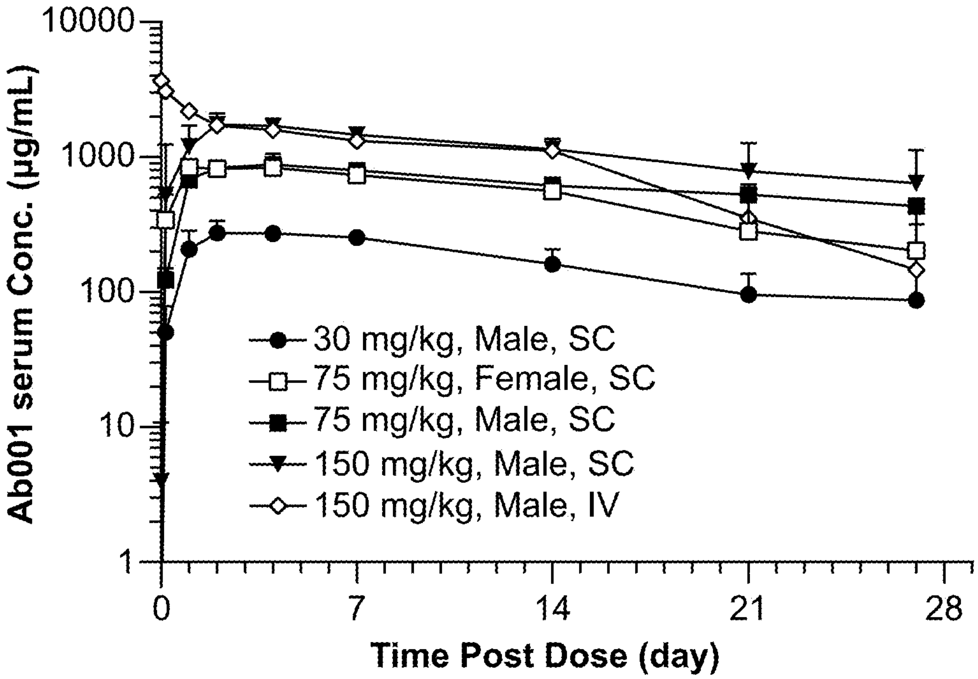

10000
1000
100
10
1
Ab001 serum Conc. (µg/mL)
0
7
14
21
28
Time Post Dose (day)
30 mg/kg, Male, SC
75 mg/kg, Female, SC
75 mg/kg, Male, SC
150 mg/kg, Male, SC
150 mg/kg, Male, IV

FIG. 9

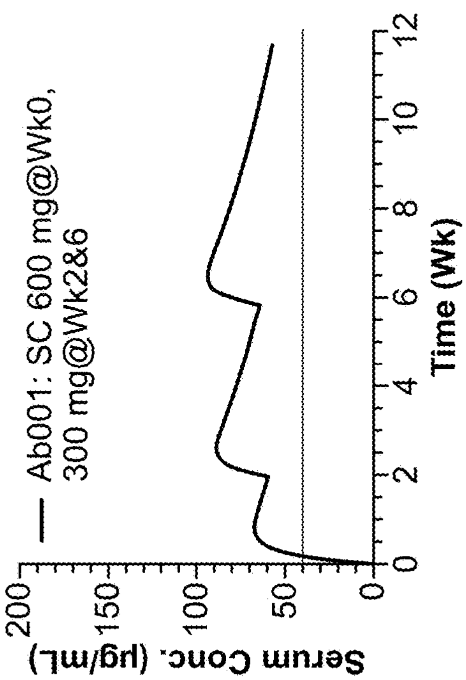


FIG. 10B

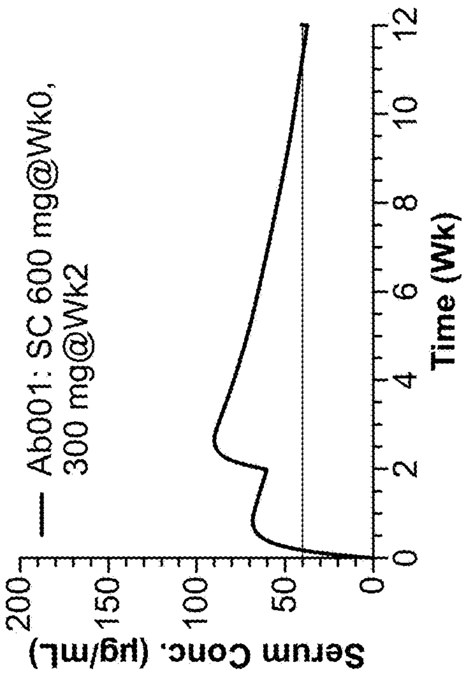


FIG. 10A

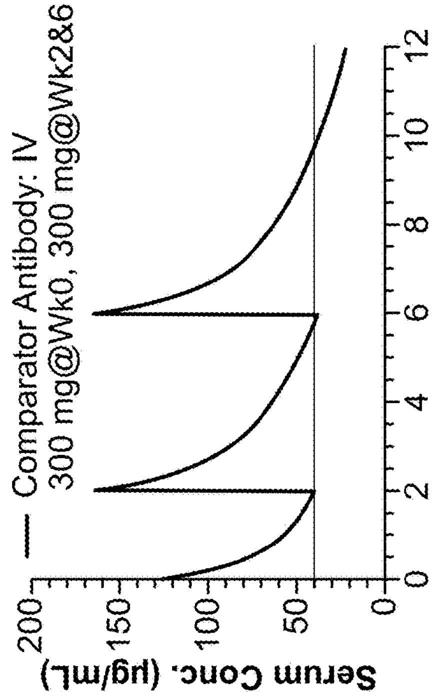


FIG. 10C

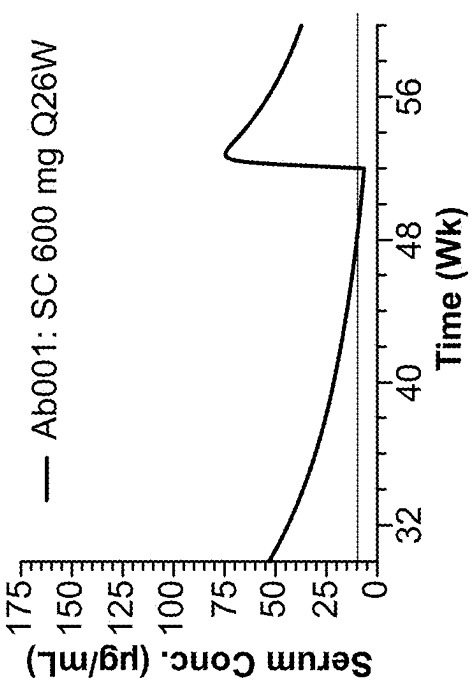


FIG. 11A

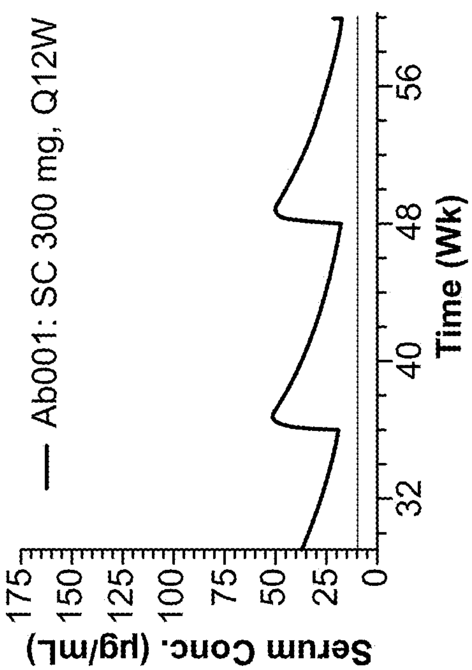


FIG. 11B

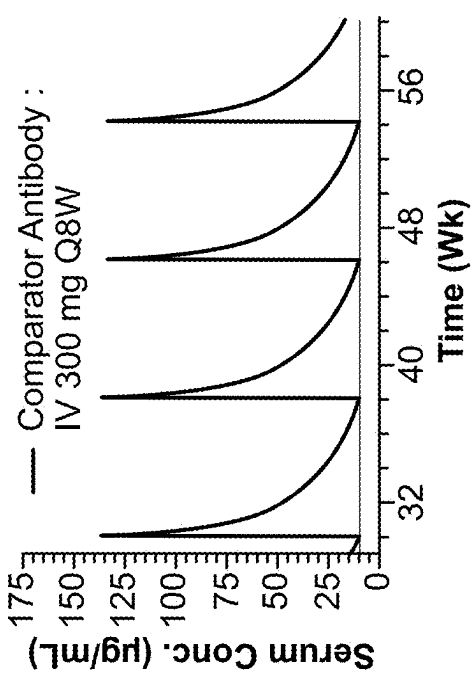


FIG. 11C

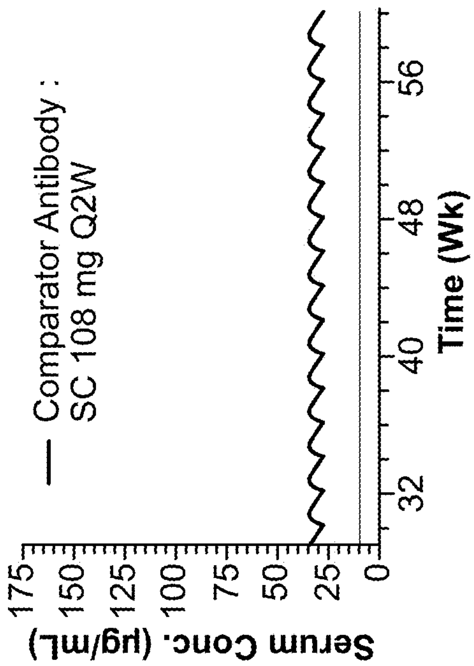


FIG. 11D

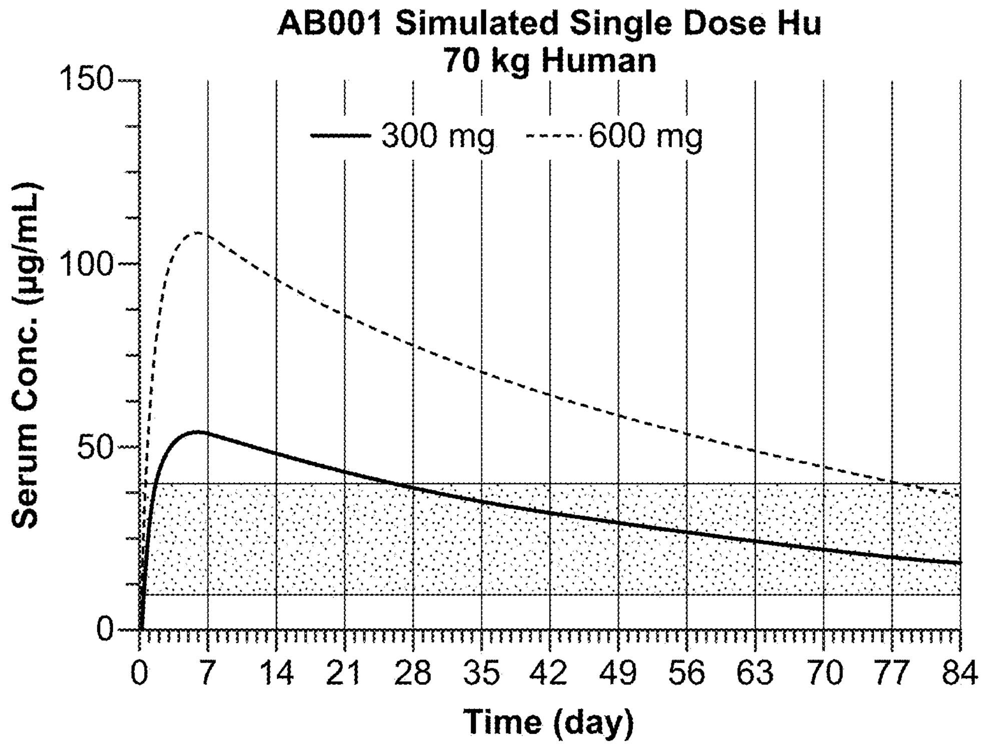

AB001 Simulated Single Dose Hu
70 kg Human
300 mg
600 mg
Serum Conc. (µg/mL)
150
100
50
0
0
7
14
21
28
35
42
49
56
63
70
77
84
Time (day)

FIG. 12

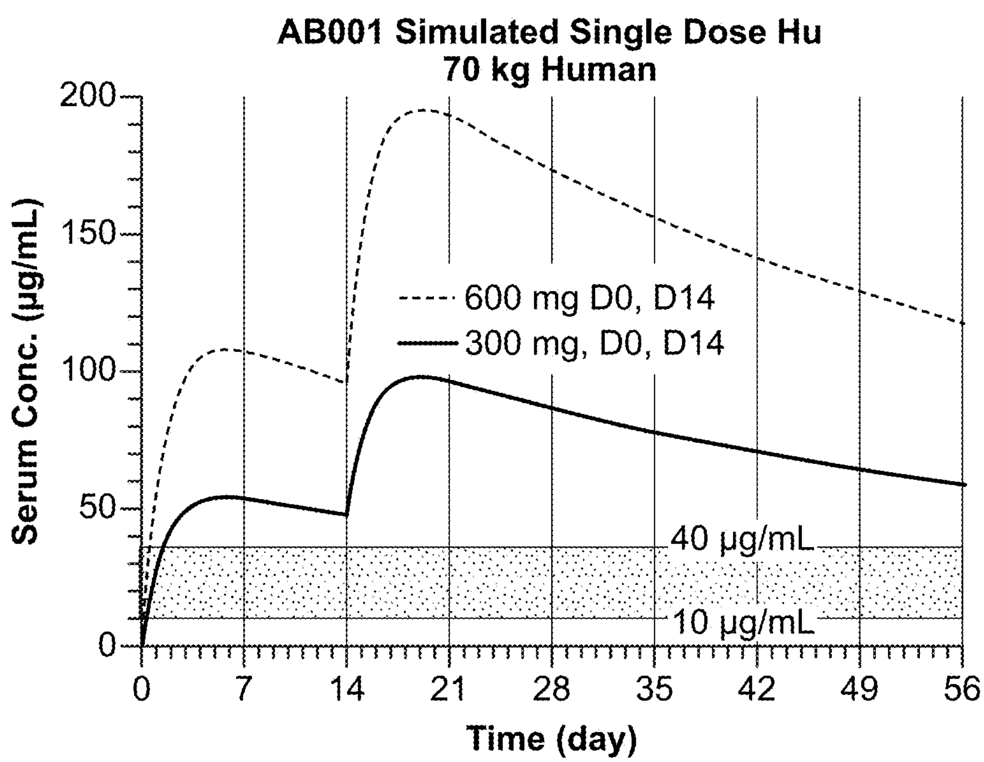

AB001 Simulated Single Dose Hu
70 kg Human
600 mg D0, D14
300 mg, D0, D14
40 µg/mL
10 µg/mL
Serum Conc. (µg/mL)
200
150
100
50
0
0
7
14
21
28
35
42
49
56
Time (day)

FIG. 13

METHODS OF TREATING GASTROINTESTINAL INFLAMMATORY DISEASE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 18/774,123, filed Jul. 16, 2024, which is a continuation of International Application No. PCT/US2024/031569, filed May 30, 2024, which claims the benefit of and priority to U.S. Provisional Application No. 63/504,966, filed on May 30, 2023; U.S. Provisional Application No. 63/505,962, filed on Jun. 2, 2023; U.S. Provisional Application No. 63/599,922, filed on Nov. 16, 2023; U.S. Provisional Application No. 63/554,886, filed on Feb. 16, 2024; and U.S. Provisional Application No. 63/559,081, filed on Feb. 28, 2024, the entire contents of each of which are incorporated herein by reference.

SEQUENCE LISTING

This application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. The XML copy, created on Apr. 14, 2025, is titled 220703-010211.xml and is 23.8 kilobytes in size.

BACKGROUND

Integrins are cell-adhesion transmembrane receptors that function as extracellular matrix (ECM)-cytoskeletal linkers and transducers of biochemical and mechanical signals between cells and their environment. Due to their exposure on the cell surface and sensitivity to molecular inhibition, integrins such as $\alpha4\beta7$ integrins have been investigated as pharmacological targets for treating various diseases including cancer and inflammatory diseases (e.g., inflammatory bowel disease). However, current integrin therapies have been associated with serious side effects given the role of integrins in important biological processes and/or require multiple and frequent doses to maintain therapeutic efficacy. As such, improved $\alpha4\beta7$ therapies are needed.

SUMMARY

The disclosure provides, in one aspect, a multidose regimen for use in treating a disease in a subject in need thereof comprising: (a) a first injectable liquid formulation comprising a total dosage amount of at least 500 mg of an $\alpha4\beta7$ binding antibody; and (b) a second injectable liquid formulation comprising a total dosage amount of at least 120 mg of the $\alpha4\beta7$ binding antibody, wherein the $\alpha4\beta7$ binding antibody consists of: (i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and (ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In certain embodiments, the first injectable liquid formulation comprises a total dosage amount of from about 500 mg to about 1200 mg of the $\alpha4\beta7$ binding antibody and the second injectable liquid formulation comprises a total dosage amount of from about 120 mg to about 450 mg of the $\alpha4\beta7$ binding antibody.

In some embodiments, the first injectable liquid formulation comprises a total dosage amount of about 600 mg of the $\alpha4\beta7$ binding antibody and the second injectable liquid

formulation comprises a total dosage amount of about 300 mg of the $\alpha4\beta7$ binding antibody.

In some embodiments, the first injectable liquid formulation is for intravenous administration.

In certain embodiments, the first injectable liquid formulation is for subcutaneous administration. In some embodiments, the first injectable liquid formulation is suitable for administration as a single injection or multiple injections.

In certain embodiments, the second injectable liquid formulation is for subcutaneous administration. In some embodiments, the second injectable liquid formulation is suitable for administration as a single injection or multiple injections.

In some embodiments, the disease is an inflammatory bowel disease. In some embodiments, the inflammatory bowel disease is Crohn's disease. In some embodiments, the inflammatory bowel disease is ulcerative colitis.

In one aspect, the disclosure provides a dosing regimen for use in treating a disease in a subject in need thereof, the dosing regimen comprising: (a) a first formulation comprising a total dosage amount of at least about 500 mg of an $\alpha4\beta7$ binding antibody for administration to the subject; (b) a second formulation comprising a total dosage amount of at least about 120 mg of the $\alpha4\beta7$ binding antibody for subcutaneous administration to the subject after the first formulation, and thereafter as a maintenance dose at least eight weeks after administration of the second formulation, wherein the $\alpha4\beta7$ binding antibody consists of: (i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and (ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, the first formulation is for subcutaneous administration.

In some embodiments, the first formulation is for intravenous administration.

In certain embodiments, the first formulation comprises a total dosage amount of from about 500 mg to about 1200 mg of the $\alpha4\beta7$ binding antibody. For example, in some embodiments, the first formulation comprises a total dosage amount of about 600 mg of the $\alpha4\beta7$ binding antibody.

In certain embodiments, the second formulation comprises a total dosage amount of from about 120 mg to about 450 mg of the $\alpha4\beta7$ binding antibody. For example, in some embodiments, the second formulation comprises a total dosage amount of about 300 mg of the $\alpha4\beta7$ binding antibody.

In certain embodiments, the second formulation is administered at least two weeks after the first formulation, and thereafter as a maintenance dose at least eight weeks after administration of the second formulation.

In some embodiments, the first formulation is suitable for administration as a single injection or multiple injections. In some embodiments, the second formulation is suitable for administration as a single injection or multiple injections.

In certain embodiments, the first formulation and the second formulation do not contain citrate.

In certain embodiments, the disease is an inflammatory bowel disease. In some embodiments, the inflammatory bowel disease is Crohn's disease. In some embodiments, the inflammatory bowel disease is ulcerative colitis.

In another aspect, the disclosure provides an injectable dosage form of an $\alpha4\beta7$ binding antibody consisting of (a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and (b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3, wherein the $\alpha4\beta7$ binding antibody has an average serum half-life greater than about 10 days in cynomolgus monkeys.

In some embodiments, the average serum half-life is about 17 days or greater in cynomolgus monkeys.

In certain embodiments, the injectable dosage form is an injectable liquid formulation.

In certain embodiments, the α4β7 binding antibody is present at a concentration of at least 180 mg/mL.

In some embodiments, the injectable liquid formulation does not comprise citrate.

In one aspect, the disclosure provides a method of treating a disease in a patient in need thereof, the method comprising: administering to a subject in need thereof (a) an effective amount of an induction dose of an α4β7 binding antibody, and (b) an effective amount of one or more maintenance doses of the α4β7 binding antibody, wherein the one or more maintenance doses are administered subcutaneously at least eight weeks apart, wherein the α4β7 binding antibody consists of: (a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and (b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In certain embodiments, the disease is an inflammatory bowel disease. In some embodiments, the inflammatory bowel disease is Crohn's disease. In some embodiments, the inflammatory bowel disease is ulcerative colitis.

In certain embodiments, the effective amount of the induction dose of the α4β7 binding antibody is at least 600 mg, and the effective amount of each of the one or more maintenance doses of the α4β7 binding antibody is at least 300 mg.

In certain embodiments, the effective amount of the induction dose of the α4β7 binding antibody is about 600 mg and the effective amount of each of the one or more maintenance doses of the α4β7 binding antibody is about 300 mg.

In some embodiments, the effective amount of the induction dose of the α4β7 binding antibody is administered subcutaneously.

In some embodiments, the effective amount of the induction dose of the α4β7 binding antibody is administered intravenously.

In some embodiments, the one or more maintenance doses are administered subcutaneously about 12 weeks apart.

In some embodiments, the one or more maintenance doses are administered subcutaneously about 26 weeks apart.

In some embodiments, the induction dose comprises a dosage amount of the α4β7 binding antibody that is about 2 or more times higher than a dosage amount of each of the one or more maintenance doses.

In one aspect, the disclosure provides a method of achieving C_{trough} of 35 μg/mL or greater for a an α4β7 binding antibody in a subject in need thereof six weeks following administration of the α4β7 binding antibody to the subject, the method comprising administering to the subject in need thereof at least about 600 mg of the α4β7 binding antibody, wherein the α4β7 binding antibody consists of: (a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and (b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In certain embodiments, from about 600 mg to about 1200 mg of the α4β7 binding antibody is administered intravenously.

In certain embodiments, from about 600 mg to about 1200 mg of the α4β7 binding antibody is administered subcutaneously.

In some embodiments, about 600 mg of the α4β7 binding antibody is administered intravenously.

In some embodiments, about 600 mg of the α4β7 binding antibody is administered subcutaneously.

In some embodiments, the method further comprises administering to the subject in need thereof at least about 300 mg of the α4β7 binding antibody, wherein the administration of the at least about 300 mg of the α4β7 binding antibody results in a steady state C_{trough} of at least 6 μg/mL. In certain embodiments, the administration results in a serum $C_{avgW0\text{-}W12}$ of at least 60 g/mL.

In some embodiments, the administration of the at least about 600 mg of the α4β7 binding antibody results in a serum $C_{avgW30\text{-}W40}$ of at least 30 g/mL.

In some embodiments, the method further comprises administering to the subject in need thereof at least about 300 mg of the α4β7 binding antibody and wherein administration results in an average serum concentration of the α4β7 binding antibody above 35 g/mL for ten weeks following the administration of the at least about 300 mg of the α4β7 binding antibody.

In some embodiments, the method further comprises administering subcutaneously to the subject in need thereof at least about 300 mg of the α4β7 binding antibody four weeks after administration of last dosage amount of the α4β7 binding antibody.

In some embodiments, the method further comprises administering subcutaneously to the subject a maintenance dose of at least about 600 mg of the α4β7 binding antibody every twenty-six weeks after administration of the last dose of the α4β7 binding antibody.

In one aspect, the disclosure provides an α4β7 binding antibody consisting of: (a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and (b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, the disclosure provides an isolated nucleic acid encoding the heavy chain and/or light chain of the α4β7 binding antibody, which antibody consists of (a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and (b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the isolated nucleic acid encodes SEQ ID NO: 1. In some embodiments, the isolated nucleic acid encodes SEQ ID NO: 3.

In certain embodiments, the disclosure provides a recombinant host cell comprising an isolated nucleic acid encoding the heavy chain and/or light chain of the α4β7 binding antibody, which antibody consists of (a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and (b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the host cell comprises an isolated nucleic acid encodes SEQ ID NO: 1. In some embodiments, the host cell comprises an isolated nucleic acid encodes SEQ ID NO: 3.

In certain embodiments, the disclosure provides an expression vector comprising an isolated nucleic acid encoding the heavy chain and/or light chain of the α4β7 binding antibody, which antibody consists of (a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and (b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the expression vector comprises an isolated nucleic acid encodes SEQ ID NO: 1. In some embodiments, the expression vector comprises an isolated nucleic acid encodes SEQ ID NO: 3.

In some embodiments, the disclosure provides a multi-dose regimen, a dosing regimen, an injectable dosage form, and/or a method, each as provided herein, wherein the disease in a gastrointestinal inflammatory disease.

In certain embodiments, the disclosure provides a multi-dose regimen, a dosing regimen, an injectable dosage form, a method, and/or an antibody, each as provided herein, wherein the α4β7 binding antibody has an average serum half-life greater than about 10 days in cynomolgus monkeys. In some embodiments, the average serum half-life is about 17 days or greater in cynomolgus monkeys.

In certain embodiments, the disclosure provides a multi-dose regimen, a dosing regimen, an injectable dosage form, a method, and/or an antibody, each as provided herein, wherein the α4β7 binding antibody has one or more of the following characteristics: (a) a melting temperature TmOnset greater than about 55° C., (b) a melting temperature TmOnset greater than 56.2° C., (c) a proportion of GOF from about 55.1% to about 60.4%, (d) a proportion of G1F from about 16.6% to about 18.8%, (e) a proportion of Man5 from about 7.3% to about 8.8%, (f) does not induce T cell activation markers CD25 or CD69, (g) does not result in release of cytokines, which cytokines are one or more of IL-6, IL-8, IL-10, IFNγ, IL-4, IL-17, IL-2, IL-23p70 and TNF, (h) does not induce complement-dependent cytotoxicity (CDC) in primary human PBMCs and in α4β7-expressing human β-lymphoid cell, (i) does not induce antibody dependent cellular cytotoxicity (ADCC) in human NK cells, and (j) does not impact suppressive activity of regulatory T cells expressing α4β7 integrin, wherein the suppressive activity is measured by presence of elevated CD71, CD25, Ki67, granzyme B, or OX40.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A depicts cell binding an α4β7 binding antibody. Cell binding was determined (Relative MFI) for increasing concentrations of α4β7 binding antibody Ab001 (dashed line, open shapes) or a comparator antibody (solid line, filled shapes) to PBMCs from three donors (Donor 1=circles; Donor 2=squares; Donor 3=triangles). EC_{50} values were calculated for each donor combination and indicated in Table 2.

FIG. **1**B and FIG. **1**C depict binding of the α4β7 binding antibody Ab001 and a comparator antibody to α4β7-expressing peripheral blood mononuclear cells. FIG. **1**B: Labeled (AlexaFluor-647) Ab001 binds to α4β7-expressing cells isolated from PBMCs. FIG. **1**C: Modification competing off AF647-Ab001 with unlabeled Ab001 demonstrates receptor occupancy. N=3 donors.

FIGS. **2**A and **2**B depict blocking of MAdCAM-1 and VCAM-1 adhesion by an α4β7 binding antibody Ab001 (i.e., α4β7 binding antibody comprising a heavy chain consisting of a sequence according to SEQ ID NO: 1 and a light chain consisting of a sequence according to SEQ ID NO: 3). FIG. **2**A depicts the percent inhibition of total adhesion of integrin-mediated adhesion by MAdCAM-1. The percent inhibition was calculated after treatment with increasing concentrations of either an α4β7 binding antibody or comparator antibody. The IC_{50} of the α4β7 binding antibody was 86 μM and the IC_{50} of the comparator antibody was 83 μM. FIG. **2**B depicts the percent inhibition of total adhesion of integrin-mediated adhesion by VCAM-1, a potential off-target interaction. The percent inhibition was calculated after treatment with increasing concentrations of either an α4β7 binding antibody, comparator antibody or a positive control antibody. The positive control antibody was able to inhibit adhesion by VCAM-1, but neither the α4β7 binding antibody nor the comparator antibody were able to.

FIG. **3** depicts the half-life of an α4β7 binding antibody Ab001 in non-human primates (NHPs). NHPs were intravenously injected with either an α4β7 binding antibody or comparator antibody and monitored out to 21 days post-injection. Normalized concentration of α4β7 binding antibody or comparator antibody was calculated at each time-point and used to determine the β-elimination half-life. Also plotted is data from an external study of the comparator antibody (dashed line) illustrating comparable results.

FIG. **4** depicts simulated maintenance of trough concentrations at 6 weeks post-induction by IV associated with higher rates of remission in ulcerative colitis. The percentage of patients with C_{trough} values ≤17, 17-25, 25-35, and ≥35 μg/mL were plotted in a stacked bar chart for patients treated with a comparator antibody during a clinical trial at 6 weeks post-induction. Also shown are the percentage of patients with C_{trough} values from a simulated trial of 1000 patients treated with the comparator antibody, and an α4β7 binding antibody described herein (e.g., Ab001) at doses of 300 mg, 350 mg, and 400 mg.

FIG. **5** depicts quarterly maintenance dosing of α4β7 binding antibody to maintain clinically relevant AUC levels. Relative $AUC_{ss,1\ year}$ values were calculated using simulated trial of 1000 patients for an α4β7 binding antibody described herein (e.g., Ab001) with a half-life of 35 or 50 days and compared to clinical trial data for intravenously (IV) and subcutaneously (SC) dosed comparator antibody. Relative $AUC_{ss,1\ year}$ values were calculated for a quarterly dosing frequency ranging from 2 weeks (Q2W) to 16 weeks (Q16W).

FIG. **6** depicts data from patients maintaining clinically relevant C_{trough} values during maintenance dosing. In a simulated trial of 1000 patients, the percent of patients with a C_{trough}-6 μg/ml was calculated for patients treated with comparator antibody dosed intravenously (IV; 300 mg dose) and subcutaneously (SC; 108 mg or 320 mg dose) and patients subcutaneously dosed with 300 mg of an α4β7 binding antibody described herein (e.g., Ab001) with a half-life of 43, 53 or 56 days. The percentage of patients with C_{trough}≥6 μg/ml were plotted for a dosing frequency ranging from Q2W to Q20W.

FIGS. **7**A and **7**B depict simulation of serum concentrations of the α4β7 binding antibody described herein (e.g., Ab001) and a comparator antibody. FIG. **7**A depicts an induction and maintenance dosing regimen by subcutaneous (SC) route of administration based on dosing at W0 and W2 during induction. FIG. **7**B depicts serum concentrations based on Q8W IV dosing of comparator antibody and Q12W SC dosing of the α4β7 binding antibody described herein (e.g., Ab001) based on the half-life of the antibodies.

FIG. **7**C (left) is a graph showing week 6 clinical remission rates in ulcerative colitis (%) (left); FIG. **7**C (right) depicts induction simulation of serum concentrations of the α4β7 binding antibody described herein (e.g., Ab001) and a comparator antibody.

FIG. **7**D is a graph showing human half-life predictions by NHP allometric scaling.

FIG. **7**E depicts maintenance simulation of serum concentrations of the α4β7 binding antibody described herein (e.g., Ab001, 300 mg Q12W) and a comparator antibody (108 mg Q2W) when administered subcutaneously (300 mg Q12W for Ab001, 108 mg Q2W for comparator antibody).

FIG. **7**F show depicts simulation of serum concentrations of the α4β7 binding antibody described herein (e.g., Ab001, SC dosing, 4-6 injections per year for maintenance) and a comparator antibody (IV/SC dosing, 26 injections per year for maintenance).

FIGS. **8**A-**8**C are graphs showing human PK simulation (Simulated Using 75 mg/kg NHP—SC Dose (n=6, M/F)). FIG. **8**A is a graph showing the α4β7 binding antibody described herein (e.g., Ab001) simulated human profiles at 600 mg. FIG. **8**B is a graph showing α4β7 binding antibody described herein (e.g., Ab001) simulated human profiles 600 mg Q26W. FIG. **8**C is a graph showing α4β7 binding antibody described herein (e.g., Ab001) simulated human profiles 400 mg, 200 mg, Q12W.

FIG. **9** is a graph showing serum concentration time profiles following subcutaneous and intravenous single-dose administration of α4β7 binding antibody Ab001 in male and female monkeys. The male and female monkeys were administered a single subcutaneous dose of either 30 mg/kg, 75 mg/kg or 150 mg/kg of Ab001, or a single intravenous dose of 150 mg/kg of Ab001.

FIGS. **10**A-**10**C are graphs showing human PK induction simulation studies for α4β7 binding antibody Ab001 relative to comparator antibody based on NHP $t_{1/2}$ data. FIG. **10**A is a graph showing induction simulation of serum concentrations of Ab001 in human at 600 mg of subcutaneous dose at week 0 and 300 mg of subcutaneous dose at week 2. FIG. **10**B is a graph showing induction simulation of serum concentrations of Ab001 in human at 600 mg of subcutaneous dose at week 0, and 300 mg of subcutaneous doses at week 2 and week 6. FIG. **10**C is a graph showing induction simulation of serum concentrations of the comparator antibody in human at 300 mg of intravenous dose at week 0, week 2 and week 6.

FIGS. **11**A-**11**D are graphs showing human PK maintenance simulation studies for α4β7 binding antibody Ab001 relative to the comparator antibody based on NHP $t_{1/2}$ data. FIG. **11**A is a graph showing maintenance simulation of serum concentrations of Ab001 in human at 600 mg of subcutaneous dose every 26 weeks. FIG. **11**B is a graph showing maintenance simulation of serum concentrations of Ab001 in human at 300 mg of subcutaneous dose every 12 weeks. FIG. **11**C is a graph showing maintenance simulation of serum concentrations of the comparator antibody in human at 300 mg of intravenous dose every 8 weeks. FIG. **11**D is a graph showing maintenance simulation of serum concentrations of the comparator antibody in human at 108 mg of subcutaneous dose every 2 weeks.

FIG. **12** is a graph showing human PK simulation profile for α4β7 binding antibody described herein such as Ab001 at 600 mg and 300 mg in a single ascending dose study.

FIG. **13** is a graph showing human PK simulation profile for α4β7 binding antibody described herein such as Ab001 at 600 mg and 300 mg in a multiple ascending dose study.

DETAILED DESCRIPTION

To facilitate an understanding of the present disclosure, a number of terms and phrases are defined below.

As used herein, unless otherwise indicated, the term "antibody" is understood to mean an intact antibody (e.g., an intact monoclonal antibody), or a fragment thereof, such as a Fc fragment of an antibody (e.g., an Fc fragment of a monoclonal antibody), or an antigen-binding fragment of an antibody (e.g., an antigen-binding fragment of a monoclonal antibody), including an intact antibody, antigen-binding fragment, or Fc fragment that has been modified, engineered, or chemically conjugated. In general, antibodies are multimeric proteins that contain four polypeptide chains. Two of the polypeptide chains are called immunoglobulin heavy chains (H chains), and two of the polypeptide chains are called immunoglobulin light chains (L chains). The immunoglobulin heavy and light chains are connected by an interchain disulfide bond. The immunoglobulin heavy chains are connected by interchain disulfide bonds. A light chain consists of one variable region (VL) and one constant region (CL). The heavy chain consists of one variable region (VH) and at least three constant regions (CH1, CH2 and CH3). The variable regions determine the binding specificity of the antibody. Each variable region contains three hypervariable regions known as complementarity determining regions (CDRs) flanked by four relatively conserved regions known as framework regions (FRs). The extent of the FRs and CDRs has been defined (Kabat, E. A., et al. (1991) Sequences of Proteins of Immunological Interest, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242; and Chothia, C. et al. (1987) J. Mol. Biol. 196:901-917). The three CDRs, referred to as CDR1, CDR2, and CDR3, contribute to the antibody binding specificity. Naturally occurring antibodies have been used as starting material for engineered antibodies, such as chimeric antibodies and humanized antibodies. Examples of antibody-based antigen-binding fragments include Fab, Fab', (Fab')2, Fv, single chain antibodies (e.g., scFv), minibodies, and diabodies. Examples of antibodies that have been modified or engineered include chimeric antibodies, humanized antibodies, and multispecific antibodies (e.g., bispecific antibodies). An example of a chemically conjugated antibody is an antibody conjugated to a toxin moiety.

The terms "variable domain" and "variable region" are used interchangeably and refer to the portions of the antibody or immunoglobulin domains that exhibit variability in their sequence and that are involved in determining the specificity and binding affinity of a particular antibody. Variability is not evenly distributed throughout the variable domains of antibodies; it is concentrated in sub-domains of each of the heavy and light chain variable regions. These sub-domains are called "hypervariable regions" or "complementarity determining regions" (CDRs). The more conserved (i.e., non-hypervariable) portions of the variable domains are called the "framework" regions (FRM or FR) and provide a scaffold for the six CDRs in three-dimensional space to form an antigen-binding surface.

An "Fc polypeptide" of a dimeric Fc as used herein refers to one of the two polypeptides forming the dimeric Fc domain, i.e. a polypeptide comprising C-terminal constant regions of an immunoglobulin heavy chain, capable of stable self-association. For example, an Fc polypeptide of a dimeric IgG Fc comprises an IgG CH2 and an IgG CH3 constant domain sequence. An Fc can be of the class IgA, IgD, IgE, IgG, and IgM. These classes are also designated α, δ, ε, γ, and μ, respectively. Several of these may be further divided into subclasses (isotypes), e.g., IgG_1, IgG_2, IgG_3, IgG_4, IgA_1, and IgA_2.

The terms "Fc receptor" and "FcR" are used to describe a receptor that binds to the Fc region of an antibody. For example, an FcR can be a native sequence human FcR. Generally, an FcR is one which binds an IgG antibody (a gamma receptor) and includes receptors of the FcγRI, FcγRII, and FcγRII subclasses, including allelic variants and alternatively spliced forms of these receptors. FcγRII receptors include FcγRIIA (an "activating receptor") and FcγRIIB (an "inhibiting receptor"), which have similar amino acid sequences that differ primarily in the cytoplasmic domains thereof. Immunoglobulins of other isotypes can also be bound by certain FcRs (see, e.g., Janeway et al., *Immuno*

Biology: the immune system in health and disease, (Elsevier Science Ltd., NY) (4th ed., 1999)). Activating receptor FcγRIIA contains an immunoreceptor tyrosine-based activation motif (ITAM) in its cytoplasmic domain. Inhibiting receptor FcγRIIB contains an immunoreceptor tyrosine-based inhibition motif (ITIM) in its cytoplasmic domain (reviewed in Daeron, *Annu. Rev. Immunol.* 15:203-234 (1997)). FcRs are reviewed in Ravetch and Kinet, *Annu. Rev. Immunol* 9:457-92 (1991); Capel et al., Immunomethods 4:25-34 (1994); and de Haas et al., J. Lab. Clin. Med. 126:330-41 (1995). Other FcRs, including those to be identified in the future, are encompassed by the term "FcR" herein. The term also includes the neonatal receptor, FcRn, which is responsible for the transfer of maternal IgGs to the fetus (Guyer et al., *J. Immunol.* 117:587 (1976); and Kim et al., *J. Immunol.* 24:249 (1994)).

The terms "recipient", "individual", "subject", "host", and "patient", are used interchangeably herein and in some embodiments, refer to any mammalian subject for whom diagnosis, treatment, or therapy is desired, particularly humans. "Mammal" for purposes of treatment refers to any animal classified as a mammal, including humans, domestic and farm animals, and laboratory, zoo, sports, or pet animals, such as dogs, horses, cats, cows, sheep, goats, pigs, mice, rats, rabbits, guinea pigs, monkeys etc. In some embodiments, the mammal is human. None of these terms require the supervision of medical personnel.

As used herein, the term "effective amount" refers to the amount of a compound (e.g., a compound of the present disclosure) sufficient to effect beneficial or desired results. An effective amount can be administered in one or more administrations, applications or dosages and is not intended to be limited to a particular formulation or administration route. As used herein, the term "treating" includes any effect, e.g., lessening, reducing, modulating, ameliorating or eliminating, that results in the improvement of the condition, disease, disorder, and the like, or ameliorating a symptom thereof.

As used herein, the term "pharmaceutical composition" refers to the combination of an active agent with a carrier, inert or active, making the composition especially suitable for diagnostic or therapeutic use in vivo or ex vivo.

As used herein, the term "pharmaceutically acceptable carrier" refers to any of the standard pharmaceutical carriers, such as a phosphate buffered saline solution, water, emulsions (e.g., such as an oil/water or water/oil emulsions), and various types of wetting agents. The compositions also can include stabilizers and preservatives. For examples of carriers, stabilizers and adjuvants, see e.g., Martin, Remington's Pharmaceutical Sciences, 15th Ed., Mack Publ. Co., Easton, PA (1975).

The terms "a" and "an" as used herein mean "one or more" and include the plural unless the context is inappropriate.

As used herein, all numerical values or numerical ranges include whole integers within or encompassing such ranges and fractions of the values or the integers within or encompassing ranges unless the context clearly indicates otherwise. Thus, for example, reference to a range of 90-100%, includes 91%, 92%, 93%, 94%, 95%, 95%, 96%, 97%, etc., as well as 91.1%, 91.2%, 91.3%, 91.4%, 91.5%, etc., 92.1%, 92.2%, 92.3%, 92.4%, 92.5%, etc., and so forth. In another example, reference to a range of 1-5,000-fold includes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, fold, etc., as well as 1.1, 1.2, 1.3, 1.4, 1.5, fold, etc., 2.1, 2.2, 2.3, 2.4, 2.5, fold, etc., and so forth.

"About" a number, as used herein, refers to range including the number and ranging from 10% below that number to 10% above that number. "About" a range refers to 10% below the lower limit of the range, spanning to 10% above the upper limit of the range.

"Percent (%) identity" refers to the extent to which two sequences (nucleotide or amino acid) have the same residue at the same positions in an alignment. For example, "an amino acid sequence is X % identical to SEQ ID NO: Y" refers to % identity of the amino acid sequence to SEQ ID NO: Y and is elaborated as X % of residues in the amino acid sequence are identical to the residues of sequence disclosed in SEQ ID NO: Y. Generally, computer programs are employed for such calculations. Exemplary programs that compare and align pairs of sequences, include ALIGN (Myers and Miller, 1988), FASTA (Pearson and Lipman, 1988; Pearson, 1990) and gapped BLAST (Altschul et al., 1997), BLASTP, BLASTN, or GCG (Devereux et al., 1984).

Throughout the description, where compositions are described as having, including, or comprising specific components, or where processes and methods are described as having, including, or comprising specific steps, it is contemplated that, additionally, there are compositions of the present disclosure that consist essentially of, or consist of, the recited components, and that there are processes and methods according to the present disclosure that consist essentially of, or consist of, the recited processing steps.

As a general matter, compositions specifying a percentage are by weight unless otherwise specified. Further, if a variable is not accompanied by a definition, then the previous definition of the variable controls.

α4β7 Integrin Binding Antibodies

Provided herein are α4β7 binding antibodies (e.g., α4β7 binding proteins, long acting α4β7 binding antibodies, long acting α4β7 binding molecules or long acting α4β7 targeting molecules. The α4β7 binding antibodies provided herein have a binding specificity for target antigen human α4β7 integrin). α4β7 binding antibodies described herein may have improved life and/or specificity. α4β7 binding antibodies described herein may have specificity for α4β7 and not related integrins including α4β1 and αEβ7.

In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 80% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 85% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 90% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 95% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 96% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 97% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 98% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 99% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having 100% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2. In a specific embodiment, α4β7 binding antibody comprises or consists of a heavy chain comprising an amino acid sequence having 100% sequence identity with an amino acid sequence according to SEQ ID NO: 1, including lacking a C-terminal lysine residue.

In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 80% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 85% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 90% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 95% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 96% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 97% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 98% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 99% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having 100% sequence identity with an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, the α4β7 binding antibody comprises a heavy chain that comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100%) identical to the heavy chain of an α4β7 binding antibody disclosed in Table 1, and a light chain that comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100%) identical to the light chain of the same α4β7 binding antibody disclosed in Table 1. In a specific embodiment of such a heavy chain, there is no C-terminal lysine residue.

In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 80% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and a light chain comprising an amino acid sequence having at least 80% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 85% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and a light chain comprising an amino acid sequence having at least 85% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 90% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and a light chain comprising an amino acid sequence having at least 90% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 95% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and a light chain comprising an amino acid sequence having at least 95% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 96% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and a light chain comprising an amino acid sequence having at least 96% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 97% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and a light chain comprising an amino acid sequence having at least 97% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 98% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and a light chain comprising an amino acid sequence having at least 98% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 99% sequence identity with an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and a light chain comprising an amino acid sequence having at least 99% sequence identity with an amino acid sequence according SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In a specific embodiment of such a heavy chain, there is no C-terminal lysine residue.

In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence having at least 80% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence having at least 85% sequence identity with an amino acid sequence according to

SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence having at least 90% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence having at least 95% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence having at least 96% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence having at least 97% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence having at least 98% sequence identity with an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence having at least 9900 sequence identity with an amino acid sequence according SEQ ID NO: 3.

In some embodiments, the antigen-binding site comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3, determined under Kabat (see Kabat et al., (1991) Sequences of Proteins of Immunological Interest, NIH Publication No. 91-3242, Bethesda), Chothia (see, e.g., Chothia C & Lesk A M, (1987), *J Mol Biol* 196: 901-917), MacCallum (see MacCallum R M et al., (1996) *J Mol Biol* 262: 732-745), IMGT (see Lefranc, (1999) *The Immunologist*, 7, 132-136), or any other CDR determination method known in the art, of the heavy chain and light chain sequences of an antibody disclosed in Table 1A.

TABLE 1A

Sequences of α4β7 binding antibodies

SEQ ID NO	Name	Sequence
1	YTE Heavy Chain	QVQLVQSGAEVKKPGASVKVSCKGSGYTFTSYWM HWVRQAPGQRLEWIGEIDPSESNTNYNQKFKGRVT LTVDISASTAYMELSSLRSEDTAVYYCARGGYDGW DYAIDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTS GGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFP AVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSN TKVDKKVEPKSCDKTHTCPPCPAPELAGAPSVFLFPP KPKDTLYITREPEVTCVVVDVSHEDPEVKFNWYVD GVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT LPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGN VFSCSVMHEALHNHYTQKSLSLSPG
2	LS Heavy Chain	QVQLVQSGAEVKKPGASVKVSCKGSGYTFTSYWM HWVRQAPGQRLEWIGEIDPSESNTNYNQKFKGRVT LTVDISASTAYMELSSLRSEDTAVYYCARGGYDGW DYAIDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTS GGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFP AVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSN TKVDKKVEPKSCDKTHTCPPCPAPELAGAPSVFLFPP KPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVD GVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT LPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGN VFSCSVLHEALHSHYTQKSLSLSPG
3	Light Chain	DVVMTQSPLSLPVTPGEPASISCRSSQSLAKSYGNTY LSWYLQKPGQSPQLLIYGISNRFSGVPDRFSGSGSGT DFTLKISRVEAEDVGVYYCLQGTHQPYTFGQGTKVE IKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPR EAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSS TLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGE C

In some embodiments, the antigen-binding site comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 disclosed in Table 1B.

TABLE 1B

SEQ ID NO	Name	Sequence
6	HCDR1	SYWMH
7	HCDR2	EIDPSESNTNYNQKFKG
8	HCDR3	GGYDGWDYAIDY
9	LCDR1	RSSQSLAKSYGNTYLS
10	LCDR2	GISNRFS
11	LCDR3	LQGTHQPYT

In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 80% sequence identity with an amino acid sequence according to SEQ ID NO: 4. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 85% sequence identity with an amino acid sequence according to SEQ ID NO: 4. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 90% sequence identity with an amino acid sequence according to SEQ ID NO: 4. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 95% sequence identity with an amino acid sequence according to SEQ ID NO: 4. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 96% sequence identity with an amino acid sequence according to SEQ ID NO: 4. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 97% sequence identity with an amino acid sequence according to SEQ ID NO: 4. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 98% sequence identity with an amino acid sequence according to SEQ ID NO: 4. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 99% sequence identity with an amino acid sequence according to SEQ ID NO: 4. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having 100% sequence identity with an amino acid sequence according to SEQ ID NO: 4.

In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 80% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 85% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 90% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 95% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 96% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 97% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 98% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having at least 99% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a light chain comprising an amino acid sequence having 100% sequence identity with an amino acid sequence according to SEQ ID NO: 5.

In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 80% sequence identity with an amino acid

sequence according to SEQ ID NO: 4; and a light chain comprising an amino acid sequence having at least 80% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 85% sequence identity with an amino acid sequence according to SEQ ID NO: 4; and a light chain comprising an amino acid sequence having at least 85% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 90% sequence identity with an amino acid sequence according to SEQ ID NO: 4; and a light chain comprising an amino acid sequence having at least 90% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 95% sequence identity with an amino acid sequence according to SEQ ID NO: 4; and a light chain comprising an amino acid sequence having at least 95% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 96% sequence identity with an amino acid sequence according to SEQ ID NO: 4; and a light chain comprising an amino acid sequence having at least 96% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 97% sequence identity with an amino acid sequence according to SEQ ID NO: 4; and a light chain comprising an amino acid sequence having at least 97% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 98% sequence identity with an amino acid sequence according to SEQ ID NO: 4; and a light chain comprising an amino acid sequence having at least 98% sequence identity with an amino acid sequence according to SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence having at least 99% sequence identity with an amino acid sequence according to SEQ ID NO: 4; and a light chain comprising an amino acid sequence having at least 99% sequence identity with an amino acid sequence according SEQ ID NO: 5. In some embodiments, the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 4; and a light chain comprising an amino acid sequence according to SEQ ID NO: 5.

In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence consisting of SEQ ID NO: 3.

In some embodiments, the α4β7 binding antibody Ab001 comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence consisting of SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody Ab001 consists of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence consisting of SEQ ID NO: 3.

The α4β7 binding antibody of the disclosure can have the following properties:		
Property	Range	Exemplary Value
Solubility (30° C.)	100 mg/ml-300 mg/ml	200 mg/ml
Solubility (5° C.)	90 mg/ml-270 mg/ml	206.5 mg/ml
Viscosity at 150 mg/ml, 25° C.	6-16 mPa	11.1 mPa (161 mg/mL, 25° C.)
pI	7.5-7.9	7.7
Expression titer	1-10	4.7 mg/mL

The α4β7 binding antibody of the disclosure can have the following N-glycan profiling (determined using enzymatically released N-glycans, labeled and monitored by hydrophilic interaction liquid chromatography (HILIC) coupled with a fluorescence detector (FLD)):

Glycan	Range	Exemplary Value
G0F	40-75%	60.4% (sample 1), 55.1% (sample 2)
G1F	10-35%	16.6% (sample 1) 18.8% (sample 2)
Man5	1-15%	7.3% (sample 1) 8.8% (sample 2)

In some embodiments, the α4β7 binding antibody comprises one or more of the following characteristics:

(a) an average serum half-life greater than about 10 days in cynomolgus monkeys,
(b) an average serum half-life of about 17 days or greater in cynomolgus monkeys,
(c) a melting temperature TmOnset greater than about 55° C.,
(d) a melting temperature TmOnset greater than 56.2° C.,
(e) does not induce T cell activation markers CD25 or CD69,
(f) does not result in release of cytokines at least 6 hours after the administration, wherein the cytokines comprise one or more of IL-6, IL-8, IL-10, IFNγ, IL-4, IL-17, IL-2, IL-23p70 and TNF,
(g) does not induce complement-dependent cytotoxicity (CDC) in primary human PBMCs and in α4β7-expressing human β-lymphoid cell,
(h) does not induce antibody dependent cellular cytotoxicity (ADCC) in human NK cells, and
(i) does not impact suppressive activity of regulatory T cells expressing α4β7 integrin, wherein the suppressive activity is measured by presence of elevated CD71, CD25, Ki67, granzyme B, or OX40.

Fc Modifications

Provided herein are α4β7 binding antibodies comprising modified Fc regions. Unless otherwise specified herein, numbering of amino acid residues in the Fc region or constant region is according to the EU numbering system, also called the EU index, as described in Kabat et al, Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD, 1991.

In some embodiments, the α4β7 binding antibodies comprise a modified Fc comprising one or more modifications. In some embodiments, the one or more modifications promote selective binding of Fc-gamma receptors.

In some embodiments, one or more modifications in the modified Fc is selected from the group consisting of: S298A, E333A, K334A, K326A, F243L, R292P, Y300L, V3051, P396L, F243L, R292P, Y300L, L235V, P396L, F243L, S239D, 1332E, A330L, S267E, L328F, D265S, S239E, K326A, A327H, G237F, K326E, G236A, D270L, H268D, S324T, L234F, N325L, V266L, and S267D. In some embodiments, one or more modifications in the modified Fc is selected from the group consisting of S228P, M252Y, S254T, T256E, T256D, T250Q, H285D, T307A, T307Q, T307R, T307W, L309D, Q411H, Q311V, A378V, E380A, M428L, N434A, N434S, N297A, D265A, L234A, L235A, and N434W.

In some embodiments, the modified Fc comprises a specific combination of amino acid substitutions selected from the group consisting of: L234A/L235A; V234A/G237A; L235A/G237A/E318A; S228P/L236E; H268Q/V309L/A330S/A331S; C220S/C226S/C229S/P238S; C226S/C229S/E3233P/L235V/L235A; L234F/L235E/P331S; C226S/P230S; L234A/G237A; L234A/L235A/G237A; L234A/L235A/P329G.

In some embodiments, the modified Fc comprises a specific combination of amino acid substitutions selected from the group consisting of M428L/N434S (LS); M252Y/S254T/T256E (YTE); T250Q/M428L; T307A/E380A/N434A; T256D/T307Q (DQ); T256D/T307W (DW); M252Y/T256D (YD); T307Q/Q311V/A378V (QVV); T256D/H285D/T307R/Q311V/A378V (DDRVV); L309D/Q311H/N434S (DHS); S228P/L235E (SPLE); L234A/L235A (LA); M428L/N434A; L234A/G237A (LAGA); L234A/L235A/G237A; L234A/L235A/P329G; D265A/YTE; LALA/YTE; LAGA/YTE; LALAGA/YTE; LALAPG/YTE; N297A/LS; D265A/LS; LALA/LS; LALAGA/LS; LALAPG/LS; N297A/DHS; D265A/DHS; LALA/DHS; LAGA/DHS; LALAGA/DHS; LALAPG/DHS; SP/YTE; SPLE/YTE; SP/LS; SPLE/LS; SP/DHS; SPLE/DHS; N297A/LA; D265A/LA; LALA/LA; LAGA/LA; LALAGA/LA; LALAPG/LA; N297A/N434A; D265A/N434A; LALA/N434A; LAGA/N434A; LALAGA/N434A; LALAPG/N434A; N297A/N434W; D265A/N434W; LALA/N434W; LAGA/N434W; LALAGA/N434W; LALAPG/N434W; N297A/DQ; D265A/DQ; LALA/DQ; LAGA/DQ; LALAGA/DQ; LALAPG/DQ; N297A/DW; D265A/DW; LALA/DW; LAGA/DW; LALAGA/DW; LALAPG/DW; N297A/YD; D265A/YD; LALA/YD; LAGA/YD; LALAGA/YD; LALAPG/YD; N297A/QVV; D265A/QVV; LALA/QVV; LAGA/QVV, LALAGA/QVV; LALAPG/QVV; DDRVV; N297A/DDRVV; D265A/DDRVV; LALA/DDRVV; LAGA/DDRVV; LALAGA/DDRVV; and LALAPG/DDRVV. In some embodiments, the modified Fc comprises a specific combination of amino acid substitutions selected from the group consisting of M428L/N434S (LS) and M252Y/S254T/T256E (YTE). In some embodiments, the modified Fc comprises M428L/N434S (LS) modifications. In some embodiments, the modified Fc comprises M252Y/S254T/T256E (YTE) modifications.

In some embodiments, the Fc of the α4β7 binding antibodies described herein lacks a C-terminal lysine residue (e.g., Ab001), the lack of which can be expected to decrease pI. In some embodiments, composition comprising the α4β7 binding antibodies wherein the Fc lacks C-terminal lysine residue (e.g., Ab001) has a homogeneous charge profile as compared to a composition comprising an α4β7 binding antibody, wherein the Fc comprises a C-terminal lysine residue. While the lack of C-terminal lysine residue can be expected to decrease solubility or decrease recombinant production yield, it has been discovered that truncation of C-terminal lysine in the α4β7 binding antibody yields a highly soluble protein and allows a relatively high production yield of α4β7 binding antibodies. In particular, Ab001 achieved surprising high yields (greater than 4.5 g/L) and high solubility (greater than 150 g/L).

In some embodiments, the methods provided herein wherein the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence according to SEQ ID NO: 3 (e.g., Ab001) may increase FcRn binding, while exhibiting satisfactory bioavailability when the α4β7 binding antibody is administered subcutaneously.

In some embodiments, the α4β7 binding antibodies described herein include modifications to improve its ability to mediate effector function. Such modifications are known in the art and include afucosylation, or engineering of the affinity of the Fc towards an activating receptor, mainly FCGR3a for antibody-dependent cellular cytotoxicity (ADCC), and towards C1q for complement-dependent cytotoxicity (CDC).

In some aspects, an antibody provided herein comprises an IgG1 domain with reduced fucose content at position Asn 297 (EU numbering) compared to a naturally occurring IgG1 domain. Such Fc domains are known to have improved ADCC. In some aspects, such antibodies do not comprise any fucose at position Asn 297.

In some embodiments, the α4β7 binding antibody provided herein comprises glycan moieties. In some embodiments, the α4β7 binding antibody provided herein comprises an IgG1 domain with high total fucose. The major N-glycan types are GOF and GIF. In some embodiments, the α4β7 binding antibody of the disclosure has one or more of the following N-glycan profiling:

- the proportion of GOF can be greater than about 40%. For example, the proportion of GOF can range from about 40% to about 75%;
- the proportion of G1F can range from about 10% to about 35%; the proportion of Man5 can be less than about 20%.

In some embodiments, the proportion of Man5 can range from about 1% to about 15%. In some embodiments, the proportion of GOF in the α4β7 binding antibody is from about 40% to about 45%, from about 45% to about 50%, from about 50% to about 55%, from about 55% to about 60%, from about 60% to about 65%, from about 65% to about 70%, from about 70% to about 75%.

In some embodiments, the proportion of G1F in the α4β7 binding antibody is from about 10% to about 15%, from about 15% to about 20%, from about 20% to about 25%, from about 25% to about 30%, from about 30% to about 35%.

In some embodiments, the proportion of Man5 in the α4β7 binding antibody is from about 1% to about 5%, from about 5% to about 10%, from about 10% to about 15%.

In some embodiments, the α4β7 binding antibodies described herein comprise an Fc region with one or more amino acid substitutions which improve ADCC, such as a substitution at one or more of positions 298, 333, and 334 of the Fc region. In some embodiments, an antibody provided herein comprises an Fc region with one or more amino acid substitutions at positions 239, 332, and 330.

In some embodiments, the α4β7 binding antibodies described herein comprise an Fc region with at least one galactose residue in the oligosaccharide attached to the Fc region. Such antibody variants may have improved CDC function.

In some embodiments, the α4β7 binding antibodies described herein comprise one or more alterations that improves or diminishes C1q binding and/or CDC.

In certain embodiments, the Fc region comprises one or more amino acid substitutions, wherein the one or more substitutions result in an increase in one or more of antibody half-life, ADCC activity, ADCP activity, or CDC activity compared with the Fc without the one or more substitutions. In certain embodiments, the one or more amino acid substitutions results in increased antibody half-life at pH 6.0 compared to an antibody comprising a wild-type Fc region. In certain embodiments, the antibody has an increased half-life that is about 10,000-fold, 1,000-fold, 500-fold, 100-fold, 50-fold, 20-fold, 10-fold, 9-fold, 8-fold, 7-fold, 6-fold, 5-fold, 4.5-fold, 4-fold, 3.5-fold, 3-fold, 2.5-fold, 2-fold, 1.95-fold, 1.9-fold, 1.85-fold, 1.8-fold, 1.75-fold, 1.7-fold, 1.65-fold, 1.6-fold, 1.55-fold, 1.50-fold, 1.45-fold, 1.4-fold, 1.35-fold, 1.3-fold, 1.25-fold, 1.2-fold, 1.15-fold, 1.1-fold, or 1.05-fold longer compared to an antibody comprising a wild-type Fc region.

In certain embodiments, the Fc region comprises one or more amino acid substitutions, wherein the one or more substitutions result in a decrease in one or more of ADCC activity, ADCP activity, or CDC activity compared with the Fc without the one or more substitutions.

In certain embodiments, the Fc region binds an Fcγ Receptor selected from the group consisting of: FcγRI, FcγRIIa, FcγRIIb, FcγRIIc, FcγRIIIa, and FcγRIIIb. In certain embodiments, the Fc region binds an Fcγ Receptor with higher affinity at pH 6.0 compared to an antibody comprising a wild-type Fc region.

In some embodiments, the α4β7 binding antibodies described herein comprise an extended half-life (i.e., serum half-life). In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of at least about 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 94, 96, or more than 96 days in humans. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life in a range of about 28 days to about 96 days, about 28 days to about 84 days, about 28 days to about 70 days, about 28 days to about 56 days, about 28 days to about 42 days, of about 35 days to about 96 days, about 35 days to about 84 days, about 35 days to about 70 days, about 35 days to about 56 days, about 35 days to about 42 days, of about 42 days to about 96 days, about 42 days to about 84 days, about 42 days to about 70 days, or about 42 days to about 56 days in humans. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life in a range of about 42 days to about 56 days in humans. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of at least about 50 days in humans. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of about 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56 or more than 56 days in humans.

In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of from about 10 days to about 25 days, from about 17 days to 23 days, from about 20 days to 22 days in non-human primates (NHPs). In some embodiments, the NHP is cynomolgus monkey. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of from about 11 days to 23 days in cynomolgus monkeys. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of from about 17 days to 23 days in cynomolgus monkeys. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of from about 19 days to 23 days in cynomolgus monkeys. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of from about 20 days to 23 days in cynomolgus monkeys. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of about 17 days in cynomolgus monkeys. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of about 20 days in cynomolgus monkeys. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of about 22 days in cynomolgus monkeys. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of from about 5 days to about 15 days, or from about 10 to 12 days in transgenic mice expressing human neonatal Fc receptor (hFcRn). In some embodiments, the transgenic mice expressing hFcRn are Tg276 mice. In some embodiments, the α4β7 binding antibodies described herein comprise a half-life of about 12 days in Tg276 mice.

Methods of measuring half-life are known in the art. In some embodiments, the half-life is measured in a rodent model, such as Tg276 mice. In some embodiments, the half-life is measured in a non-human primate, such as cynomolgus monkeys. Other animals can also be used to measure PK of α4β7 binding antibodies. In some embodiments, the half-life is measured in human. In some embodiments, the half-life is following intravenous administration. In some embodiments, the half-life is following subcutaneous administration.

In some embodiments, the α4β7 binding antibodies described herein have a half-life that is at least 20% longer than a comparator antibody. In some embodiments, the comparator antibody comprises the same complementarity determining regions and variable regions but different Fc regions. In some embodiments, the half-life of the α4β7 binding antibodies described herein protein is at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90% longer, at least 95% longer, at least 100% longer, at least 150% longer, at last 200% longer, at least 250% longer, at least 300% longer or at least 350% longer than the half-life of the comparator antibody. In some embodiments, the half-life of the α4β7 binding antibodies described herein is longer than the half-life of the comparator antibody by at least 2 fold, at least 3 fold, at least 4 fold, at least 5 fold, at least 6 fold, at least 7 fold, at least 8 fold, at least 9 fold, or at least 10 fold.

In some embodiments, the α4β7 binding antibodies described herein bind α4β7 with a K_D lower than or equal to 20 nM, 15 nM, 10 nM, 9 nM, 8 nM, 7 nM, 6 nM, 5 nM, 4 nM, 3 nM, 2 nM, 1 nM, 0.9 nM, 0.8 nM, 0.7 nM, 0.6 nM, 0.5 nM, 0.4 nM, 0.3 nM, 0.2 nM, 0.1 nM, 90 pM, 80 pM, 70 pM, 60 pM, 50 pM, 40 pM, 30 pM, 20 pM, or 10 pM. For example, in certain embodiments, the first antigen-binding site binds α4β7 with a K_D within the range of about 10 pM-about 1 nM, about 10 pM-about 0.9 nM, about 10 pM-about 0.8 nM, about 10 pM-about 0.7 nM, about 10 pM-about 0.6 nM, about 10 pM-about 0.5 nM, about 10 pM-about 0.4 nM, about 10 pM-about 0.3 nM, about 10 pM-about 0.2 nM, about 10 pM-about 0.1 nM, about 10 pM-about 50 pM, 0.1 nM-about 10 nM, about 0.1 nM-about 9 nM, about 0.1 nM-about 8 nM, about 0.1 nM-about 7 nM, about 0.1 nM-about 6 nM, about 0.1 nM-about 5 nM, about 0.1 nM-about 4 nM, about 0.1 nM-about 3 nM, about 0.1 nM-about 2 nM, about 0.1 nM-about 1 nM, about 0.1 nM-about 0.5 nM, about 0.5 nM-about 10 nM, about 0.5 nM-about 9 nM, about 0.5 nM-about 8 nM, about 0.5 nM-about 7 nM, about 0.5 nM-about 6 nM, about 0.5 nM-about 5 nM, about 0.5 nM-about 4 nM, about 0.5 nM-about 3 nM, about 0.5 nM-about 2 nM, about 0.5 nM-about 1 nM, about 1 nM-about 10 nM, about 1 nM-about 9 nM, about 1 nM-about 8 nM, about 1 nM-about 7 nM, about 1 nM-about 6 nM, about 1 nM-about 5 nM, about 1 nM-about 4 nM, about 1 nM-about 3 nM, about 1 nM-about 2 nM, about 2 nM-about 10 nM, about 3 nM-about 10 nM, about 4 nM-about 10 nM, about 5 nM-about 10 nM, about 6 nM-about 10 nM, about 7 nM-about 10 nM, about 8 nM-about 10 nM, or about 9 nM-about 10 nM. In some embodiments, the affinity is measured by SPR. In some embodiments, the affinity is measured by BLI. In some embodiments, the affinity is measured by KinExA.

Methods of Treatment and Dosage Regimens

Described herein, in certain embodiments, are methods of treating a disease or disorder in a patient in need thereof, the method comprising subcutaneously or intravenously administering to the patient an effective amount of an α4β7 binding antibody disclosed herein.

Provided herein are methods of treating a disease or disorder in a patient in need thereof comprising subcutaneously administering to the patient one or more initial doses comprising an effective amount of the α4β7 binding antibody (also referred to herein as "induction dose", "induction regimen" or "induction therapy") and one or more subsequent doses comprising an effective amount of the α4β7 binding antibody (also referred to herein as "maintenance dose").

Also provided herein are methods of treating a disease or disorder in a patient in need thereof comprising intravenously administering to the patient one or more initial doses comprising an effective amount of the α4β7 binding antibody (also referred to herein as "induction dose", "induction regimen" or "induction therapy") and one or more subsequent doses comprising an effective amount of the α4β7 binding antibody (also referred to herein as "maintenance dose"). As used herein, "maintenance dose", "maintenance regimen" and "maintenance therapy" are used interchangeably and is administered after induction dose(s) to continue the response achieved by induction dose(s) of α4β7 binding antibody.

In some embodiments, a single or more than one induction dose is administered, followed by maintenance doses. The disclosure contemplates one, two, three, or more induction doses. An induction dose can be administered intravenously or subcutaneously.

The induction dosing can involve administration of a higher dose than a maintenance dose, more frequent administration than a maintenance dose, or both, of the α4β7 binding antibody, e.g., in case potentially by a factor of 2. Alternatively, the induction dosing amount is the same as the maintenance dosing amount.

In some embodiments, one or more induction dose(s) is greater than the maintenance dose. In some embodiments, the method comprises administering a single induction dose that is greater than the maintenance dose. For example, the induction dose can be 1.5, 2, 3, 4, 5 times higher than the maintenance dose. Alternatively, the one or more induction dose amount(s) is the same amount as the maintenance dose. In some embodiments, one or more induction dose(s) is administered subcutaneously to the patient in need thereof. Yet, in other embodiments, the one or more induction dose(s) is administered intravenously. In some embodiments, an induction dose may require one or more than one subcutaneous injection or infusion on the same day.

In some embodiments, the method comprises administering one or more induction doses, wherein the induction doses are the same or greater than the maintenance dose and/or are administered at a frequency that is greater than the administration of the maintenance dose. For example, the method comprises administering one, two, three or four induction doses of the α4β7 binding antibody. In some embodiments, a first dose is administered at day 0 and a second dose at a subsequent day, for example, day 1 to week 6, e.g. day 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, day 14, days 21, day 28, day 35, day 42. In some embodiments, the one or more induction dose(s) may require one or more than one subcutaneous injection, or one or more than one infusion on the same day.

In some embodiments, a total induction dosage amount corresponding to the sum of the one or more administrations is from about 400 mg to about 1000 mg of the α4β7 binding antibody. In some embodiments, a total induction dosage amount corresponding to the sum of the one or more administrations is from about 500 mg to about 1200 mg of the α4β7 binding antibody. In some embodiments, the total dosage amount is about 400 mg of the α4β7 binding antibody. In some embodiments, the total dosage amount is about 500 mg of the α4β7 binding antibody. In some embodiments, the total dosage amount about 600 mg of the α4β7 binding antibody. In some embodiments, the total dosage amount is about 700 mg of the α4β7 binding antibody. In some embodiments, the dosage amount is about 800 mg of the α4β7 binding antibody. In some embodiments, the total dosage amount is about 900 mg of the α4β7 binding antibody. In some embodiments, the total dosage amount is about 1000 mg of the α4β7 binding antibody. In some embodiments, the total dosage amount is about 1100 mg of the α4β7 binding antibody. In some embodiments, the total dosage amount is about 1200 mg of the α4β7 binding antibody.

In some embodiments, the one or more induction dose(s) is administered subcutaneously to the patient in need thereof. Yet, in other embodiments, the one or more induction dose(s) is administered intravenously.

In some embodiments, the maintenance regimen comprises lower doses than the induction regimen and/or further apart than in the induction regimen. In some embodiments, the maintenance dose is administered every 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23, 24, 25, 26 or more weeks. In some embodiments, an induction dose of the α4β7 binding antibody is administered subcutaneously at day zero, and a maintenance dose is administered every 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23, 24, 25, 26 or more weeks thereafter.

In some embodiments, one or more induction dose(s) of from about 150 mg to about 1200 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of from about 300 mg to about 900 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of from about 300 mg to about 400 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of from about 400 mg to about 500 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of from about 500 mg to about 600 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of from about 600 mg to about 700 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of from about 700 mg to about 800 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of from about 800 mg to about 900 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of from about 900 mg to about 1000 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of from about 1000 mg to about 1100 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of from about 1100 mg to about 1200 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 150 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 200 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 250 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 300 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 350 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 400 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 450 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 500 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 550 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 600 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 650 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 700 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 750 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 800 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 850 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 900 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 950 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 1000 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 1050 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 1100 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 1150 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, one or more induction dose(s) of about 1200 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, the one or more induction dose(s) is administered subcutaneously.

In some embodiments, the method comprises administering two or more induction doses, wherein the first induction dose is greater than the second induction dose, and wherein the two or more induction doses are administered at a frequency that is greater than the administration of the maintenance dose. In some embodiments, the first induction dose is greater than second induction dose and greater than the maintenance dose, and the second induction is the same or greater than the maintenance dose. In some embodiments, one or more induction dose(s) of from about 150 mg to about 1200 mg of the α4β7 binding antibody is administered intravenously or subcutaneously at day zero. In some embodiments, the method comprises administering intravenously or subcutaneously a first induction dose is that greater than the second induction dose of the α4β7 binding antibody (e.g. a first induction dose of about 600 mg) at day zero, an administering a second induction of the α4β7 binding antibody (e.g. a second induction dose of about 150 mg) at day 14, and administering a maintenance dose that is the same or lower than the second inductions dose (e.g. about 150 mg) every two weeks thereafter. In one example, the induction doses are about 600 mg/mL at week 0 and about 300 mg/mL at week 2, with maintenance doses of about 300 mg/mL every 8 to 12 weeks after the second induction dose. In one example, the induction doses are about 600 mg/mL at week 0 and about 300 mg/mL at week 2, and about 300 mg/mL at week 6. In some embodiments, administration of the induction doses can result in plasma or serum $C_{avg W0-W6}$ of at least about 65 μg/mL, e.g. from about 65 μg/mL to about 75 μg/mL or greater, about 65 μg/mL, about 70 μg/mL, about 75 μg/mL or more. In some embodiments, administration of the induction doses can result in plasma or serum $C_{avg W0-W6}$ of about 70 μg/mL. In some embodiments, administration of the induction doses can result in plasma or serum $C_{avg W0-W12}$ of at least about 60 μg/mL, e.g. from about 60 μg/mL to about 70 μg/mL or greater, about 60 μg/mL, about 65 μg/mL, about 70 μg/mL or more. In some embodiments, administration of the induction doses can result in plasma or serum $C_{avg W0-W12}$ of about 65 μg/mL. Maintenance doses can be about 300 mg/mL at least every 8 weeks, for example every 12 weeks or every 24 weeks after the final induction dose. In some embodiments, administration of the induction doses can result in a plasma or serum $C_{avg W30-W40}$ of at least about 30 μg/mL, e.g. about 30 μg/mL, about 40 μg/mL, about 50 μg/mL, about 60 μg/mL, about 70 μg/mL or more.

In some embodiments, the plasma or serum concentration of at least about 35 μg/mL (e.g. from 35 to 40 μg/mL, from 40 to 45 μg/mL, from 45 to 50 μg/mL, from 50 to 55 μg/mL, from 55 to 60 μg/mL, e.g. 36 μg/mL, 41 μg/mL, 46 μg/mL, 51 μg/mL, 56 μg/mL, 60 μg/mL or more) is achieved in the patient for at least 4, at least 8, at least 12, at least 16, at least 20, at least 24 weeks or more (e.g. from 4 weeks to 6 weeks, from 6 weeks to 8 weeks, from 8 weeks to 10 weeks, from 10 weeks to 12 weeks, from 12 weeks to 14 weeks, from 14 weeks to 16 weeks, from 16 weeks to 18 weeks, from 18 weeks to 20 weeks, from 20 weeks to 22 weeks, from 22 weeks to 24 weeks or more) upon administration of an effective amount of the α4β7 binding antibody during the induction phase.

In some embodiments, the average plasma or serum concentration of at least about 35 μg/mL (e.g. from 35 to 40 μg/mL, from 40 to 45 μg/mL, from 45 to 50 μg/mL, from 50 to 55 μg/mL, from 55 to 60 μg/mL, e.g. 36 μg/mL, 41 μg/mL, 46 μg/mL, 51 μg/mL, 56 μg/mL, 60 μg/mL or more) is achieved in the patient over the course of the induction period. In some embodiments, the average plasma or serum concentration of at least about 45 μg/mL is achieved in the patient over the course of the induction period. In some embodiments, the average plasma or serum concentration of at least about 60 μg/mL is achieved in the patient over the course of the induction period.

In some embodiments, a maintenance dose of the α4β7 binding antibody is administered subcutaneously at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24 weeks or more (e.g. from 4 weeks to 6 weeks, from 6 weeks to 8 weeks, from 8 weeks to 10 weeks, from 10 weeks to 12 weeks, from 12 weeks to 14 weeks, from 14 weeks to 16 weeks, from 16 weeks to 18 weeks, from 18 weeks to 20 weeks, from 20 weeks to 22 weeks, from 22 weeks to 24 weeks or more) after the administration of the final induction dose. In some embodiments, the maintenance dose is lower than the induction dose (e.g. twice, 3 times or more lower than the induction dose). In some embodiments, the maintenance dose is the same as the induction dose. In some embodiments, a maintenance dose of the α4β7 binding antibody is administered subcutaneously every 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23, 24, 25, 26 or more weeks. In some embodiments, a maintenance dose of the α4β7 binding antibody is administered subcutaneously every 12, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23, 24, 25, 26 or more weeks. In some embodiments, the maintenance dose comprises an effective amount of the α4β7 binding antibody. In some embodiments, the maintenance dose comprises from about 150 mg to about 300 mg of the α4β7 binding antibody. In some embodiments, upon administration of the maintenance dose results, an effective minimum serum or plasma concentration of the α4β7 binding antibody is achieved. In some embodiments, the plasma or serum concentration of at least about 6 μg/mL (e.g. from about 6 μg/mL to about 10 μg/mL, from about 6 μg/mL to about 20 μg/mL, from about 6 μg/mL to about 30 μg/mL e.g. 6 μg/mL, 10 μg/mL, 15 μg/mL, 20 μg/mL, 30 μg/mL or more) is achieved in the patient for at least 4, at least 8, at least 12, at least 16, at least 20, at least 24 weeks or more (e.g. from 4 weeks to 6 weeks, from 6 weeks to 8 weeks, from 8 weeks to 10 weeks, from 10 weeks to 12 weeks, from 12 weeks to 14 weeks, from 14 weeks to 16 weeks, from 16 weeks to 18 weeks, from 18 weeks to 20 weeks, from 20 weeks to 22 weeks, from 22 weeks to 24 weeks or more) upon the administration upon administration of the α4β7 binding antibody. In some embodiments, the plasma or serum concentration does not fall below 6 μg/mL between maintenance dosing intervals. In some embodiments, the plasma or serum concentration does not fall below 10 μg/mL between maintenance dosing intervals. In some embodiments, the plasma or serum concentration does not fall below 20 μg/mL between maintenance dosing intervals. In some embodiments, the plasma or serum concentration does not fall below 30 μg/mL between maintenance dosing intervals.

In some embodiments, the method comprises administering subcutaneously to the subject in need thereof one or more induction dose(s) of an α4β7 binding antibody, and administering subcutaneously to the subject in need thereof one or more maintenance dose(s) of the α4β7 binding antibody. In some embodiments, the induction dose comprises an amount to the α4β7 binding antibody about 2 or more times higher than the maintenance dose. In some embodiments, the induction dose comprises from about 150 mg to about 1200 mg of the α4β7 binding antibody. For example, the induction dose comprises from about 400 mg to about 600 mg of the α4β7 binding antibody and the maintenance dose of from about 150 mg to about 300 mg of the α4β7 binding antibody. In another example, the induction dose comprises about 400 mg of the α4β7 binding antibody and wherein the maintenance dose of about 200 mg of the α4β7 binding antibody. In some embodiments, the induction dose comprises from about 500 mg to about 1200 mg of the α4β7 binding antibody. For example, in some embodiments, the induction dose comprises about 600 mg of the α4β7 binding antibody and the maintenance dose of about 300 mg of the α4β7 binding antibody. In another example, the induction dose comprises about 1000 mg of the α4β7 binding antibody and the maintenance dose of about 300 mg of the α4β7 binding antibody. In some embodiments, the one or more induction dose comprises a total dosage amount of from about 150 mg to about 1200 mg of the α4β7 binding antibody. For example, the induction dose comprises one or more induction dose comprises a total dosage amount of from about 400 mg to about 600 mg of the α4β7 binding antibody and the maintenance dose of from about 150 mg to about 300 mg of the α4β7 binding antibody. In another example, the one or more induction dose comprises a total dosage amount of about 400 mg of the α4β7 binding antibody and the maintenance dose of about 200 mg of the α4β7 binding antibody. In some embodiments, the one or more induction dose comprises a total dosage amount of from about 500 mg to about 1200 mg of the α4β7 binding antibody. In another example the one or more induction dose comprises a total dosage amount of about 600 mg of the α4β7 binding antibody and the maintenance dose of about 300 mg of the α4β7 binding antibody. In another example, the one or more induction dose comprises a total dosage amount of about 1000 mg of the α4β7 binding antibody and the maintenance dose of about 300 mg of the α4β7 binding antibody.

In some embodiments, upon administration of the one or more induction dose(s), an effective minimum average serum or plasma concentration of the α4β7 binding antibody is achieved over the course of the induction period. In some embodiments, the average serum or plasma concentration of the α4β7 binding antibody of at least about 35 μg/mL is achieved in the patient over the course of the induction period, for example 36 μg/mL, is achieved in the patient over the course of the induction period. In some embodiments, the serum or plasma concentration of the α4β7 binding antibody of at least about 35 μg/mL is achieved in the patient for at least 12 weeks or more upon the administration of the induction dose. In some embodiments, the serum or plasma concentration of the α4β7 binding antibody of at least about 6 μg/mL (e.g. from about 6 μg/mL to about 10 μg/mL from about 6 μg/mL to about 20 μg/mL, from about 6 μg/mL to about 30 μg/mL e.g. 6 μg/mL, 10 μg/mL, 15 μg/mL, 20 μg/mL, 30 μg/mL or more) is achieved in the patient for at least 12 weeks upon the administration of the maintenance dose. In some embodiments, the serum or plasma concentration of the α4β7 binding antibody of at least about 10 μg/mL is achieved in the patient for at least 12 weeks upon the administration of the maintenance dose.

Further described herein, in certain embodiments, are methods of treating a disease or disorder in a patient in need thereof comprising subcutaneously administering to the patient about 108 mg or about 300 mg of an α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2 and a light chain comprising an amino acid sequence according to SEQ ID NO: 3; and wherein a plasma or serum concentration of the α4β7 binding antibody of at least about 35 μg/mL is achieved in the patient for at least 2 weeks or more upon the administration.

In some embodiments, the method of treating a disease or disorder in a patient in need thereof comprises subcutaneously administering to the patient one or more induction doses and one or more maintenance doses of an α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence according to SEQ ID NO: 3; and wherein an average serum or plasma concentration of the α4β7 binding antibody of at least about 60 μg/mL is achieved in the patient during the induction period, and wherein a serum or plasma concentration of the α4β7 binding antibody of at least about 10 μg/mL is achieved in the patient for at least 12 weeks upon the administration of the maintenance dose.

In some embodiments, the average plasma or serum concentration of at least about 35 μg/mL is achieved in the patient during the induction period, for example from about 35 to about 60 μg/mL.

In some embodiments, the plasma or serum concentration of at least about 35 μg/mL is achieved in the patient for at least 2 weeks or more upon the administration of the one or more induction doses(s). In some embodiments, the plasma or serum concentration of at least about 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 48, 59, 60, or more than 60 μg/mL is achieved in the patient for at least 2 weeks or more upon the administration of the one or more induction doses(s).

In some embodiments, the plasma or serum concentration of at least about 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, or more than 60 μg/mL is achieved in the patient for at least 2 weeks or more upon the administration of the one or more induction dose(s) and/or one or more of the maintenance dose(s) of the α4β7 binding antibody. In some embodiments, the plasma or serum concentration of at least about 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 100, 150, or more than 150 μg/mL is achieved in the patient for at least 4 weeks or more upon administration of the one or more induction dose(s) and/or one or more of the maintenance dose(s) of the α4β7 binding antibody. In some embodiments, the plasma or serum concentration of at least about 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 100, 150, or more than 150 μg/mL is achieved in the patient for at least 6 weeks or more upon the administration of the one or more induction dose(s) and/or one or more of the maintenance dose(s) of the α4β7 binding antibody. In some embodiments, the plasma or serum concentration of at least about 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 100, 150, or more than 150 μg/mL is achieved in the patient for at least 8 weeks or more upon the administration of the one or more induction dose(s) and/or one or more of the maintenance dose(s) of the α4β7 binding antibody. In some embodiments, the plasma or serum concentration of at least about 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 100, 150, or more than 150 μg/mL is achieved in the patient for at least 10 weeks or more upon the administration of the one or more induction dose(s) and/or one or more of the maintenance dose(s) of the α4β7 binding antibody.

In some embodiments, the plasma or serum concentration of at least about 35 μg/mL is achieved in at least about 80% of the patients for at least 2 weeks or more upon the administration of the one or more induction dose(s) of the α4β7 binding antibody. In some embodiments, the plasma or serum concentration of at least about 35 μg/mL is achieved in at least about 85% of the patients for at least 2 weeks or more upon the administration of the one or more induction dose(s) of the α4β7 binding antibody. In some embodiments, the plasma or serum concentration of at least about 35 μg/mL is achieved in at least about 90% of the patients for at least 2 weeks or more upon the administration of the one or more induction dose(s) of the α4β7 binding antibody. In some embodiments, the plasma or serum concentration of at least about 35 μg/mL is achieved in at least about 95% of the patients for at least 2 weeks or more upon the administration of the one or more induction dose(s) of the α4β7 binding antibody. In some embodiments, the plasma or serum concentration of at least about 35 μg/mL is achieved in at least about 99% of the patients for at least 2 weeks or more upon the administration of the one or more induction dose(s) of the α4β7 binding antibody.

In some embodiments, the plasma or serum concentration of at least about 60 μg/mL is achieved in at least about 80% of the patients for at least 2 weeks or more upon the administration of the one or more induction doses(s). In some embodiments, the plasma or serum concentration of at least about 60 μg/mL is achieved in at least about 85% of the patients for at least 2 weeks or more upon the administration of the one or more induction doses(s). In some embodiments, the plasma or serum concentration of at least about 60 μg/mL is achieved in at least about 90% of the patients for at least 2 weeks or more upon the administration of the one or more induction doses(s). In some embodiments, the plasma or serum concentration of at least about 60 μg/mL is achieved in at least about 95% of the patients for at least 2 weeks or more upon the administration of the one or more induction doses(s). In some embodiments, the plasma or serum concentration of at least about 60 μg/mL is achieved in at least about 99% of the patients for at least 2 weeks or more upon the administration of the one or more induction doses(s).

In some embodiments, the average plasma or serum concentration of at least about 35 μg/mL is achieved in at least about 80% of the patients during the induction period. In some embodiments, the average plasma or serum concentration of at least about 35 μg/mL is achieved in at least about 85% of the patients during the induction period. In some embodiments, the average plasma or serum concentration of at least about 35 μg/mL is achieved in at least about 90% of the patients during the induction period. In some embodiments, the average plasma or serum concentration of at least about 35 μg/mL is achieved in at least about 95% of the patients during the induction period. In some embodiments, the average plasma or serum concentration of at least about 35 μg/mL is achieved in at least about 99% of the patients during the induction period.

In some embodiments, the average plasma or serum concentration of at least about 60 μg/mL is achieved in at least about 80% of the patients during the induction period. In some embodiments, the average plasma or serum concentration of at least about 60 μg/mL is achieved in at least about 85% of the patients during the induction period. In some embodiments, the average plasma or serum concentration of at least about 60 μg/mL is achieved in at least about 90% of the patients during the induction period. In some embodiments, the average plasma or serum concentration of at least about 60 μg/mL is achieved in at least about 95% of the patients during the induction period. In some embodiments, the average plasma or serum concentration of at least about 60 μg/mL is achieved in at least about 99% of the patients during the induction period.

Further described herein, in certain embodiments, are methods of treating a disease or disorder in a patient in need thereof comprising subcutaneously administering to the patient from about 75 mg to about 300 mg of an α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2 and a light chain comprising an amino acid sequence according to SEQ ID NO: 3; and wherein a maintenance C_{trough} of the α4β7 binding antibody of at least about 6 μg/mL is achieved in the patient for at least 2 weeks or more upon the administration. In some embodiments, the α4β7 antibody is a long acting engineered α4β7 binding antibody consisting of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In a further embodiment, the dose and administration frequency provide for a maintenance C_{trough} of the α4β7 binding antibody of at least about 6 μg/mL in the patient perpetually upon the administration, and the dosing is no more frequent than every 4 weeks, or alternatively every 8 weeks, or alternatively every 12 weeks, or alternatively every 16 weeks, or alternatively every 26 weeks. In an embodiment, the dose is sufficient for annual (Q52) dosing. In a further embodiment, the dose and administration frequency provide for a maintenance C_{trough} of the α4β7 binding antibody of at least about 10 μg/mL in the patient perpetually upon the administration, and the dosing is no more frequent than every 4 weeks, or alternatively every 8 weeks, or alternatively every 12 weeks, or alternatively every 16 weeks, or alternatively every 26 weeks. The methods may be by intravenous (IV) or subcutaneous (SC) administration, or both (an IV induction dose or doses and SC maintenance dose or doses); a particular advantage of the disclosure is that is provides for subcutaneous administration of both the induction and maintenance doses.

In some embodiments, the maintenance C_{trough} of at least about 6 μg/mL (for example 10 μg/mL) is achieved in the patient upon administration of the one or more maintenance dose(s). In some embodiments, the maintenance C_{trough} of at least about 2, 4, 6, 8, 10, 12, 14, 16, or more than 16 μg/mL is achieved in the patient for at least 2 weeks or more upon the administration. In some embodiments, the maintenance C_{trough} of at least about 2, 4, 6, 8, 10, 12, 14, 16, or more than 16 μg/mL is achieved in the patient for at least 4 weeks or more upon administration. In some embodiments, the maintenance C_{trough} of at least about 2, 4, 6, 8, 10, 12, 14, 16, or more than 16 μg/mL is achieved in the patient for at least 6 weeks or more upon the administration. In some embodiments, the maintenance of at least about 2, 4, 6, 8, 10, 12, 14, 16, or more than 16 μg/mL is achieved in the patient for at least 8 weeks or more upon the administration. In some embodiments, the maintenance C_{trough} of at least about 2, 4, 6, 8, 10, 12, 14, 16, or more than 16 μg/mL is achieved in the patient for at least 10 weeks or more upon the administration.

In some embodiments, the maintenance C_{trough} of at least about 6 μg/mL is achieved in at least about 80% of the patients for at least 2 weeks or more upon the administration. In some embodiments, the maintenance C_{trough} of at least about 6 μg/mL is achieved in at least about 85% of the patients for at least 2 weeks or more upon the administration. In some embodiments, the maintenance C_{trough} of at least about 6 μg/mL is achieved in at least about 90% of the patients for at least 2 weeks or more upon the administration. In some embodiments, the maintenance C_{trough} of at least about 6 μg/mL is achieved in at least about 95% of the patients for at least 2 weeks or more upon the administration. In some embodiments, the maintenance C_{trough} of at least about 6 μg/mL is achieved in at least about 99% of the patients for at least 2 weeks or more upon the administration.

In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 30 μg/mL is achieved in the patient. In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 35 μg/mL is achieved in the patient. In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 20, 25, 30, 35, 40, 45, 50, 55, 60 or more than 60 μg/mL is achieved in the patient. In some embodiments, the $C_{7\text{-}wks,\ induction}$ of at least about 20, 25, 30, 35, 40, 45, 50, 55, 60 or more than 60 μg/mL is achieved in the patient. In some embodiments, the $C_{8\text{-}wks,\ induction}$ of at least about 20, 25, 30, 35, 40, 45, 50, 55, 60 or more than 60 μg/mL is achieved in the patient. In some embodiments, the $C_{9\text{-}wks,\ induction}$ of at least 20, 25, 30, 35, 40, 45, 50, 55, 60 or more than 60 μg/mL is achieved in the patient. In some embodiments, the $C_{10\text{-}wks,\ induction}$ of at least about 20, 25, 30, 35, 40, 45, 50, 55, 60 or more than 60 45 μg/mL is achieved in the patient. In some embodiments, the $C_{11\text{-}wks,\ induction}$ of at least about 20, 25, 30, 35, 40, 45, 50, 55, 60 or more than 60 μg/mL is achieved in the patient. In some embodiments, the $C_{12\text{-}wks,\ induction}$ of at least about 20, 25, 30, 35, 40, 45, 50, 55, 60 or more than 60 μg/mL is achieved in the patient.

In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 35 μg/mL is achieved in at least about 80% of the patients. In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 35 μg/mL is achieved in at least about 85% of the patients. In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 35 μg/mL is achieved in at least about 90% of the patients. In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 35 μg/mL is achieved in at least about 95% of the patients. In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 35 μg/mL is achieved in at least about 99% of the patients.

In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 60 μg/mL is achieved in at least about 80% of the patients. In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 60 μg/mL is achieved in at least about 85% of the patients. In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 60 μg/mL is achieved in at least about 90% of the patients. In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 60 μg/mL is achieved in at least about 95% of the patients. In some embodiments, the $C_{6\text{-}wks,\ induction}$ of at least about 60 μg/mL is achieved in at least about 99% of the patients. Provided herein is a method of achieving $C_{avg\,W0\text{-}W12}$ of from 60 μg/mL to 70 μg/mL or greater for an α4β7 binding antibody in a subject in need thereof, the method comprising administering to the subject in need thereof a first dosage amount of at least 600 mg of the α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody consists of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3

In some embodiments, administration results in a $C_{avg\,W0\text{-}W12}$ of at least 60 μg/mL. In some embodiments, administration results in a $C_{avg\,W0\text{-}W12}$ of about 65 μg/mL.

In some embodiments, administration comprises administering to the subject in need thereof the first dosage amount of at least 600 mg of the α4β7 binding antibody and a second dosage amount of at least 300 mg of the α4β7 binding antibody at least two weeks after administration of the first dosage amount of the α4β7 binding antibody.

In some embodiments, the method comprises administering the first dosage amount of at least 600 mg intravenously and the second dosage amount of at least 300 mg of the α4β7 binding antibody subcutaneously.

In some embodiments, the method comprises the first dosage amount of at least 600 mg subcutaneously and the second dosage amount of at least 300 mg of the α4β7 binding antibody subcutaneously.

In some embodiments, the method further comprises comprising administering to the subject in need thereof one or more dosage amount of at least 300 mg of the α4β7 binding antibody at least four weeks after administration of the second dosage amount of the α4β7 binding antibody. In some embodiments, the one or more dosage amount of the α4β7 binding antibody is administered subcutaneously.

In some embodiments, the method comprises administering to the subject in need thereof at least 600 mg of the α4β7 binding antibody every twenty-six weeks after administration of the second dosage amount of the α4β7 binding antibody.

Provided herein is a method of achieving $C_{avg\,W0\text{-}W6}$ of from 65 μg/mL to 75 μg/mL or greater for an α4β7 binding antibody in a subject in need thereof, the method comprising administering to the subject in need thereof a first dosage amount of at least 600 mg of the α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody consists of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, administration results in a $C_{avg\,W0\text{-}W6}$ of about 70 μg/mL. In some embodiments, administration results in a $C_{avg\,W0\text{-}W12}$ of about 65 μg/mL.

In some embodiments, the administering comprises administering to the subject in need thereof a first dosage amount of at least 600 mg of the α4β7 binding antibody and a second dosage amount of at least 300 mg of the α4β7 binding antibody at least two weeks after administration of the first dosage amount of at least 600 mg of the α4β7 binding antibody.

In some embodiments, the method comprises administering the first dosage amount of the α4β7 binding antibody intravenously and the second dosage amount of the α4β7 binding antibody subcutaneously.

In some embodiments, the method comprises administering the first dosage amount and the second dosage amount of the α4β7 binding antibody subcutaneously.

In some embodiments, the method further comprises administering to the subject in need thereof one or more additional dosage amount of at least 300 mg of the α4β7 binding antibody at least four weeks after administration of the second dosage amount of the α4β7 binding antibody.

In some embodiments, the method comprises administering the one or more additional dosage amount of the α4β7 binding antibody subcutaneously.

In some embodiments, the method further comprises administering to the subject in need thereof at least 600 mg of the α4β7 binding antibody every twenty-six weeks after administration of second dosage amount of the α4β7 binding antibody.

In some embodiments, the administration of the one or more dosage amount of the α4β7 binding antibody results in C_{avg} of from 40 μg/mL to 50 μg/mL or greater. In some embodiments, the administration of the one or more dose of the α4β7 binding antibody results in C_{avg} of about 45 μg/mL.

Provided herein is a method of achieving C_{trough} of 35 μg/mL or greater for an α4β7 binding antibody in a subject in need thereof six weeks following administration of the α4β7 binding antibody to the subject, the method comprising administering to the subject in need thereof at least 600 mg of the α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody consists of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, the administration of the at least 600 mg of the α4β7 binding antibody results in a serum $C_{avgW0\text{-}W12}$ of at least 60 μg/mL. In some embodiments, the administration of the at least 600 mg of the α4β7 binding antibody results in a serum $C_{avgW30\text{-}W40}$ of at least 30 μg/mL.

In some embodiments, the at least 600 mg of the α4β7 binding antibody is administered intravenously. In other embodiments, the at least 600 mg of the α4β7 binding antibody is administered subcutaneously.

In some embodiments, the method further comprises administering to the subject in need thereof at from about 120 to about 450 mg of the α4β7 binding antibody, wherein the administration of the from about 120 to about 450 mg of the α4β7 binding antibody results in a steady state C_{trough} of at least 6 μg/mL. In some embodiments, the method further comprises administering to the subject in need thereof at least 120 mg of the α4β7 binding antibody, wherein the administration of the at least 120 mg of the α4β7 binding antibody results in a steady state C_{trough} of at least 6 μg/mL. In some embodiments, the method further comprises administering to the subject in need thereof at least 300 mg of the α4β7 binding antibody, wherein the administration of the at least 300 mg of the α4β7 binding antibody results in a steady state C_{trough} of at least 6 μg/mL.

In some embodiments, the method further comprises administering to the subject in need thereof at least 300 mg of the α4β7 binding antibody resulting in an average serum concentration of the α4β7 binding antibody above 35 μg/mL for ten weeks following the administration of the at least 300 mg of the α4β7 binding antibody. In some embodiments, the method results in a serum $C_{avgW0\text{-}W12}$ of at least 60 g/mL. In some embodiments, the administration of the at least 300 mg of the α4β7 binding antibody results in a serum $C_{avgW30\text{-}W40}$ of at least 30 g/mL.

In some embodiments, the method further comprises administering to the subject in need thereof at least 300 mg of the α4β7 binding antibody and results in an average serum concentration of the α4β7 binding antibody above 35 g/mL for ten weeks following the administration of the at least 300 mg of the α4β7 binding antibody.

In some embodiments, the method further comprises administering subcutaneously to the subject in need thereof at least 300 mg of the α4β7 binding antibody four weeks after administration of last dosage amount of the α4β7 binding antibody.

In some embodiments, the method further comprises administering subcutaneously to the subject a maintenance dose of at least 600 mg of the α4β7 binding antibody every twenty-six weeks after administration of the last dose of the α4β7 binding antibody.

Provided herein are multidose regimen for use in treating a disease in a subject in need thereof comprising a first injectable liquid formulation comprising a total dosage amount of at least 500 mg of an α4β7 binding antibody; and a second injectable liquid formulation comprising a total dosage amount of at least 120 mg of the α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody consists of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the second injectable liquid formulation comprises a total dosage amount of about 300 mg of the α4β7 binding antibody. In some embodiments, the first injectable liquid formulation is for intravenous administration. In some embodiments, the first injectable liquid formulation is a single dose or a multidose formulation. In some embodiments, the second injectable liquid formulation is for subcutaneous administration. In some embodiments, the second injectable liquid formulation is a single dose or a multidose formulation.

Provided herein are multidose regimen for use in treating a disease in a subject in need thereof comprising (a) a first injectable liquid formulation comprising a total dosage amount of at least 500 mg of a α4β7 binding antibody; and (b) a second injectable liquid formulation comprising a total dosage amount of at least 300 mg of the α4β7 binding antibody, wherein the α4β7 binding antibody comprises: (i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and (ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody consists of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, the first injectable liquid formulation comprises a total dosage amount of about 600 mg of the α4β7 binding antibody. In some embodiments, the first injectable liquid formulation comprises a total dosage amount of about 1000 mg of the α4β7 binding antibody.

In some embodiments, the second injectable liquid formulation comprises a total dosage amount of about 300 mg of the α4β7 binding antibody.

In some embodiments, the first injectable liquid formulation is for intravenous administration. In some embodiments, the first injectable liquid formulation is a single dose or a multidose formulation. In some embodiments, the second injectable liquid formulation is for subcutaneous administration. In some embodiments, the second injectable liquid formulation is a single dose or a multidose formulation.

Provided herein is a dosing regimen for use in treating a disease in a subject in need thereof, the dosing regimen comprising a first formulation comprising a total dosage amount of at least about 5500 mg of an α4β7 binding antibody for administration to the subject; a second formulation comprising a total dosage amount of at least about 120 mg of the α4β7 binding antibody for subcutaneous administration to the subject after the first formulation, and thereafter as a maintenance dose at least eight weeks after administration of the second formulation, wherein the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody consists of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the first injectable liquid formulation comprises a total dosage amount of from about 500 mg to about 1200 mg of the α4β7 binding antibody. In some embodiments, the first injectable liquid formulation comprises a total dosage amount of about 600 mg of the α4β7 binding antibody. In some embodiments, the first injectable liquid formulation comprises a total dosage amount of about 1000 mg of the α4β7 binding antibody. In some embodiments, the second injectable liquid formulation comprises a total dosage amount of about 120 mg to about 450 mg of the α4β7 binding antibody. In some embodiments, the second injectable liquid formulation comprises a total dosage amount of about 300 mg of the α4β7 binding antibody. In some embodiments, the second formulation is administered at least two weeks after the first formulation, and thereafter as a maintenance dose at least eight weeks after administration of the second formulation.

Provided herein is a dosing regimen for use in treating a disease in a subject in need thereof, the dosing regimen comprising: (a) a first formulation comprising a total dosage amount of at least about 600 mg of an α4β7 binding antibody for administration to the subject; (b) a second formulation comprising a total dosage amount of at least about 300 mg of the α4β7 binding antibody for subcutaneous administration to the subject at least two weeks after the first formulation, and thereafter as a maintenance dose at least eight weeks after administration of the second formulation, wherein the α4β7 binding antibody comprises: (i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and (ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody consists of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, the first formulation is for subcutaneous administration. On other embodiments, the first formulation is for intravenous administration.

In some embodiments, the first formulation comprises a total dosage amount of about 600 mg of the α4β7 binding antibody. In some embodiments, the second formulation comprises a total dosage amount of about 300 mg of the α4β7 binding antibody.

In some embodiments, the first formulation dosing regimen comprises a single or multiple injections. In some embodiments, the second formulation dosing regimen comprises a single or multiple injections.

In some embodiments, the first formulation and the second formulation do not contain citrate. In some embodiments, the first formulation or second formulation or both formulations comprise one or more of: histidine; arginine or a salt thereof; ethylenediaminetetraacetic acid (EDTA); and polysorbate 80.

Provided herein is an injectable dosage form of an α4β7 binding antibody comprising: a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3, wherein the α4β7 binding antibody has an average serum half-life greater than 6 days in cynomolgus monkeys. In some embodiments, the α4β7 binding antibody has an average serum half-life greater than 10 days in cynomolgus monkeys. In some embodiments, the α4β7 binding antibody consists of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, the average serum half-life is about 17 days or greater in cynomolgus monkeys. In some embodiments, the average serum half-life is 22 days in cynomolgus monkeys.

In some embodiments, the injectable dosage form comprises at least about 100 mg/ml, at least about 150 mg/ml, at least about 180 mg/ml, or at least about 200 mg/ml of the α4β7 binding antibody. For example, the injectable dosage form comprises from at least about 100 mg/ml to at least 110 mg/ml, from at least about 110 mg/ml to at least 120 mg/ml, from at least about 120 mg/ml to at least 130 mg/ml, from at least about 130 mg/ml to at least 140 mg/ml, from at least about 140 mg/ml to at least 150 mg/ml, from at least about 150 mg/ml to at least 160 mg/ml, from at least about 160 mg/ml to at least 170 mg/ml, from at least about 170 mg/ml to at least 180 mg/ml, from at least about 180 mg/ml to at least 190 mg/ml, from at least about 190 mg/ml to at least 200 mg/ml, from at least about 200 mg/ml to at least 210 mg/ml or more of the α4β7 binding antibody.

In some embodiments, the injectable dosage form comprises about 600 mg of the α4β7 binding antibody. In other embodiments, the injectable dosage form comprises about 1000 mg of the α4β7 binding antibody.

In some embodiments, the injectable dosage form is an injectable liquid formulation. In some embodiments, the injectable liquid formulation does not comprise citrate. In some embodiments, the injectable liquid formulation comprises one or more of: histidine; arginine or a salt thereof; ethylenediaminetetraacetic acid (EDTA); and polysorbate 80.

Provided herein are methods of treating a disease in a subject in need thereof, the method comprising administering to the subject in need thereof (a) an effective amount of an induction dose of an α4β7 binding antibody, and (b) an effective amount of one or more maintenance doses of the α4β7 binding antibody, wherein the one or more maintenance doses are administered subcutaneously at least eight weeks apart. In some embodiments, the α4β7 binding antibody comprises: (i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and (ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody consists of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, the effective amount of each of the one or more maintenance doses of the α4β7 binding antibody is at least about 120 mg. In some embodiments, the effective amount of each of the one or more maintenance doses of the α4β7 binding antibody is at least about 300 mg.

In some embodiments, the effective amount of the induction dose of the α4β7 binding antibody is about 600 mg. In some embodiments, the effective amount of the induction dose of the α4β7 binding antibody is about 1000 mg. In some embodiments, the effective amount of the induction dose of the α4β7 binding antibody is administered subcutaneously. In some embodiments, the effective amount of the induction dose of the α4β7 binding antibody is administered intravenously. In some embodiments, the one or more maintenance doses are administered subcutaneously about 12 weeks apart. In some embodiments, the one or more maintenance doses are administered subcutaneously about 26 weeks apart. In some embodiments, the induction dose comprises a dosage amount of the α4β7 binding antibody that is about 2 or more times higher than a dosage amount of each of the one or more maintenance doses.

In some embodiments, the administration of the α4β7 binding antibody follows a biphasic decline in serum concentration. In some embodiments, repeated dosing of α4β7 binding antibody does not have a significant influence on clearance (CL) and steady-state volume of distribution (Vss). In some embodiments, pharmacokinetic is not impacted by disease state of the patient.

In some embodiments, the α4β7 binding antibody provided herein and the α4β7 binding antibody in the regimen, injectable dosage form, method provided herein has one or more of the following characteristics:

(a) an average serum half-life greater than about 6 days in cynomolgus monkeys,
(b) an average serum half-life of 22 days in cynomolgus monkeys,
(c) a melting temperature TmOnset greater than about 55° C.,
(d) a melting temperature TmOnset greater than 56.2° C.,
(e) a proportion of GOF from about 55.1% to about 60.4%,
(f) a proportion of G1F from about 16.6% to about 18.8%,
(g) a proportion of Man5 from about 7.3% to about 8.8%,
(h) does not induce T cell activation markers CD25 or CD69,
(i) does not result in release of cytokines at least 6 hours after the administration, wherein the cytokines comprise one or more of IL-6, IL-8, IL-10, IFNγ, IL-4, IL-17, IL-2, IL-23p70 and TNF,
(j) does not induce complement-dependent cytotoxicity (CDC) in primary human PBMCs and in α4β7-expressing human β-lymphoid cell,
(k) does not induce antibody dependent cellular cytotoxicity (ADCC) in human NK cells, and
(l) does not impact suppressive activity of regulatory T cells expressing α4β7 integrin, wherein the suppressive activity is measured by presence of elevated CD71, CD25, Ki67, granzyme B, or OX40.

Described herein, in certain embodiments, are methods of treating a gastrointestinal inflammatory disease in a patient in need thereof. As used herein, the term "gastrointestinal inflammatory disease" refers to a disease of the gastrointestinal tract that involves inflammatory pathways. For example, the gastrointestinal inflammatory disease includes, but is not limited to, inflammatory bowel disease, ulcerative colitis (with or without exposure to anti-tumor necrosis factor (anti-TNF), Crohn's disease (including fistulizing Crohn's Disease), chronic pouchitis, collagenous gastritis, microscopic or collagenous colitis, colitis (including immune mediated colitis), sclerosing cholangitis (including in subjects with underlying inflammatory bowel disease, celiac enteritis, ileitis. In other aspects of the disclosure, provided herein are methods of treating Intestinal Acute Graft Versus Host Disease (aGVHD) (e.g. in subjects undergoing allogeneic hematopoietic stem cell transplantation (Allo-HSCT)), steroid-refractory acute intestinal graft-versus-host disease (GvHD) (e.g. in subjects who have undergone Allo-HSCT), Type 1 diabetes (TlD) (e.g. with or without anti-TNF pre-treatment), immune checkpoint inhibitor-related colitis in subjects with genitourinary cancer or melanoma.

In some embodiments, the disease is a gastrointestinal inflammatory disease. In some embodiments, the disease is an inflammatory bowel disease. In some embodiments, the inflammatory bowel disease is Crohn's disease. In some embodiments, the inflammatory bowel disease is ulcerative colitis.

In some embodiments, the subject in need thereof has moderately to severely active ulcerative colitis. The Mayo score is a commonly used disease activity index in UC. The complete Mayo score is composed of four parts: rectal bleeding, stool frequency, physician assessment, and endoscopy appearance, each part being rated from 0 to 3, giving a total score of 0 to 12. In some embodiments, the subject in need thereof has a Mayo score 6 to 12 with endoscopic sub score ≥2. In some embodiments, administration of the α4β7 binding antibody results in clinical response defined as a reduction in complete Mayo score of 3 or more points and 30% from baseline, (or a partial Mayo score reduction of 2 or more points and 25% or greater from baseline, if the complete Mayo score was not performed at the visit) with an accompanying decrease in rectal bleeding subscore of 1 or 0. In some embodiments, administration of the α4β7 binding antibody results in clinical remission, for example results in a complete Mayo score of 2 or less points and no individual subscore greater than 1 point. Alternatively, a modified or adapted Mayo score can be used for inclusion and for the clinical remission endpoint for UC. Clinical remission by the modified or adapted Mayo score is: stool frequency subscore 0 or 1 and not greater than baseline, rectal bleeding subscore of 0, and endoscopic subscore 0 or 1 without friability.

In some embodiments, the subject in need thereof has moderately to severely active Crohn's disease. The Crohn's disease activity index (CDAI) is a commonly used disease activity index in Crohn's disease (CD) and is calculated from eight independent variables. Crohn's disease "clinical remission" refers to a CDAI score of 150 points or less. The American College of Gastroenterology and the European Crohn's and Colitis Organization define CDAI<150, 250-220, 220-450, and >450 as reflecting remission, mild disease activity, moderate disease activity, and severe disease activity, respectively. In some embodiments, administration of the α4β7 binding antibody results in clinical response defined as a ≥100-point decrease in the CDAI score from baseline. In some embodiments, administration of the α4β7 binding antibody results in clinical remission defined as a CDAI score of 150 points or less.

Described herein, in certain embodiments, are methods of treating an inflammatory bowel disease in a patient in need thereof, the method comprising subcutaneously or intravenously administering to the patient an effective amount of an α4β7 binding antibody comprising: a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, the inflammatory bowel disease is Crohn's disease or ulcerative colitis. In some embodiments, the inflammatory bowel disease is ulcerative colitis.

Described herein, in certain embodiments, are methods of treating an inflammatory bowel disease in a patient in need thereof, the method comprising subcutaneously or intravenously administering to the patient an effective amount of an α4β7 binding antibody comprising: a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the 407 binding antibody is a long acting engineered α4β7 binding antibody, consisting of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, the α4β7 binding antibody is administered at a dose of about 75 mg to about 150 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 150 mg to about 300 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 250 mg to about 750 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg to about 700 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 700 mg to about 900 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg to about 600 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg to about 500 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg to about 400 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 400 mg to about 700 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 400 mg to about 600 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg to about 500 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 500 mg to about 700 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 700 mg to about 900 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 500 mg to about 600 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 600 mg to about 700 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 75 mg, about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg, about 350 mg, about 400 mg, about 450 mg, about 500 mg, about 550 mg, about 600 mg, about 650 mg, about 700 mg, about 750 mg, about 800 mg, about 850 mg, about 900 mg. In some embodiments, the α4β7 binding antibody is administered 30 at a dose of about 300 mg.

In some embodiments, administration of the α4β7 binding antibody is intravenous, intratumoral, intramuscular, subcutaneous, intralesional, intraintestinal, intracolonic, intrarectal, intrapouch, or intraperitoneal. In some embodiments, administration of the α4β7 binding antibody is through a parenteral route such as intravenous, intramuscular, subcutaneous, intraarterial, or intraperitoneal administration. In some embodiments, administration of the α4β7 binding antibody is intravenous or subcutaneous. In some embodiments, administration of the α4β7 binding antibody is intravenous. In some embodiments, administration of the α4β7 binding antibody is subcutaneous.

Administration of the α4β7 binding antibody can occur at various intervals. In some embodiments, the α4β7 binding antibody is administered to the patient at least once at an interval more than 8 weeks. In some embodiments, the interval is about 8 to about 12 weeks. In some embodiments, the interval is about 8 to about 16 weeks. In some embodiments, the interval is about 8 to about 20 weeks. In some embodiments, the interval is about 8 to about 24 weeks. In some embodiments, the interval is about 12 to about 26 weeks. In some embodiments, the interval is about 12 to about 22 weeks. In some embodiments, the interval is about 12 to about 18 weeks. In some embodiments, the interval is about 12 to about 14 weeks. In some embodiments, the interval is about 16 to about 26 weeks. In some embodiments, the interval is about 16 to about 22 weeks. In some embodiments, the interval is about 16 to about 18 weeks. In some embodiments, the interval is about 20 to about 26 weeks. In some embodiments, the interval is about 20 to about 22 weeks. In some embodiments, the interval is about 22 to about 26 weeks. In some embodiments, the interval is about 24 to about 26 weeks. In some embodiments, the interval is about 12 weeks. In some embodiments, the interval is about 14 weeks. In some embodiments, the interval is about 13 weeks. In some embodiments, the interval is about 16 weeks. In some embodiments, the interval is about 17 weeks. In some embodiments, the interval is about 18 weeks. In some embodiments, the interval is about 19 weeks. In some embodiments, the interval is about 20 weeks. In some embodiments, the interval is about 21 weeks. In some embodiments, the interval is about 22 weeks. In some embodiments, the interval is about 23 weeks. In some embodiments, the interval is about 24 weeks. In some embodiments, the interval is about 25 weeks. In some embodiments, the interval is about 26 weeks. In an embodiment, the dose is sufficient for annual (Q52) dosing.

Described herein, in certain embodiments, are methods of treating a disease or disorder in a patient in need thereof comprising subcutaneously administering to the patient about 108 mg or about 300 mg of an α4β7 binding antibody every four or six weeks, wherein the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 4 and a light chain comprising an amino acid sequence according to SEQ ID NO: 5; and wherein no intervening subcutaneous administration of the α4β7 binding antibody occurs during the four or six weeks. In some embodiments, the α4β7 binding antibody is administered at a dose of about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg. In some embodiments, the α4β7 binding antibody is administered every one week, two weeks, three weeks, four weeks, five weeks, six week, 7 weeks, 8 weeks, 10 weeks, 12 weeks, 14 weeks, 16 weeks, 20 weeks, 22 weeks, 24 weeks, 26 weeks, or more than 26 weeks.

In some embodiments, the induction dose comprises from about 400 mg to about 1000 mg of the α4β7 binding antibody and wherein the maintenance dose of from about 150 to about 300 mg of the α4β7 binding antibody. In some embodiments, the induction dose comprises about 600 mg of the α4β7 binding antibody and wherein the maintenance dose of about 300 mg of the α4β7 binding antibody. In some embodiments, the induction dose comprises about 1000 mg of the α4β7 binding antibody and wherein the maintenance dose of about 300 mg of the α4β7 binding antibody. In some embodiments, the induction dose comprises about 400 mg of the α4β7 binding antibody and wherein the maintenance dose of about 200 mg of the α4β7 binding antibody. In some embodiments, the induction dose comprises about 600 mg of the α4β7 binding antibody and wherein the maintenance dose of about 600 mg of the α4β7 binding antibody.

In some embodiments, the method comprises administering subcutaneously to the subject an induction dose of about 600 mg of the α4β7 binding antibody at week 0 and administering subcutaneously to the subject a maintenance dose of about 600 mg of the α4β7 binding antibody at week 26.

In some embodiments, upon administration of the one or more induction dose(s) an average serum or plasma concentration at least about 35 μg/mL of the α4β7 binding antibody is achieved in the patient during an induction period. In some embodiments, upon administration of the one or more maintenance dose(s) a serum or plasma concentration at least about 6 μg/mL, at least about 10 μg/mL, of the α4β7 binding antibody is achieved in the patient in the patient for at least 2 weeks or more.

Pharmaceutical Compositions

The present disclosure also features pharmaceutical compositions that contain a therapeutically effective amount of the α4β7 binding antibodies described herein. The composition can be formulated for use in a variety of drug delivery systems. One or more physiologically acceptable excipients or carriers can also be included in the composition for proper formulation. Suitable formulations for use in the present disclosure are found in Remington's Pharmaceutical Sciences, Mack Publishing Company, Philadelphia, Pa., 17th ed., 1985. For a brief review of methods for drug delivery, see, e.g., Langer (Science 249:1527-1533, 1990).

In some embodiments, the pharmaceutical composition is a liquid composition. In some embodiments, the pharmaceutical composition is formulated for subcutaneous administration. In some embodiments, the pharmaceutical composition is formulated for intravenous administration. In some embodiments, the liquid composition is a colorless to brown; in some embodiments, the liquid composition is clear to opalescent. In some embodiments, the liquid composition contains at least 150 mg/ml, or at least 180 mg/ml, or at least 200 mg/ml of an α4β7 binding antibody. In some embodiments, the liquid composition contains from about 150 mg/ml to about 160 mg/ml, from about 160 mg/ml to about 170 mg/ml, from about 170 mg/ml to about 10 mg/ml, from about 180 mg/ml to about 190 mg/ml, from about 190 mg/ml to about 200 mg/ml or more of a α4β7 binding antibody.

In some embodiments, a pharmaceutical composition may contain formulation materials for modifying, maintaining or preserving, for example, the pH, osmolarity, viscosity, clarity, color, isotonicity, odor, sterility, stability, rate of dissolution or release, adsorption or penetration of the composition. In such embodiments, suitable formulation materials include, but are not limited to, amino acids or salt thereof (such as glycine, glutamine, asparagine, arginine (e.g., arginine-HCl), histidine, or lysine); antimicrobials; antioxidants (such as ascorbic acid, sodium sulfite or sodium hydrogen-sulfite); buffers (such as borate, bicarbonate, Tris-HCl, citrates, phosphates or other organic acids); bulking agents (such as mannitol or glycine); chelating agents (such as ethylenediamine tetraacetic acid (EDTA), pentetic acid (DTPA)); complexing agents (such as caffeine, polyvinylpyrrolidone, beta-cyclodextrin or hydroxypropyl-beta-cyclodextrin); fillers; monosaccharides; disaccharides; and other carbohydrates (such as glucose, mannose or dextrins); proteins (such as serum albumin, gelatin or immunoglobulins); coloring, flavoring and diluting agents; emulsifying agents; hydrophilic polymers (such as polyvinylpyrrolidone); low molecular weight polypeptides; salt-forming counterions (such as sodium); preservatives (such as benzalkonium chloride, benzoic acid, salicylic acid, thimerosal, phenethyl alcohol, methylparaben, propylparaben, chlorhexidine, sorbic acid or hydrogen peroxide); solvents (such as glycerin, propylene glycol or polyethylene glycol); sugar alcohols (such as mannitol or sorbitol); suspending agents; surfactants or wetting agents (such as pluronics, PEG, sorbitan esters, polysorbates such as polysorbate 20, polysorbate 80, triton, tromethamine, lecithin, cholesterol, tyloxapal); stability enhancing agents (such as sucrose or sorbitol); tonicity enhancing agents (such as alkali metal halides, preferably sodium or potassium chloride, mannitol sorbitol); delivery vehicles; diluents; excipients and/or pharmaceutical adjuvants (see, Remington's Pharmaceutical Sciences, 18th ed. (Mack Publishing Company, 1990).

In some embodiments, a pharmaceutical composition is citrate-free. In some embodiments, a pharmaceutical composition may contain nanoparticles, e.g., polymeric nanoparticles, liposomes, or micelles.

In some embodiments, a pharmaceutical composition may contain a sustained- or controlled-delivery formulation. Techniques for formulating sustained- or controlled-delivery means, such as liposome carriers, bio-erodible microparticles or porous beads and depot injections, are also known to those skilled in the art. Sustained-release preparations may include, e.g., porous polymeric microparticles or semi-permeable polymer matrices in the form of shaped articles, e.g., films, or microcapsules. Sustained release matrices may include polyesters, hydrogels, polylactides, copolymers of L-glutamic acid and gamma ethyl-L-glutamate, poly (2-hydroxyethyl-methacrylate), ethylene vinyl acetate, or poly-D (−)-3-hydroxybutyric acid. Sustained release compositions may also include liposomes that can be prepared by any of several methods known in the art.

Pharmaceutical compositions containing an α4β7 binding antibody disclosed herein can be presented in a dosage unit form and can be prepared by any suitable method. A pharmaceutical composition should be formulated to be compatible with its intended route of administration. Examples of routes of administration are intravenous (IV), subcutaneous (SC), intradermal, inhalation, transdermal, topical, transmucosal, intrathecal and rectal administration. In some embodiments, the α4β7 binding antibody disclosed herein is administered by intravenously. In some embodiments, the α4β7 binding antibody disclosed herein is administered intravenously or subcutaneously. In specific embodiments, the α4β7 binding antibody disclosed herein is administered subcutaneously.

Aspects of the disclosure relate to injectable dosage forms of α4β7 binding antibody. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody consists of a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the injectable dosage form is a liquid formulation. In some embodiments, the liquid formulation comprises one or more of histidine, arginine or a salt thereof, EDTA and polysorbate 80. In some embodiments, the injectable liquid formulation does not comprise citrate.

Described herein, in certain embodiments, are subcutaneous dosages form comprising at least about 150 mg of an α4β7 binding antibody comprising: a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the subcutaneous dosages form comprising at least about 300 mg of an $\alpha 4\beta 7$ binding antibody comprising: a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, provided herein are subcutaneous dosage forms comprising at least about 150 mg of an $\alpha 4\beta 7$ binding antibody comprising: a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1; and b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In some embodiments, provided herein are subcutaneous dosage forms comprising at least about 150 mg of an $\alpha 4\beta 7$ binding antibody comprising: a) a heavy chain consisting an amino acid sequence according to SEQ ID NO: 1; and b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3.

In some embodiments, provided herein are subcutaneous dosage forms comprising at least about 300 mg of an $\alpha 4\beta 7$ binding antibody comprising: a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1; and b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In some embodiments, provided herein are subcutaneous dosage forms comprising at least about 300 mg of an $\alpha 4\beta 7$ binding antibody comprising: a) a heavy chain consisting an amino acid sequence according to SEQ ID NO: 1; and b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3.

Useful formulations can be prepared by methods known in the pharmaceutical art. For example, see *Remington's Pharmaceutical Sciences,* 18th ed. (Mack Publishing Company, 1990). Formulation components suitable for parenteral administration include a sterile diluent such as water for injection, saline solution, fixed oils, polyethylene glycols, glycerin, propylene glycol or other synthetic solvents; antibacterial agents such as benzyl alcohol or methyl parabens; antioxidants such as ascorbic acid or sodium bisulfite; chelating agents such as EDTA; buffers such as acetates, citrates or phosphates; and agents for the adjustment of tonicity such as sodium chloride or dextrose. In some embodiments, the formulation for parenteral administration is citrate-free.

For intravenous or subcutaneous administration, suitable carriers include physiological saline, bacteriostatic water, Cremophor ELTM (BASF, Parsippany, NJ) or phosphate buffered saline (PBS). The carrier should be stable under the conditions of manufacture and storage, and should be preserved against microorganisms. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol), and suitable mixtures thereof.

An intravenous or subcutaneous drug delivery formulation may be contained in a syringe, pen, or bag. In some embodiments, the bag is connected to a channel comprising a tube and/or a needle. In some embodiments, the formulation is a lyophilized formulation or a liquid formulation. In some embodiments, the formulation is a liquid formulation. Various devices can be used to deliver liquid formulations by subcutaneous route of administration, including on-body infusion devices, autoinjector devices, prefilled syringes, and syringes. Generally, administration time depends on volume and device, and can range from seconds to minutes.

These compositions may be sterilized by conventional sterilization techniques, or may be sterile filtered. The resulting aqueous solutions may be packaged for use as-is, or lyophilized, the lyophilized preparation being combined with a sterile aqueous carrier prior to administration.

A polyol, which acts as a tonicifier and may stabilize the $\alpha 4\beta 7$ binding antibody, may also be included in the formulation. The polyol is added to the formulation in an amount which may vary with respect to the desired isotonicity of the formulation. In some embodiments, the aqueous formulation is isotonic. The amount of polyol added may also be altered with respect to the molecular weight of the polyol. For example, a lower amount of a monosaccharide (e.g., mannitol) is added, compared to a disaccharide (such as trehalose). In some embodiments, the polyol which is used in the formulation as a tonicity agent is mannitol.

A detergent or surfactant may also be added to the formulation. Exemplary detergents include nonionic detergents such as polysorbates (e.g., polysorbates 20, 80 etc.) or poloxamers (e.g., poloxamer 188). The amount of detergent added is such that it reduces aggregation of the formulated antibody and/or minimizes the formation of particulates in the formulation and/or reduces adsorption. In some embodiments, the formulation may include a surfactant which is a polysorbate. In some embodiments, the formulation may contain the detergent polysorbate 80 or Tween 80. Tween 80 is a term used to describe polyoxyethylene (20) sorbitanmonooleate (see Fiedler, Lexikon der Hifsstoffe, Editio Cantor Verlag Aulendorf, 4th edi., 1996).

In embodiments, the protein product of the present disclosure is formulated as a liquid formulation. In some embodiments, the liquid formulation is prepared in combination with a sugar at stabilizing levels. In some embodiments, the liquid formulation is prepared in an aqueous carrier. In some embodiments, a stabilizer is added in an amount no greater than that which may result in a viscosity undesirable or unsuitable for intravenous administration. In some embodiments, the sugar is disaccharides, e.g., sucrose. In some embodiments, the liquid formulation may also include one or more of a buffering agent, a surfactant, and a preservative.

In some embodiments, the pH of the liquid formulation is set by addition of a pharmaceutically acceptable acid and/or base. In some embodiments, the pharmaceutically acceptable acid is hydrochloric acid. In some embodiments, the base is sodium hydroxide. In some embodiments, the pH of the liquid formulation is from about to about 3.5 to 9, 5.5 to 6.5, for example 6.0. In some embodiments, the pH of the liquid formulation is about 6.

The aqueous carrier of interest herein is one which is pharmaceutically acceptable (safe and non-toxic for administration to a human) and is useful for the preparation of a liquid formulation. Illustrative carriers include sterile water for injection (SWFI), bacteriostatic water for injection (BWFI), a pH buffered solution (e.g., phosphate-buffered saline), sterile saline solution, Ringer's solution or dextrose solution.

A preservative may be optionally added to the formulations herein to reduce bacterial action. The addition of a preservative may, for example, facilitate the production of a multi-use (multiple-dose) formulation.

The $\alpha 4\beta 7$ binding antibody may be lyophilized to produce a lyophilized formulation including the proteins and a lyoprotectant. The lyoprotectant may be sugar, e.g., disaccharides. In some embodiments, the lyoprotectant is sucrose or maltose. The lyophilized formulation may also include one or more of a buffering agent, a surfactant, a bulking agent, and/or a preservative.

The amount of sucrose or maltose useful for stabilization of the lyophilized drug product may be in a weight ratio of at least 1:2 protein to sucrose or maltose. In some embodiments, the protein to sucrose or maltose weight ratio is of from 1:2 to 1:5. In some embodiments, the pH of the formulation, prior to lyophilization, is set by addition of a pharmaceutically acceptable acid and/or base. In some embodiments, the pharmaceutically acceptable acid is hydrochloric acid. In some embodiments, the pharmaceutically acceptable base is sodium hydroxide.

In some embodiments, the α4β7 binding antibody is administered at a dose of about 75 mg to about 150 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 250 mg to about 750 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg to about 700 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg to about 600 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 700 mg to about 900 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 600 mg to about 1200 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg to about 500 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg to about 400 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 400 mg to about 700 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 400 mg to about 600 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg to about 500 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 500 mg to about 700 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 500 mg to about 600 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 600 mg to about 700 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 75 mg, about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg, about 350 mg, about 400 mg, about 450 mg, about 500 mg, about 550 mg, about 600 mg, about 650 mg, about 700 mg, about 750 mg, about 800 mg, about 850 mg, about 900 mg or more. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg. Actual dosage levels of the active ingredients in the pharmaceutical compositions of this disclosure may be about 250 mg, 350 mg, 400 mg, 450 mg, 500 mg, 550 mg, 600 mg, 650 mg, 700 mg 750 mg, 800 mg, 850 mg, 900 mg, 950 mg, 1000 mg, 1050 mg, 1100 mg, 1150 mg, or 1200 mg so as to obtain an amount of the active ingredient which is effective to achieve the desired therapeutic response for a particular patient, composition, and mode of administration, without being toxic to the patient.

The specific dose can be a uniform dose for each patient of about 150 mg, of about 200 mg, of about 250 mg, about 300 mg, about 350 mg, about 400 mg, about 450 mg, about 500 mg, about 550 mg, about 600 mg, about 650 mg, about 700 mg, about 750 mg, about 800 mg, about 850 mg, about 900 mg, about 1000 mg, about 1100 mg, about 1200 mg or more of α4β7 binding antibody. Alternatively, a patient's dose can be tailored to the approximate body weight or surface area of the patient. Other factors in determining the appropriate dosage can include the disease or condition to be treated or prevented, the severity of the disease, the route of administration, and the age, sex and medical condition of the patient. Further refinement of the calculations necessary to determine the appropriate dosage for treatment is routinely made by those skilled in the art, especially in light of the dosage information and assays disclosed herein. The dosage can also be determined through the use of known assays for determining dosages used in conjunction with appropriate dose-response data. An individual patient's dosage can be adjusted as the progress of the disease is monitored. Blood levels of the targetable construct or complex in a patient can be measured to see if the dosage needs to be adjusted to reach or maintain an effective concentration. Pharmacogenomics may be used to determine which targetable constructs and/or complexes, and dosages thereof, are most likely to be effective for a given individual (Schmitz et al., *Clinica Chimica Acta* 308: 43-53, 2001; Steimer et al., *Clinica Chimica Acta* 308: 33-41, 2001).

Kits

Provided herein is a kit for treating a disease in a subject in need thereof comprising: (a) an injectable liquid formulation comprising an α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3, and (b) instructions to administer from about 500 mg to about 1200 mg of the α4β7 binding antibody, instructions to administer from about 120 mg to about 450 mg of the α4β7 binding antibody at least two weeks after the administration of the about 500 mg to about 1200 mg of the α4β7 binding antibody, and instructions to administer from about 120 mg to about 450 mg of the α4β7 binding antibody at least four weeks thereafter. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Provided herein is a kit for treating a disease in a subject in need thereof comprising: (a) an injectable liquid formulation comprising an α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3, and (b) instructions to administer at least 600 mg of the α4β7 binding antibody, at least 300 mg of the α4β7 binding antibody at least two weeks after the administration of the 600 mg of the α4β7 binding antibody, and at least 300 mg of the α4β7 binding antibody at least four weeks thereafter. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Provided herein is a kit for treating a disease in a subject in need thereof comprising: (a) an injectable subcutaneous liquid formulation comprising an α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3, and (b) instructions to administer subcutaneously at least 600 mg of α4β7 binding antibody, at least 300 mg of the α4β7 binding antibody at least two weeks after the administration of the 600 mg of α4β7 binding antibody, and at least 300 mg of α4β7 binding antibody at least four weeks thereafter. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Methods of Preparation

The α4β7 binding antibodies described above can be made using recombinant DNA technology well known to a skilled person in the art. For example, one or more isolated polynucleotides encoding the α4β7 binding antibody can be ligated to other appropriate nucleotide sequences, including, for example, constant region coding sequences, and expression control sequences, to produce conventional gene expression constructs (i.e., expression vectors) encoding the desired α4β7 binding antibodies. Production of defined gene constructs is within routine skill in the art.

Nucleic acids encoding desired α4β7 binding antibodies can be incorporated (ligated) into expression vectors, which can be introduced into host cells through conventional transfection or transformation techniques. Exemplary host cells are *E. coli* cells, Chinese hamster ovary (CHO) cells, human embryonic kidney 293 (HEK 293) cells, HeLa cells, baby hamster kidney (BHK) cells, monkey kidney cells (COS), human hepatocellular carcinoma cells (e.g., Hep G2), and myeloma cells that do not otherwise produce IgG protein. Transformed host cells can be grown under conditions that permit the host cells to express the genes that encode α4β7 binding antibodies. In some embodiments, the nucleic acid comprises a sequence as set forth in SEQ ID NOs: 12-17.

Specific expression and purification conditions will vary depending upon the expression system employed. For example, if a gene is to be expressed in *E. coli*, it is first cloned into an expression vector by positioning the engineered gene downstream from a suitable bacterial promoter, e.g., Trp or Tac, and a prokaryotic signal sequence. The expressed protein may be secreted. The expressed protein may accumulate in refractile or inclusion bodies, which can be harvested after disruption of the cells by French press or sonication. The refractile bodies then are solubilized, and the protein may be refolded and/or cleaved by methods known in the art.

If the engineered gene is to be expressed in eukaryotic host cells, e.g., CHO cells, it is first inserted into an expression vector containing a suitable eukaryotic promoter, a secretion signal, a poly A sequence, and a stop codon. Optionally, the vector or gene construct may contain enhancers and introns. In embodiments involving fusion proteins comprising an α4β7 binding antibody or portion thereof, the expression vector optionally contains sequences encoding all or part of a constant region, enabling an entire, or a part of, a heavy or light chain to be expressed. The gene construct can be introduced into eukaryotic host cells using conventional techniques.

In some embodiments, in order to express an α4β7 binding antibody, an N-terminal signal sequence is included in the protein construct. Exemplary N-terminal signal sequences include signal sequences from interleukin-2, CD-5, IgG kappa light chain, trypsinogen, serum albumin, and prolactin.

After transfection, single clones can be isolated for cell bank generation using methods known in the art, such as limited dilution, ELISA, FACS, microscopy, or Clonepix. Clones can be cultured under conditions suitable for bioreactor scale-up and maintained expression of the α4β7 binding antibodies.

The α4β7 binding antibodies can be isolated and purified using methods known in the art including centrifugation, depth filtration, cell lysis, homogenization, freeze-thawing, affinity purification, gel filtration, ion exchange chromatography, hydrophobic interaction exchange chromatography, and mixed-mode chromatography.

Additional Embodiments

Described herein, in certain embodiments, are subcutaneous dosages form comprising an effective amount of an α4β7 binding antibody comprising: a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the effective amount achieves a serum or plasma average concentration of about 35 μg/mL and a trough serum or plasma concentration of at least about 6 μg/mL, e.g., about 10 μg/mL. In some embodiments, the heavy chain comprises an amino acid sequence according to SEQ ID NO: 1 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the heavy chain comprises an amino acid sequence according to SEQ ID NO: 2 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the subcutaneous dosage forms comprise from about 75 mg to about 900 mg or more of the α4β7 binding antibody. In some embodiments, the subcutaneous dosage forms comprise about 75 mg, 100 mg, 150 mg, 200 mg, 350 mg, 400 mg, 450 mg, 500 mg, 550 mg, 600 mg, 650 mg, 700 mg, 750 mg, 800 mg, 850 mg, 900 mg, 1000 mg, 1100 mg, 1200 mg or more of the α4β7 binding antibody. In some embodiments, the subcutaneous dosage forms comprise about 300 mg to about 1200 mg for an induction dose and about 150 mg to 350 mg for a maintenance dose of the α4β7 binding antibody.

Described herein, in certain embodiments, are methods of administering one or more of the dosage described herein. More particularly, the methods of administering one or more of the dosage forms described herein provide for dosing no more frequently than every eight weeks (Q8 dosing), or no more frequently than every twelve weeks (Q12 dosing), or no more frequently than every sixteen weeks (Q16 dosing), and in some embodiments no more frequently than twice per year (Q26 dosing) for maintenance therapy after one to three doses for induction therapy. In some embodiments, the dose is sufficient for annual (Q52) dosing. The methods of administering may further include an induction dose containing 2 or more times the dose of an α4β7 binding antibody of the disclosure. The methods may be by intravenous (IV) or subcutaneous (SC) administration, or both (an IV induction dose or doses and SC maintenance dose); a particular advantage of the disclosure is that is provides for subcutaneous administration of both the induction and maintenance doses.

Described herein, in certain embodiments, are methods of treating an inflammatory bowel disease in a patient in need thereof, the method comprising subcutaneously or intravenously administering to the patient an effective amount of an α4β7 binding antibody comprising: a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the inflammatory bowel disease is Crohn's disease or ulcerative colitis. In some embodiments, the inflammatory bowel disease is ulcerative colitis. In some embodiments, the heavy chain comprises an amino acid sequence according to SEQ ID NO: 1 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the heavy chain comprises an amino acid sequence according to SEQ ID NO: 2 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3. In some embodiments, administration of the α4β7 binding antibody is subcutaneous. In some embodiments, administration of the α4β7 binding antibody is intravenous. In some embodiments, the α4β7 binding antibody is administered at a dose of from about 100 mg to about 900 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of from about 100 mg to about 600 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of from about 100 mg to about 300 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 350 mg, 400 mg, 450 mg, 500 mg, 550 mg, 600 mg, 650 mg, 700 mg, 750 mg, 800 mg, 850 mg, 900 mg, 1000 mg, 1100 mg, 1200 mg or more.

Described herein, in certain embodiments, are methods of treating a disease or disorder in a patient in need thereof, the method comprising administering to the patient at least once at an interval more than 8 weeks, an effective amount of an α4β7 binding antibody comprising: a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the interval is about 12 to about 26 weeks. In some embodiments, the interval is about 12 weeks. In some embodiments, the interval is about 16 weeks. In some embodiments, the interval is about 26 weeks. In some embodiments, the disease or disorder is an inflammatory bowel disease. In some embodiments, the inflammatory bowel disease is Crohn's disease or ulcerative colitis. In some embodiments, the inflammatory bowel disease is ulcerative colitis. In some embodiments, the heavy chain comprises an amino acid sequence according to SEQ ID NO: 1 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the heavy chain comprises an amino acid sequence according to SEQ ID NO: 2 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3. In some embodiments, administration of the α4β7 binding antibody is subcutaneous. In some embodiments, administration of the α4β7 binding antibody is intravenous. In some embodiments, the α4β7 binding antibody is administered at a dose of about 300 mg. In some embodiments, the α4β7 binding antibody is administered at a dose of about 350 mg, 400 mg, 450 mg, 500 mg, 550 mg, 600 mg, 650 mg, 700 mg, 750 mg, 800 mg, 850 mg, 900 mg, 1000 mg, 1100 mg, 1200 mg or more. In some embodiments, the disease or disorder is an inflammatory bowel disease and administration of the α4β7 binding antibody is at an induction dose of about 600 mg, either by intravenous or subcutaneous administration, and a maintenance dose of about 300 mg and is subcutaneous.

Described herein, in certain embodiments, are methods of treating a disease or disorder in a patient in need thereof comprising administering to the patient about 108 mg or about 300 mg of an α4β7 binding antibody every four or six weeks, wherein the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 4 and a light chain comprising an amino acid sequence according to SEQ ID NO: 5; and wherein no intervening subcutaneous administration of the α4β7 binding antibody occurs during the four or six weeks.

Described herein, in certain embodiments, are methods of treating a disease or disorder in a patient in need thereof comprising administering to the patient about 108 mg or about 300 mg of an α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2 and a light chain comprising an amino acid sequence according to SEQ ID NO: 3; and wherein a serum or plasma concentration of the α4β7 binding antibody of at least about 35 μg/mL is achieved in the patient for at least 2 weeks or more upon the administration.

Described herein, in certain embodiments, are methods of treating a disease or disorder in a patient in need thereof comprising subcutaneously administering to the patient an induction dose of about 150 mg to about 1200 mg, for example about 300 mg to about 1000 mg, of an α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2 and a light chain comprising an amino acid sequence according to SEQ ID NO: 3; and wherein a serum or plasma concentration of the α4β7 binding antibody of at least about 35 μg/mL is achieved in the patient for at least 2 weeks or more upon the administration and a trough serum or plasma concentration is at least about 6 μg/mL, for example 10 μg/mL.

Described herein, in certain embodiments, are methods of treating a disease or disorder in a patient in need thereof comprising subcutaneously administering to the patient about 108 mg or about 300 mg of an α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2 and a light chain comprising an amino acid sequence according to SEQ ID NO: 3; and wherein a maintenance C_{trough} of the α4β7 binding antibody of at least about 6 μg/mL, e.g., about 10 μg/mL, is achieved in the patient upon the administration. In a further embodiment, the dose and administration frequency provide for a C_{trough} of the α4β7 binding antibody of at least about 6 μg/mL, e.g., about 10 μg/mL, in the patient perpetually upon the administration, and the dosing is no more frequent than every 4 weeks, or alternatively every 8 weeks, or alternatively every 12 weeks, or alternatively every 16 weeks, or alternatively every 26 weeks.

In one embodiment, a single or more than one induction dose is administered, followed by maintenance doses. An induction dose can be administered intravenously or subcutaneously. The induction dose may be higher than a maintenance dose or more frequent than a maintenance dose, e.g., in case potentially by a factor of 2. Aspects of the disclosure contemplate one, two, three, or more induction doses. In one example, the induction doses are 600 mg/mL at week 0 and 300 mg/mL at week 2, with maintenance doses of 300 mg/mL every 12 weeks after the second induction dose.

Aspects of the disclosure relate to isolated nucleic acids, expression vector comprising the nucleic acid sequences or recombinant host cell comprising the nucleic acid sequences encoding the heavy chain and/or light chain of the α4β7 binding antibodies described herein. In some embodiments, the nucleic acid sequence encodes SEQ ID NO: 1. In some embodiments, the nucleic acid sequence encodes SEQ ID NO: 3.

Aspects of the disclosure relate to an α4β7 binding antibody comprising a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the α4β7 binding antibody comprises a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Aspects of the disclosure relate to an α4β7 binding antibody comprising a heavy chain and a light chain, wherein the heavy chain consists of an amino acid sequence according to SEQ ID NO: 1. In some embodiments, the light chain consists of an amino acid sequence according to SEQ ID NO: 3.

According to aspects of the disclosure, the α4β7 binding antibody has an increased half-life as compared to an α4β7 binding antibody comprising a heavy chain sequence consisting of SEQ ID NO: 4. In some embodiments, the α4β7 binding antibody has a half-life of from 42 days to 56 days or more in humans.

Aspects of the disclosure relate to a pharmaceutical composition comprising an α4β7 binding antibody and a pharmaceutically acceptable carrier, the α4β7 binding antibody comprising a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In some embodiments, pharmaceutical composition comprises an α4β7 binding antibody and a pharmaceutically acceptable carrier, the α4β7 binding antibody comprising a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the pharmaceutical composition is citrate free.

Aspects of the disclosure relate to a subcutaneous dosage form comprising at least about 150 mg of an α4β7 binding antibody comprising a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain comprising an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the subcutaneous dosage form comprises at least about 150 mg of an α4β7 binding antibody comprising a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and a light chain consisting of an amino acid sequence according to SEQ ID NO: 3. In some embodiments, the dosage form is citrate free. In some embodiments, the subcutaneous dosage comprises from about 150 mg to about 900 mg of the α4β7 binding antibody.

Methods for treating inflammatory bowel disease in a subject in need thereof are provided. In some embodiments, the method comprises administering subcutaneously to the subject one or more of the dosage forms of aspects the disclosure. In some embodiments, the inflammatory bowel disease is Crohn's disease or ulcerative colitis. In some embodiments, the inflammatory bowel disease is ulcerative colitis.

In some embodiments, a method of treating an inflammatory bowel disease in a subject in need thereof comprises administering subcutaneously to the subject an effective amount of an α4β7 binding antibody comprising a heavy chain and a light chain, wherein the heavy chain consists of an amino acid sequence according to SEQ ID NO: 1; and the light chain comprises the variable region of SEQ ID NO: 3.

In some embodiments, a method for treating inflammatory bowel disease in a subject in need thereof, the method comprising administering subcutaneously to the subject an effective amount of an α4β7 binding antibody comprising a heavy chain and a light chain, wherein the heavy chain consists of an amino acid sequence according to SEQ ID NO: 1; and the light chain consists of the variable region of SEQ ID NO: 3.

In some embodiments, the α4β7 binding antibody is administered at a dose of from about 150 mg to about 900 mg. In some embodiments, the method comprises administering to the subject the effective amount of an α4β7 binding antibody at an interval of more than 8 weeks, for example about 12 to about 26 weeks, about 12 weeks, about 16 weeks, about 26 weeks.

In some embodiments, the inflammatory bowel disease is Crohn's disease or ulcerative colitis. In some embodiments, the inflammatory bowel disease is ulcerative colitis.

In some embodiments, an average serum or plasma or serum concentration of the α4β7 binding antibody of at least about 35 μg/mL is achieved in the subject during an induction period.

In some embodiments, a maintenance C_{trough} of the α4β7 binding antibody of at least about 6 μg/mL is achieved in the subject.

Aspects of the disclosure relate to a method for treating inflammatory bowel disease in a subject in need thereof, the method comprising (a) administering subcutaneously to the subject one or more induction dose comprising an effective amount of an α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain comprising a heavy chain and a light chain, wherein the heavy chain consists of an amino acid sequence according to SEQ ID NO: 1; and the light chain consists of the variable region of SEQ ID NO: 3; (b) administering subcutaneously to the subject one or more maintenance dose comprising an effective amount of the α4β7 binding antibody.

In some embodiments, the one or more maintenance dose of the α4β7 binding antibody is administered at an interval of more than 8 weeks.

In some embodiments, the induction dose comprises an amount to the α4β7 binding antibody about 2 or more times higher than the maintenance dose.

In some embodiments, the induction dose comprises from about 400 mg to about 600 mg of the α4β7 binding antibody and wherein the maintenance dose of from about 150 to about 300 mg of the α4β7 binding antibody. In some embodiments, the induction dose comprises about 600 mg of the α4β7 binding antibody and wherein the maintenance dose of about 300 mg of the α4β7 binding antibody. In some embodiments, the induction dose comprises about 400 mg of the α4β7 binding antibody and wherein the maintenance dose of about 200 mg of the α4β7 binding antibody. In some embodiments, the induction dose comprises about 600 mg of the α4β7 binding antibody and wherein the maintenance dose of about 600 mg of the α4β7 binding antibody.

In some embodiments, the method comprises administering subcutaneously to the subject an induction dose of about 600 mg of the α4β7 binding antibody at week 0 and administering subcutaneously to the subject a maintenance dose of about 600 mg of the α4β7 binding antibody at week 26.

In some embodiments, upon administration of the one or more induction dose an average serum or plasma concentration at least about 35 μg/mL of the α4β7 binding antibody is achieved in the patient during an induction period. In some embodiments, upon administration of the one or more maintenance dose a serum or plasma concentration at least about 6 μg/mL of the α4β7 binding antibody is achieved in the patient in the patient for at least 2 weeks or more.

While the present disclosure provides for subcutaneous administration of the α4β7 binding antibody, it is also possible to administer the α4β7 binding antibody by the intravenous route, i.e., by infusion. It is further contemplated that induction or loading dose or doses may be administered by the subcutaneous (SC) or intravenous (IV) route of administration.

Specific Embodiments

Non-limiting specific embodiments are described below, each of which is considered to be within the present disclosure.

Embodiment 1. A multidose regimen comprising liquid formulations for use in treating a disease in a subject in need thereof comprising:

(a) a first injectable liquid formulation comprising a total dosage amount of at least 500 mg of a long acting α4β7 targeting molecule; and

(b) a second injectable liquid formulation comprising a total dosage amount of at least 300 mg of the long acting α4β7 targeting molecule,

wherein the long acting α4β7 targeting molecule comprises:

(i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and

(ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Embodiment 2. The multidose regimen of Embodiment 1, wherein the first injectable liquid formulation comprises a total dosage amount of from about 500 mg to about 1200 mg of the long acting α4β7 targeting molecule.

Embodiment 3. The multidose regimen of Embodiment 1, wherein the first injectable liquid formulation comprises a total dosage amount of about 600 mg of the long acting α4β7 targeting molecule.

Embodiment 4. The multidose regimen of Embodiment 1, wherein the first injectable liquid formulation comprises a total dosage amount of about 1000 mg of the long acting α4β7 targeting molecule.

Embodiment 5. The multidose regimen of anyone of Embodiments 1-4, wherein the second injectable liquid formulation comprises a total dosage amount of from about 300 mg to about 450 mg of the long acting α4β7 targeting molecule.

Embodiment 6. The multidose regimen of anyone of Embodiments 1-4, wherein the second injectable liquid formulation comprises a total dosage amount of about 300 mg of the long acting α4β7 targeting molecule.

Embodiment 7. The multidose regimen of anyone of Embodiments 1-6, wherein the first injectable liquid formulation is for intravenous administration.

Embodiment 8. The multidose regimen of anyone of Embodiments 1-6, wherein the first injectable liquid formulation is for subcutaneous administration.

Embodiment 9. The multidose regimen of Embodiment 8, wherein the first injectable liquid formulation is a single dose or a multidose formulation.

Embodiment 10. The multidose regimen of anyone of Embodiments 1-9, wherein the second injectable liquid formulation is for subcutaneous administration.

Embodiment 11. The multidose regimen of Embodiment 10, wherein the second injectable liquid formulation is suitable for administration as a single injection or multiple injections.

Embodiment 12. The multidose regimen of any one of Embodiments 1-11, wherein the disease is a gastrointestinal inflammatory disease.

Embodiment 13. The multidose regimen of any one of Embodiments 1-11, wherein the disease is an inflammatory bowel disease.

Embodiment 14. The multidose regimen of Embodiment 13, wherein the inflammatory bowel disease is Crohn's disease.

Embodiment 15. The multidose regimen of Embodiment 13, wherein the inflammatory bowel disease is ulcerative colitis.

Embodiment 16. A dosing regimen for use in treating a disease in a subject in need thereof, the dosing regimen comprising:

(a) a first formulation comprising a total dosage amount of at least about 600 mg of a long acting α4β7 targeting molecule for administration to the subject;

(b) a second formulation comprising a total dosage amount of at least about 300 mg of the long acting α4β7 targeting molecule for subcutaneous administration to the subject at least two weeks after the first formulation, and thereafter as a maintenance dose at least eight weeks after administration of the second formulation,

wherein the long acting α4β7 targeting molecule comprises:

(i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and

(ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Embodiment 17. The dosing regimen of Embodiment 16, wherein the first formulation is for subcutaneous administration.

Embodiment 18. The dosing regimen of Embodiment 16, wherein the first formulation is for intravenous administration.

Embodiment 19. The dosing regimen of any one of Embodiments 16-18, wherein the first formulation comprises a total dosage amount of from about 600 mg to about 1200 mg of the long acting α4β7 targeting molecule.

Embodiment 20. The dosing regimen of any one of Embodiments 16-18, wherein the first formulation comprises a total dosage amount of about 600 mg of the long acting α4β7 targeting molecule.

Embodiment 21. The dosing regimen of any one of Embodiments 16-20, wherein the second formulation comprises a total dosage amount of from about 300 mg to about 450 mg of the long acting α4β7 targeting molecule.

Embodiment 22. The dosing regimen of any one of Embodiments 16-20, wherein the second formulation comprises a total dosage amount of about 300 mg of the long acting α4β7 targeting molecule.

Embodiment 23. The dosing regimen of Embodiment 19 or Embodiment 20, wherein the first formulation is suitable for administration as a single injection or multiple injections.

Embodiment 24. The dosing regimen of any one of Embodiments 16-23, wherein the second formulation is a single dose or multidose injectable formulation.

Embodiment 25. The dosing regimen of any one of Embodiments 16-24, wherein the first formulation and the second formulation do not contain citrate.

Embodiment 26. The dosing regimen of Embodiment 25, wherein the first formulation or second formulation or both formulations comprise one or more of:

(a) histidine;

(b) arginine or a salt thereof;

(c) ethylenediaminetetraacetic acid (EDTA); and

(d) polysorbate 80.

Embodiment 27. The dosing regimen of any one of Embodiments 16-26, wherein the disease is a gastrointestinal inflammatory disease.

Embodiment 28. The dosing regimen of any one of Embodiments 16-26, wherein the disease is an inflammatory bowel disease.

Embodiment 29. The dosing regimen of Embodiment 28, wherein the inflammatory bowel disease is Crohn's disease.

Embodiment 30. The dosing regimen of Embodiment 28, wherein the inflammatory bowel disease is ulcerative colitis.

Embodiment 31. An injectable dosage form of an α4β7 binding antibody comprising:

a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and

a) a light chain consisting of an amino acid sequence according to SEQ ID NO,

wherein the α4β7 binding antibody has an average serum half-life greater than 6 days in cynomolgus monkeys.

Embodiment 32. The injectable dosage form of Embodiment 31, wherein the average serum half-life is about 17 days or greater in cynomolgus monkeys.

Embodiment 33. The injectable dosage form of Embodiments 31 or 32 comprising at least 100 mg/ml of the long acting α4β7 targeting molecule.

Embodiment 34. The injectable dosage form of Embodiments 31 or 32 comprising at least 180 mg/ml of the α4β7 binding antibody.

Embodiment 35. The injectable dosage form of Embodiments 31 or 32 comprising at least 150 mg/ml of the long acting α4β7 targeting molecule.

Embodiment 36. The injectable dosage form of Embodiment 31 or Embodiment 32 comprising at least 200 mg/ml of the long acting α4β7 targeting molecule.

Embodiment 37. The injectable dosage form of any one of Embodiments 31-32, comprising from about 600 mg to about 1200 mg of the long acting α4β7 targeting molecule.

Embodiment 38. The injectable dosage form of any one of Embodiments 31-32, comprising about 600 mg of the long acting α4β7 targeting molecule.

Embodiment 39. The injectable dosage form of any one of Embodiments 31-32, comprising about 1000 mg of the long acting α4β7 targeting molecule.

Embodiment 40. The injectable dosage form of any one of Embodiments 31-39, wherein the injectable dosage form is an injectable liquid formulation.

Embodiment 41. The injectable dosage form of Embodiment 40, wherein the injectable liquid formulation does not comprise citrate.

Embodiment 42. The injectable dosage form of Embodiment 41, wherein the injectable liquid formulation comprises one or more of:

(a) histidine;

(b) arginine or a salt thereof;

(c) ethylenediaminetetraacetic acid (EDTA); and

(d) polysorbate 80.

Embodiment 43. The injectable liquid formulation of any one of Embodiments 1-30 or the Embodiments 40-42, wherein the injectable liquid formulation is a pharmaceutical composition.

Embodiment 44. A method of treating a disease in a patient in need thereof, the method comprising: administering to a subject in need thereof (a) an effective amount of an induction dose of a long acting α4β7 targeting molecule, and (b) an effective amount of one or more maintenance doses of the long acting α4β7 targeting molecule, wherein the one or more maintenance doses are administered subcutaneously at least eight weeks apart,

wherein the long acting α4β7 targeting molecule comprises:

(i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and (ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Embodiment 45. The method of Embodiment 44, wherein the disease is a gastrointestinal inflammatory disease.

Embodiment 46. The method of Embodiment 44, wherein the disease is an inflammatory bowel disease.

Embodiment 47. The method of Embodiment 45, wherein the inflammatory bowel disease is Crohn's disease.

Embodiment 48. The method of Embodiment 45, wherein the inflammatory bowel disease is ulcerative colitis.

Embodiment 49. The method of any one of Embodiments 44-48, wherein the effective amount of each of the one or more maintenance doses of the long acting α4β7 targeting molecule is at least about 300 mg.

Embodiment 50. The method of any one of Embodiments 44-48, wherein the effective amount of each of the one or more maintenance doses of the long acting α4β7 targeting molecule is from about 300 mg to about 450 mg.

Embodiment 51. The method of any one of Embodiments 44-48, wherein the effective amount of the induction dose of the long acting α4β7 targeting molecule is about 600 mg.

Embodiment 52. The method of any one of Embodiments 44-48, wherein the effective amount of the induction dose of the long acting α4β7 targeting molecule is about 1000 mg.

Embodiment 53. The method of any one of Embodiments 44-48, wherein the effective amount of the induction dose of the long acting α4β7 targeting molecule is from 600 mg to about 1200 mg.

Embodiment 54. The method of any one of Embodiments 51-53 comprising administering the effective amount of the induction dose in a single or multiple injections.

Embodiment 55. The method of any one of Embodiments 44-54, wherein the effective amount of the induction dose of the long acting α4β7 targeting molecule is administered subcutaneously.

Embodiment 56. The method of any one of Embodiments 44-54, wherein the effective amount of the induction dose of the long acting α4β7 targeting molecule is administered intravenously.

Embodiment 57. The method of any one of Embodiments 44-56, wherein the one or more maintenance doses are administered subcutaneously about 12 weeks apart.

Embodiment 58. The method of any one of Embodiments 44-56, wherein the one or more maintenance doses are administered subcutaneously about 26 weeks apart.

Embodiment 59. The method of any one of Embodiments 44-58, wherein the induction dose comprises a dosage amount of the long acting α4β7 targeting molecule that is about 2 or more times higher than a dosage amount of each of the one or more maintenance doses.

Embodiment 60. The method of any one of Embodiments 44-59, wherein:

(a) the long acting α4β7 targeting molecule follows a biphasic decline in serum concentration;

(b) repeated dosing of the long acting α4β7 targeting molecule does not have a significant influence on clearance (CL) and steady-state volume of distribution (Vss);

(c) pharmacokinetic is not impacted by disease state of the patient; or

(d) a combination thereof.

Embodiment 61. A method of achieving $C_{avg W0-W12}$ of from 60 μg/ml to 70 μg/ml or greater for a long acting α4β7 targeting molecule in a subject in need thereof, the method comprising administering to the subject in need thereof a first dosage amount of at least 600 mg of the α4β7 binding antibody, wherein the long acting α4β7 targeting molecule comprises:

(a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and

(b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Embodiment 62. The method of Embodiment 61, wherein the administering results in a $C_{avgW0-W12}$ of about 65 μg/ml.

Embodiment 63. The method of Embodiment 62, wherein the administering comprises administering to the subject in need thereof the first dosage amount of at least 600 mg of the long acting α4β7 targeting molecule and a second dosage amount of at least 300 mg of the long acting α4β7 targeting molecule at least two weeks after administration of the first dosage amount of the long acting α4β7 targeting molecule.

Embodiment 64. The method of Embodiment 63, comprising administering the first dosage amount of at least 600 mg intravenously and the second dosage amount of at least 300 mg of the long acting α4β7 targeting molecule subcutaneously.

Embodiment 65. The method of Embodiment 63, comprising administering the first dosage amount of at least 600 mg subcutaneously and the second dosage amount of at least 300 mg of the long acting α4β7 targeting molecule subcutaneously.

Embodiment 66. The method of any one of Embodiments 61-65 further comprising administering to the subject in need thereof one or more dosage amount of at least 300 mg of the long acting α4β7 targeting molecule at least four weeks after administration of the second dosage amount of the long acting α4β7 targeting molecule.

Embodiment 67. The method of Embodiment 66 comprising administering the one or more dosage amount of the long acting α4β7 targeting molecule subcutaneously.

Embodiment 68. The method of any one of Embodiments 61-67 comprising administering to the subject in need thereof at least 600 mg of the long acting α4β7 targeting molecule every twenty-six weeks after administration of the second dosage amount of the long acting α4β7 targeting molecule.

Embodiment 69. A method of achieving $C_{avgW0-W6}$ of from 65 μg/ml to 75 μg/ml or greater for a long acting α4β7 targeting molecule in a subject in need thereof, the method comprising administering to the subject in need thereof a first dosage amount of at least 600 mg of the long acting α4β7 targeting molecule, wherein the long acting α4β7 targeting molecule comprises:

(a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and

(b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Embodiment 70. The method of Embodiment 69, wherein the administering results in a $C_{avgW0-W6}$ of about 70 μg/ml.

Embodiment 71. The method of Embodiment 69, wherein the administering results in a $C_{avgW0-W6}$ of about 65 μg/ml.

Embodiment 72. The method of any one of Embodiments 69-71, wherein the administering comprises administering to the subject in need thereof a first dosage amount of at least 600 mg of the long acting α4β7 targeting molecule and a second dosage amount of at least 300 mg of long acting α4β7 targeting molecule at least two weeks after administration of the first dosage amount of at least 600 mg of the long acting α4β7 targeting molecule.

Embodiment 73. The method of Embodiment 72, comprising administering the first dosage amount of the long acting α4β7 targeting molecule intravenously and the second dosage amount of the long acting α4β7 targeting molecule subcutaneously.

Embodiment 74. The method of Embodiment 72, comprising administering the first dosage amount and the second dosage amount of the long acting α4β7 targeting molecule subcutaneously.

Embodiment 75. The method of any one of Embodiments 72-74 further comprising administering to the subject in need thereof one or more additional dosage amount of at least 300 mg of the long acting α4β7 targeting molecule at least four weeks after administration of the second dosage amount of the long acting α4β7 targeting molecule.

Embodiment 76. The method of Embodiment 75 comprising administering the one or more additional dosage amount of the long acting α4β7 targeting molecule subcutaneously.

Embodiment 77. The method of Embodiment 75 or Embodiment 68 further comprising administering to the subject in need thereof at least 600 mg of the long acting α4β7 targeting molecule every twenty-six weeks after administration of second dosage amount of the long acting α4β7 targeting molecule.

Embodiment 78. The method of any one of Embodiments 66-68 or 75-77, wherein the administration of the one or more dosage amount of the long acting α4β7 targeting molecule results in C_{avg} of from 40 μg/ml to 50 μg/ml or greater.

Embodiment 79. The method of any one of Embodiments 66-68 or 75-77, wherein the administration of the one or more dose of the long acting α4β7 targeting molecule results in C_{avg} of about 45 μg/ml.

Embodiment 80. A method of achieving C_{trough} of 35 μg/ml or greater for a long acting α4β7 targeting molecule in a subject in need thereof six weeks following administration of the long acting α4β7 targeting molecule to the patient, the method comprising administering to the patient in need thereof at least 600 mg of the long acting α4β7 targeting molecule, wherein the long acting α4β7 targeting molecule comprises:

(a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and

(b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Embodiment 81. The method of Embodiment 80, wherein the administration of the at least 600 mg of the long acting α4β7 targeting molecule results in a serum $C_{avgW0-W12}$ of at least 60 μg/ml.

Embodiment 82. The method of Embodiment 80, wherein the administration of the at least 600 mg of the long acting α4β7 targeting molecule results in a serum $C_{avgW30-W40}$ of at least 30 μg/ml.

Embodiment 83. The method of Embodiment 80, wherein the at least 600 mg of the long acting α4β7 targeting molecule is administered intravenously.

Embodiment 84. The method of Embodiment 80, wherein the at least 600 mg of the long acting α4β7 targeting molecule is administered subcutaneously.

Embodiment 85. The method of any one of Embodiments 80-84 further comprising administering to the subject in need thereof at least 300 mg of the long acting α4β7 targeting molecule, wherein the administration of the at least 300 mg of the long acting α4β7 targeting molecule results in a steady state C_{trough} of at least 6 μg/ml.

Embodiment 86. The method of any one of Embodiments 80-84, further comprising administering to the subject in need thereof at least 300 mg of the long acting $\alpha4\beta7$ targeting molecule results in an average serum concentration of the $\alpha4\beta7$ binding antibody above 35 μg/ml for ten weeks following the administration of the at least 300 mg of the long acting $\alpha4\beta7$ targeting molecule.

Embodiment 87. The method of any one of Embodiments 85-86 further comprising administering subcutaneously to the subject in need thereof at least 300 mg of long acting $\alpha4\beta7$ targeting molecule four weeks after administration of last dosage amount of the long acting $\alpha4\beta7$ targeting molecule.

Embodiment 88. The method of any one of Embodiments 85-86 further comprising administering subcutaneously to the patient a maintenance dose of at least 600 mg of the long acting $\alpha4\beta7$ targeting molecule every twenty-six weeks after administration of the last dose of the $\alpha4\beta7$ binding antibody.

Embodiment 89. A long acting $\alpha4\beta7$ targeting molecule comprising:

(a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and

(b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Embodiment 90. An isolated nucleic acid encoding the heavy chain and/or light chain of the long acting $\alpha4\beta7$ targeting molecule of Embodiment 89.

Embodiment 91. The isolated nucleic acid of Embodiment 90 encoding SEQ ID NO: 1.

Embodiment 92. The isolated nucleic acid of Embodiment 90 encoding SEQ ID NO: 3.

Embodiment 93. A recombinant host cell comprising the isolated nucleic acid of any one of Embodiments 90-92, or an expression vector comprising the isolated nucleic acid of any one of Embodiments 90-92.

Embodiment 95. A kit for treating a disease in a subject in need thereof comprising:

(a) an injectable liquid formulation comprising a long acting $\alpha4\beta7$ targeting molecule,

wherein the long acting $\alpha4\beta7$ targeting molecule comprises:

(i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and

(ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3, and

(b) instructions to administer at least 600 mg of the long acting $\alpha4\beta7$ targeting molecule, at least 300 mg of the long acting $\alpha4\beta7$ targeting molecule at least two weeks after the administration of the 600 mg of the long acting $\alpha4\beta7$ targeting molecule, and at least 300 mg of the long acting $\alpha4\beta7$ targeting molecule at least four weeks thereafter.

Embodiment 96. A kit for treating a disease in a subject in need thereof comprising:

(a) an injectable subcutaneous liquid formulation comprising a long acting $\alpha4\beta7$ targeting molecule,

wherein the long acting $\alpha4\beta7$ targeting molecule comprises:

(i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and

(ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3, and

(b) instructions to administer subcutaneously at least 600 mg of the long acting $\alpha4\beta7$ targeting molecule, at least 300 mg of the long acting $\alpha4\beta7$ targeting molecule at least two weeks after the administration of the 600 mg of the long acting $\alpha4\beta7$ targeting molecule, and at least 300 mg of the long acting $\alpha4\beta7$ targeting molecule at least four weeks thereafter.

Embodiment 97. The kit of Embodiment 95 or Embodiment 96, wherein the disease is a gastrointestinal inflammatory disease.

Embodiment 98. The kit of Embodiment 95 or Embodiment 96, wherein the disease is an inflammatory bowel disease, optionally Crohn's disease or ulcerative colitis.

Embodiment 99. The long acting $\alpha4\beta7$ targeting molecule of Embodiment 89, the long acting $\alpha4\beta7$ targeting molecule of the regimen of any one of Embodiments 1-30, long acting $\alpha4\beta7$ targeting molecule of the injectable dosage form of any one of Embodiments 31-42, the long acting $\alpha4\beta7$ targeting molecule of the method of any one of Embodiments 44-88, the long acting $\alpha4\beta7$ targeting molecule of the kit of any one of Embodiments 95-98, wherein the long acting $\alpha4\beta7$ targeting molecule comprises one or more of the following characteristics:

i. an average serum half-life greater than about 6 days in cynomolgus monkeys,

ii. an average serum half-life of about 17 days or greater in cynomolgus monkeys,

iii. a melting temperature TmOnset greater than about 55° C.,

iv. a melting temperature TmOnset greater than 56.2° C.,

v. does not induce T cell activation markers CD25 or CD69,

vi. does not result in release of cytokines at least 6 hours after the administration, wherein the cytokines comprise one or more of IL-6, IL-8, IL-10, IFNγ, IL-4, IL-17, IL-2, IL-23p70 and TNF,

vii. does not induce complement-dependent cytotoxicity (CDC) in primary human PBMCs and in $\alpha4\beta7$-expressing human β-lymphoid cell,

viii. does not induce antibody dependent cellular cytotoxicity (ADCC) in human NK cells,

ix. does not impact suppressive activity of regulatory T cells expressing $\alpha4\beta7$ integrin, wherein the suppressive activity is measured by presence of elevated CD71, CD25, Ki67, granzyme B, or OX40,

x. a proportion of GOF from about 55.10% to about 60.4%,

xi. a proportion of G1F from about 16.6% to about 18.8%,

xii. a proportion of Man5 from about 7.3% to about 8.8%,

Embodiment 100. A subcutaneous dosage form comprising at least about 300 mg of an $\alpha4\beta7$ binding antibody comprising:

a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and

b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3.

Embodiment 101. The dosage form of Embodiment 100, wherein the heavy chain comprises an amino acid sequence according to SEQ ID NO: 1 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3.

Embodiment 102. The dosage form of Embodiment 100, wherein the heavy chain comprises an amino acid sequence according to SEQ ID NO: 2 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3.

Embodiment 103. The dosage form of any one of Embodiments 100-102, comprising about 350 mg, 400 mg, 450 mg, 500 mg, 550 mg, 600 mg, 650 mg, or 700 mg of the α4β7 binding antibody.

Embodiment 104. A method of administering one or more of the dosage forms of any one of Embodiments 100-103.

Embodiment 105. A method of treating an inflammatory bowel disease in a patient in need thereof, the method comprising subcutaneously or intravenously administering to the patient an effective amount of an α4β7 binding antibody comprising:

a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and

b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3.

Embodiment 106. The method of Embodiment 105, wherein the inflammatory bowel disease is Crohn's disease or ulcerative colitis.

Embodiment 107. The method of Embodiment 105, wherein the inflammatory bowel disease is ulcerative colitis.

Embodiment 108. The method of any one of Embodiments 105-107, wherein the heavy chain comprises an amino acid sequence according to SEQ ID NO: 1 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3.

Embodiment 109. The method of any one of Embodiments 105-108, wherein the heavy chain comprises an amino acid sequence according to SEQ ID NO: 2 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3.

Embodiment 110. The method of any one of claims Embodiments 105-109, wherein administration of the α4β7 binding antibody is subcutaneous.

Embodiment 111. The method of any one of Embodiments 105-110, wherein administration of the α4β7 binding antibody is intravenous.

Embodiment 112. The method of any one of Embodiments 105-111, wherein the α4β7 binding antibody is administered at a dose of about 300 mg.

Embodiment 113. The method of any one of Embodiments 105-112, wherein the α4β7 binding antibody is administered at a dose of about 350 mg, 400 mg, 450 mg, 500 mg, 550 mg, 600 mg, 650 mg, or 700 mg.

Embodiment 114. A method of treating a disease or disorder in a patient in need thereof, the method comprising administering to the patient at least once at an interval more than 8 weeks, an effective amount of an α4β7 binding antibody comprising:

a) a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2; and

b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3.

Embodiment 115. The method of Embodiment 114, wherein the interval is about 12 to about 26 weeks.

Embodiment 116. The method of Embodiment 115, wherein the interval is about 12 weeks.

Embodiment 117. The method of Embodiment 115, wherein the interval is about 16 weeks.

Embodiment 118. The method of Embodiment 115, wherein the interval is about 26 weeks.

Embodiment 119. The method of Embodiment 115, wherein the disease or disorder is an inflammatory bowel disease.

Embodiment 120. The method of Embodiment 119, wherein the inflammatory bowel disease is Crohn's disease or ulcerative colitis.

Embodiment 121. The method of Embodiment 119, wherein the inflammatory bowel disease is ulcerative colitis.

Embodiment 122. The method of any one of Embodiments 115-121, wherein the heavy chain comprises an amino acid sequence according to SEQ ID NO: 1 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3.

Embodiment 123. The method of any one of Embodiments 115-121, wherein the heavy chain comprises an amino acid sequence according to SEQ ID NO: 2 and the light chain comprises an amino acid sequence according to SEQ ID NO: 3.

Embodiment 124. The method of any one of Embodiments 115-123, wherein administration of the α4β7 binding antibody is subcutaneous.

Embodiment 125. The method of any one of Embodiments 115-123, wherein administration of the α4β7 binding antibody is intravenous.

Embodiment 126. The method of any one of Embodiments 115-125, wherein the α4β7 binding antibody is administered at a dose of about 300 mg.

Embodiment 127. The method of any one of Embodiments 115-125, wherein the α4β7 binding antibody is administered at a dose of about 350 mg, 400 mg, 450 mg, 500 mg, 550 mg, 600 mg, 650 mg, or 700 mg.

Embodiment 128. The method of any one of Embodiments 115-127, wherein the disease or disorder is an inflammatory bowel disease and administration of the α4β7 binding antibody is at a dose of about 300 mg and is subcutaneous.

Embodiment 129. A method of treating a disease or disorder in a patient in need thereof comprising subcutaneously administering to the patient about 108 mg or about 300 mg of an α4β7 binding antibody every four or six weeks,

wherein the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 4 and a light chain comprising an amino acid sequence according to SEQ ID NO: 5; and

wherein no intervening subcutaneous administration of the α4β7 binding antibody occurs during the four or six weeks.

Embodiment 130. A method of treating a disease or disorder in a patient in need thereof comprising subcutaneously administering to the patient about 108 mg or about 300 mg of an α4β7 binding antibody,

wherein the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2 and a light chain comprising an amino acid sequence according to SEQ ID NO: 3; and

wherein a plasma concentration of the α4β7 binding antibody of at least about 35 μg/mL is achieved in the patient for at least 2 weeks or more upon the administration.

Embodiment 131. A method of treating a disease or disorder in a patient in need thereof comprising subcutaneously administering to the patient about 108 mg or about 300 mg of an α4β7 binding antibody,

wherein the α4β7 binding antibody comprises a heavy chain comprising an amino acid sequence according to SEQ ID NO: 1 or SEQ ID NO: 2 and a light chain comprising an amino acid sequence according to SEQ ID NO: 3; and

wherein a C_{trough} of the α4β7 binding antibody of at least about 6 μg/mL is achieved in the patient for at least 2 weeks or more upon the administration.

Embodiment 132. The method of any one of Embodiments 130-131, wherein the disorder is inflammatory bowel disease.

Embodiment 133. The method of any one of Embodiments 130-131, wherein the disorder is Crohn's disease or ulcerative colitis.

Embodiment 134. An isolated nucleic acid, wherein the nucleic acid encodes the heavy chain and/or light chain of the α4β7 binding antibody of any one of Embodiments 100-134.

Embodiment 135. The isolated nucleic acid of Embodiment 134, wherein the nucleic acid sequence encodes SEQ ID NO: 1.

Embodiment 136. The isolated nucleic acid of Embodiment 134, wherein the nucleic acid encodes SEQ ID NO: 3.

Embodiment 137. A recombinant host cell comprising the nucleic acid sequences of any one of Embodiments 135-136 or an expression vector comprising the nucleic acid sequences of any one of Embodiments 135-136.

Embodiment 138. An α4β7 binding antibody comprising:

a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and

b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3.

Embodiment 139. An α4β7 binding antibody comprising:

a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and

b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Embodiment 140. An α4β7 binding antibody comprising a heavy chain and a light chain, wherein the heavy chain consists of an amino acid sequence according to SEQ ID NO: 1.

Embodiment 141. The α4β7 binding antibody of Embodiment 140, wherein the light chain consists of an amino acid sequence according to SEQ ID NO: 3.

Embodiment 142. The α4β7 binding antibody of any one of Embodiments 138-141, wherein the α4β7 binding antibody has an increased half-life as compared to an α4β7 binding antibody comprising a heavy chain sequence consisting of SEQ ID NO: 4.

Embodiment 143. The α4β7 binding antibody of any one of Embodiments 138-142, wherein the α4β7 binding antibody has a half-life of from 42 days to 56 days or more in humans.

Embodiment 144. A pharmaceutical composition comprising an α4β7 binding antibody and a pharmaceutically acceptable carrier, the α4β7 binding antibody comprising:

a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and

b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3.

Embodiment 145. A pharmaceutical composition comprising an α4β7 binding antibody and a pharmaceutically acceptable carrier, the α4β7 binding antibody comprising:

a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and

b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Embodiment 146. The pharmaceutical composition of any one of Embodiment 144-145, wherein the composition is citrate free.

Embodiment 147. A subcutaneous dosage form comprising at least about 150 mg of an α4β7 binding antibody comprising:

a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and

b) a light chain comprising an amino acid sequence according to SEQ ID NO: 3.

Embodiment 148. A subcutaneous dosage form comprising at least about 150 mg of an α4β7 binding antibody comprising:

a) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and

b) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

Embodiment 149. The subcutaneous dosage form of Embodiment 147 or Embodiment 148, wherein the dosage form is citrate free.

Embodiment 150. The subcutaneous dosage form of Embodiment 147, Embodiment 148, or Embodiment 149, comprising from about 150 mg to about 900 mg of the α4β7 binding antibody.

Embodiment 151. A method for treating inflammatory bowel disease in a subject in need thereof, the method comprising administering subcutaneously to the subject one or more of the dosage forms of any one of Embodiments 147-150.

Embodiment 152. The method of Embodiment 151, wherein the inflammatory bowel disease is Crohn's disease or ulcerative colitis.

Embodiment 153. The method of Embodiment 151, wherein the inflammatory bowel disease is ulcerative colitis.

Embodiment 154. A method of treating an inflammatory bowel disease in a subject in need thereof, the method comprising administering subcutaneously to the subject an effective amount of an α4β7 binding antibody comprising a heavy chain and a light chain, wherein the heavy chain consists of an amino acid sequence according to SEQ ID NO: 1; and the light chain comprises the variable region of SEQ ID NO: 3.

Embodiment 155. A method for treating inflammatory bowel disease in a subject in need thereof, the method comprising administering subcutaneously to the subject an effective amount of an α4β7 binding antibody comprising a heavy chain and a light chain, wherein the heavy chain consists of an amino acid sequence according to SEQ ID NO: 1; and the light chain consists of the variable region of SEQ ID NO: 3.

Embodiment 156. The method of Embodiment 154 or Embodiment 155, wherein the α4β7 binding antibody is administered at a dose of from about 150 mg to about 900 mg.

Embodiment 157. The method of Embodiment 154 or Embodiment 155, comprising administering to the subject the effective amount of an α4β7 binding antibody at an interval of more than 8 weeks.

Embodiment 158. The method of Embodiment 157, wherein the interval is about 12 to about 26 weeks.

Embodiment 159. The method of Embodiment 157, wherein the interval is about 12 weeks.

Embodiment 160. The method of Embodiment 157, wherein the interval is about 16 weeks.

Embodiment 161. The method of Embodiment 157, wherein the interval is about 26 weeks.

Embodiment 162. The method of Embodiment 157, wherein the inflammatory bowel disease is Crohn's disease or ulcerative colitis.

Embodiment 163. The method of Embodiment 157, wherein the inflammatory bowel disease is ulcerative colitis.

Embodiment 164. The method of Embodiment 155 or Embodiment 156, wherein an average plasma or serum concentration of the α4β7 binding antibody of at least about 35 μg/mL is achieved in the subject during an induction period.

Embodiment 165. The method of Embodiment 154 or Embodiment 155, wherein a maintenance C_{trough} of the α4β7 binding antibody of at least about 6 μg/mL is achieved in the subject upon the administration.

Embodiment 166. A method for treating inflammatory bowel disease in a subject in need thereof, the method comprising (a) administering subcutaneously to the subject one or more induction dose comprising an effective amount of an α4β7 binding antibody, wherein the α4β7 binding antibody comprises a heavy chain comprising a heavy chain and a light chain, wherein the heavy chain consists of an amino acid sequence according to SEQ ID NO: 1; and the light chain consists of the variable region of SEQ ID NO: 3; (b) administering subcutaneously to the subject one or more maintenance dose comprising an effective amount of the α4β7 binding antibody.

Embodiment 167. The method of Embodiment 166, wherein the induction dose comprises an amount to the α4β7 binding antibody about 2 or more times higher than the maintenance dose.

Embodiment 168. The method of Embodiment 166, wherein the induction dose comprises from about 400 mg to about 600 mg of the α4β7 binding antibody and wherein the maintenance dose of from about 150 to about 300 mg of the α4β7 binding antibody.

Embodiment 169. The method of Embodiment 166, wherein the induction dose comprises about 600 mg of the α4β7 binding antibody and wherein the maintenance dose of about 300 mg of the α4β7 binding antibody.

Embodiment 170. The method of Embodiment 166, wherein the induction dose comprises about 400 mg of the α4β7 binding antibody and wherein the maintenance dose of about 200 mg of the α4β7 binding antibody.

Embodiment 171. The method of Embodiment 166, wherein the induction dose comprises about 600 mg of the α4β7 binding antibody and wherein the maintenance dose of about 600 mg of the α4β7 binding antibody.

Embodiment 172. The method of any one of Embodiments 166-171, wherein upon administration of the one or more induction dose an average serum or plasma concentration at least about 35 μg/mL of the α4β7 binding antibody is achieved in the patient during an induction period.

Embodiment 173. The method of any one of Embodiments 166-171, wherein upon administration of the one or more maintenance dose a serum or plasma concentration at least about 6 μg/mL of the α4β7 binding antibody is achieved in the patient in the patient for at least 2 weeks or more.

Embodiment 174. The method of any one of Embodiments 166-171, wherein the one or more maintenance dose of the α4β7 binding antibody is administered at an interval of more than 8 weeks.

EXAMPLES

The disclosure now being generally described, will be more readily understood by reference to the following examples, which are included merely for purposes of illustration of certain aspects and embodiments of the present disclosure, and is not intended to limit the disclosure.

Example 1: α4β7 Binding Antibodies Efficacy

This Example describes the efficacy of the α4β7 binding antibodies described herein. PBMCs contain cell types that express α4β7 integrin and CD4+ helper T cells are associated with inflammatory bowel diseases. The ability of α4β7 binding antibody described herein (i.e., α4β7 binding antibody Ab001 comprising a heavy chain comprising a sequence according to SEQ ID NO: 1 and a light chain comprising a sequence according to SEQ ID NO: 3) to bind PBMCs was analyzed. Cell binding to PBMCs isolated from three human donors was determined by staining with either comparator antibody (solid line; filled shapes) or Ab001 (dashed line; open shapes) at various increasing concentrations (FIG. **1**A and FIG. **1**B). FIG. **1**C shows the receptor occupancy on α4β7-expressing peripheral blood mononuclear cells. The comparator antibody described in this Example and the remaining Examples comprises a heavy chain comprising a sequence according to SEQ ID NO: 4 and a light chain comprising a sequence according to SEQ ID NO: 5. A basic immunophenotyping T-cell panel indicated that both the comparator antibody and α4β7 binding antibody exhibit binding on the memory T-helper CD3+ CD4+CD45RO+ population. EC_{50} values for the comparator antibody and α4β7 binding antibody is seen in (Table 2A) and EC_{50} values for both antibodies correspond well with what has previously been reported for binding to the same population: EC_{50} 0.3 nM.

TABLE 2A

EC50 data for binding to PBMCs

	Comparator Antibody EC_{50} (nM)	α4β7 binding antibody (Ab001) EC_{50} (nM)
Donor 1	0.22	0.21
Donor 2	0.27	0.30
Donor 3	0.24	0.22

The IC_{50} values for competitive inhibition were determined by titrating unlabeled Ab001 into samples equilibrated with EC50 concentrations of fluorescently labeled Ab001. Ab001 and Comparator Antibody Receptor Occupancy (IC_{50}) on CD4+, CD20+, and CD45RO+ Cells from Human and Monkey Donors are shown in Table 2B.

TABLE 2B

Receptor Occupancy by Cell Type							
		Human whole blood		Human PBMCs		Monkey PBMCs	
Cell Type	Attribute	Ab001	Comparator Antibody	Ab001	Comparator Antibody	Ab001	Comparator Antibody
CD4 + T_h Cells	IC 50 (nM, range, n = 3 donors)	0.41-0.64	0.40-0.46	0.37-0.39	0.35-0.40	0.49-0.54	0.48-0.60
	Maximum inhibition (%, range, n = 3 donors)	85-97%	86-97%	68-84%	88-91%	86-88%	88-93%
CD20 + B Cells	IC_{50} (nM, range, n = 3 donors)	0.44-0.60	0.40-0.49	0.35-0.38	0.36-0.38	0.46-0.50	0.42-0.66
	Maximum inhibition (%, range, n = 3 donors)	85-98%	83-98%	68-83%	90-91%	76-83%	80-85%
CD45RO + Memory T Cells	IC_{50} (nM, range, n = 1 donor)	0.41-0.64	0.40-0.47	0.37-0.40	0.31-0.44	0.53	0.58
	Maximum inhibition (%, range, n = 1 donor)	86-97%	87-97%	67-84%	84-89%	89%	93%
All cell types	IC_{50} (nM, mean, n = 7)	0.53 ± 0.08	0.44 ± 0.03	0.38 ± 0.02	0.38 ± 0.03	0.50 ± 0.03	0.56 ± 0.08
All cell types	Maximum inhibition (%, mean, n = 7)	90 ± 5%	90 ± 5%	81 ± 3%	90 ± 1%	84% ± 4%	87% ± 5%

α4β7 integrin is a ligand for mucosal addressin cell adhesion molecule 1 (MAdCAM1) and the interaction of these two molecules leads to gut homing of lymphocytes. Blocking this interaction is a therapeutic target for inflammatory bowel diseases including ulcerative colitis and Crohn's disease. α4β7-mediated adhesion of MAdCAM-1 was tracked in cells treated with various increasing concentrations of α4β7 binding antibody Ab001 or comparator antibody to determine percent inhibition of total adhesion of MAdCAM-1 (FIG. **2**A). α4β7 binding antibody Ab001 preformed as effectively as the comparator antibody. α4β1 integrin is a related integrin that is a ligand for VCAM-1 that mediates homing of lymphocytes to vascular endothelium. Importantly, in a similar assay tracking VCAM-1 adhesion the α4β7 binding antibody Ab001 did not block VCAM-1 adhesion. As a positive control an antibody that can target α4β1 integrin did inhibit VCAM-1 adhesion (FIG. **2**B). α4β7 binding antibody Ab001 was evaluated in human whole blood assays designed to detect effects on cytokine release and T cell activation. In whole blood assays from three human donors, α4β7 binding antibody Ab001 did not induce T cell activation markers CD25 or CD69 as measured by FACS. Furthermore, no release of cytokines (Luminex assays of IL-6, IL-8, IL-1, IFNγ, IL-4, IL-17, IL-2, IL-23p70 and TNF) was observed at time points taken 6 hours or 24 hours after addition of α4β7 binding antibody Ab001 (400 μg/mL) to the samples. In contrast, the above markers were induced when human whole blood was incubated with lipopolysaccharide (LPS) or phytohemagglutinin (PHA), known mitogens that target T and B cells and activate cytokine release.

The ability of α4β7 binding antibody Ab001 to induce CDC was evaluated in primary human PBMCs and in the α4β7-expressing human B-lymphoid cell line RPMI-8866. PBMCs were incubated (4 hr) with (i) α4β7 binding antibody Ab001, (ii) comparator antibody, (iii) OKT3 (known to induce CDC in T cells), and (iv) an IgG1 isotype negative control antibody at a series of concentrations up to 10 μg/mL. RPMI-8866 cells were incubated (4 hr) with (i) α4β7 binding antibody Ab001, (ii) comparator antibody, (iii) rituximab (known to induce CDC in B cells), and (iv) the same IgG1 isotype negative control antibody as in the PBMC cell assay. In both the PBMC and RPMI-8866 assays, cell viability was measured and complement-induced specific death (%) was reported versus test antibody concentration for all experimental conditions. Ab001, like the comparator antibody, did not induce CDC in either PBMCs or RPMI-8866 cells.

α4β7 binding antibody Ab001 was evaluated for its ability to induce ADCC in human NK cells. Primary human NK cells were isolated and used as effector cells along with an α4β7 expressing human B-lymphoid cell line (RPMI-8866) used as the target cell population. Effector NK cells and RPMI-8866 cells together were incubated (3 hr) with (i) α4β7 binding antibody Ab001, (ii) comparator antibody, (iii) rituximab (known to induce ADCC), and (iv) an IgG1 isotype negative control antibody at a series of concentrations from 300 μg/mL to 10 g/mL. Cell viability was measured and ADCC-induced specific death (%) was reported versus test antibody concentration for all experimental conditions. Neither α4β7 binding antibody Ab001 nor the comparator antibody induced NK cell-mediated

ADCC in RPMI-8866 cells. α4β7 binding antibody Ab001 showed similar cell death ranges as comparator antibody.

The ability of α4β7 binding antibody Ab001 to impact the normal activity of regulatory T cells (Treg) was tested using human PBMCs. Regulatory T cells expressing α4β7 integrin were isolated by FACS and co-cultured with CD8+T responder cells in the presence or absence of test antibodies and a T cell stimulation reagent. Markers of T cell activation (CD71, CD25, Ki67, granzyme B, or OX40) were measured. α4β7 binding antibody Ab001 did not impact Treg suppressive activity at concentrations more than 10-fold above its known EC50. As Treg cell numbers increased from a ratio of 0:16 (None) to 8:16 Treg: CD8+, stimulated CD8+ cells were suppressed based on one marker of T cell activation (CD25+). Similar results were observed when treated with α4β7 binding antibody Ab001, media alone, the IgG1 isotype negative control, and the comparator antibody. Similar results were also seen for Treg suppressive activity when measured by the markers CD71, Ki67, granzyme B, and OX40.

The ability of α4β7 binding antibody Ab001 to bind and induce internalization of α4β7 integrin was measured using CD4+ memory T Cells. Human memory CD4+ T cells were exposed to fluorescently labeled test antibodies, including α4β7 binding antibody Ab001-A647 and the comparator antibody, at concentrations ranging from 0 nM to 100 nM, then incubated (24 hr) at temperatures that either allow (37° C.) or restrict (4° C.) receptor internalization. Cells were then washed with either buffer (FACS), or a gentle acid (0.5 M NaCl+0.2 M acetic acid), the latter to remove membrane-bound antibodies and associated fluorescence signal. Cells incubated with A647-labeled Ab001 under internalization conditions (37° C.) retained greater fluorescence after acid wash than cells stained and incubated under conditions when internalization was restricted (4° C.). This indicated that α4β7 integrin was internalized after binding to fluorescently comparator antibody.

The ability for functional α4β7 to be rapidly expressed after antibody-induced internalization was determined by incubating α4β7 binding antibody Ab001 or the comparator antibody (24 hr at 37° C.) at concentrations of 0 nM (control), 0.11 nM, 0.33 nM, or 1.0 nM with human PBMCs. After incubation, to allow internalization, cells were washed twice with PBS to remove unbound, unlabeled test antibodies. The cells were then resuspended in suitable media (RPMI 1640+10% fetal bovine serum), incubated for 96 hr with fluorescently-labeled α4β7 binding antibody Ab001 or comparator antibody and monitored for fluorescence at 0 hr, 24 hr, and 96 hr. The results were similar for both α4β7 binding antibody Ab001 and comparator antibody, demonstrating that functional α4β7 could be re-expressed on lymphocytes within 24 to 96 hours of antibody-induced internalization.

Using non-human primates (NHPs) the half-life of Ab001 was studied. NHPs were intravenously injected with comparator antibody or Ab001 and normalized concentration of each protein was calculated at multiple timepoints out to 21 days post injection (FIG. **3**). Data from an external study using the comparator antibody was also plotted ($t_{1/2}$≈10 days). The β-elimination half-life of α4β7 binding antibody Ab001 ($t_{1/2}$≈17 days) was 70% longer than observed for the comparator antibody. YTE modification used for the α4β7 binding antibody Ab001 described herein typically increases mAb half-life by 2-4× versus wild-type in both NHP and human. Human mAb half-life is at least ~2×NHP half-life. Therefore the human half-life based on initial allometric scaling based on these data is projected to be 43-56 days.

In additional experiments, half-life extension was evaluated via pharmacokinetic analysis in Tg276 transgenic mice (hemizygous for human FcRn) following a single intravenous bolus dose of the α4β7 binding antibody Ab001 (10 mg/kg, 5 mg/kg, 1 mg/kg) and in cynomolgus monkeys following a single bolus dose of the α4β7 binding antibody Ab001 (150 mg/kg), given by either intravenous (IV) and subcutaneous (SC) administration. PK simulations in humans were based on allometric scaling of the α4β7 binding antibody Ab001 clearance observed in cynomolgus monkey and conducted in MATLAB.

In Tg276 mice, the half-life of the α4β7 binding antibody Ab001 was 10-11 days across all dose cohorts compared to approximately 3 days for the comparator antibody. In cynomolgus monkeys, the half-life of the α4β7 binding antibody Ab001 was 19-23 days following a single IV or SC infusion, a significant increase over the reported half-life of 10 days for the comparator antibody in cynomolgus monkeys. Based on allometric scaling of the clearance of the α4β7 binding antibody Ab001 observed in this study, predictive simulations of the α4β7 binding antibody Ab001 PK in humans suggested that Q8W-Q12W SC dosing at 300 mg would be able to achieve a $C_{6\text{-}wks,\ induction}$≥35 μg/mL and $C_{trough,\ ss}$≥6 μg/mL for simulated patients.

Example 2: Subcutaneous Dose Regimens for α4β7 Binding Antibodies

Pharmacokinetic (PK)-pharmacodynamic (PD) modeling and simulation was conducted to determine dose and regimens of subcutaneous doses that maintain certain serum concentrations above a specific level at a given time during the regimen or at steady state. Achieving these thresholds has been demonstrated to be an important metric for patients to achieve clinical remission in ulcerative colitis. Specifically, maintaining a trough concentration C_{trough}≥35 μg/mL after 6 weeks post-induction is associated with significantly higher rates of remission in ulcerative colitis.

A two compartment linear model with parameters for central volume of distribution (Vc), peripheral volume of distribution (Vp), clearance from the central volume (Cl), first-order absorption rate constant from a subcutaneous depot into the central volume (ka), and subcutaneous bioavailability relative to the intravenous administration was used to identify dosing regimens. Given the expectation of the α4β7 binding antibodies disclosed herein to have a longer serum half-life than normal IgG, the effective human half-life used in these models was 53 days.

In order to achieve higher clinical remission overall, dosing regimens where >95% of patients achieved a minimum concentration of ≥35 μg/mL at 6 weeks (Table 3) or C_{trough} Of ≥6 μg/mL at steady state (Table 4) were identified. The model was further refined by adding the effect of covariates including body weight and serum albumin.

To simulate the serum concentrations of the α4β7 binding antibodies over time with specified dosing regimens in a large population of diverse patients, 1000 patients were uniquely assigned a body weight and serum albumin concentration. These two covariates were assigned independently and randomly chosen using a cumulative distribution function (CDF) of body weights and a separate CDF for serum albumin concentration. To derive each CDF, the percentiles for body weight and serum albumin concentration reported in a clinical trial of a comparator antibody were used to fit a generalized extreme value distribution to allow for a variety of probability distribution behaviors that could be expected in a real-world clinical trial. Body weights values were log-transformed while serum albumin concentration values were used directly.

TABLE 3

Minimum induction regimens allowing for >95% of patients to achieve a serum concentration of ≥35 μg/mL at 6 weeks

Strategy	IV (Dose/Time)	Minimum x Dose (mg)	SC (Dose/Time)	Minimum y Dose (mg)
Strategy 1	x W0	550	None	N/A
Strategy 2	300 mg W0, x W2	200	None	N/A
Strategy 3	600 mg W0, x W2	0	None	N/A
Strategy 4	None	N/A	y W0	620
Strategy 5	None	N/A	300 mg W0, y W2	280
Strategy 6	None	NA	600 mg W0, y W2	100
Strategy 7	None	N/A	300 mg W0, y W1, W2, W3	150

TABLE 4

Minimum maintenance regimens allowing for >90% of patients to achieve a serum C_{trough} of ≥6 μg/mL at steady-state

Frequency	Route	Dose (mg)
Every 4 weeks	IV	40
	SQ	45
Every 8 weeks	IV	110
	SQ	120
Every 12 weeks	IV	220
	SQ	275
Every 26 weeks	IV	1800
	SQ	2000

Example 3: Dose Responses of α4β7 Binding Antibodies

This Example describes the dose responses of the α4β7 binding antibodies described herein.

A simulated trial of 1000 patients was simulated using induction scenarios with an α4β7 binding antibody with YTE modifications such as α4β7 binding antibody Ab001 at intravenous doses of 300 mg, 350 mg, and 400 mg to determine the C_{trough} values at 6 weeks post-induction (FIG. **4**). Data generated in a clinical trial of a comparator antibody dosed intravenously at 300 mg is also shown. The comparator antibody was also run through the same simulation. A human half-life of 25.5 days was used for the comparator antibody and a human half-life of 50 days was used for Ab001 in these simulations. At most, 25% of patients dosed with the comparator antibody met the C_{trough}≥35 μg/ml threshold at 6 weeks post-induction. In comparison, the vast majority of simulated patients dosed with all concentrations of Ab001 tested maintained a C_{trough}≥35 μg/ml.

The best in class efficacy maintenance dosing of the comparator antibody is based on area under the curve at steady state (AUC_{ss}) data of Q8W IV maintenance dosing at 300 mg (FIG. **5**). When dosed subcutaneously the comparator antibody can only be dosed at 108 mg Q2W without dropping below the clinically relevant AUC. In contrast, Ab001 is predicted to maintain clinically relevant AUC with quarterly maintenance subcutaneous dosing. These data were generated using a human half-life of 35 and 50 days for Ab001 and plotted with the comparator antibody data (FIG. **5**). Depending on human half-life, this could be achieved even as infrequently as Q16W.

A post-hoc analysis of the comparator antibody trial data suggests a conservative target of C_{trough}≥6 μg/mL during treatment maintains long-term efficacy. Best-in-class 300 mg IV Q8W dosing of the comparator antibody showed 75% of patients had C_{trough}≥6 μg/mL. The comparator antibody dosed subcutaneously at 108 mg with a half-life of 25.5 days, dosed IV at 300 mg with a half-life of 25.5 days, and Ab001 dosed subcutaneously at 300 mg with a half-life of 43, 53, or 56 days was used to simulate a trial of 1000 patients (FIG. **6**). Similar to clinical results, 65% of simulated patients with Q8W IV achieve C_{trough}≥6 μg/mL (65% is marked with a dashed line). Ab001 with a minimum 43-day half-life enables Q12W SC dosing. This matches the simulated comparator antibody IV Q8W dosing. Ab001 with a 50-day half-life at Q12W SC dosing results in more than 90% of patient with C_{trough}≥6 μg/mL. Dosing frequencies that maintained a clinically relevant AUC_{ss} in FIG. **5** are indicated with stars.

Taken together, these data indicate Ab001 with a half-life of at least 35 days can be dosed subcutaneously with a Q12W frequency and maintain both a clinically relevant C_{trough} and AUC_{ss}.

Example 4: Binding Affinity of α4β7 Binding Antibodies

This Example describes the binding affinity of the α4β7 binding α4β7 binding antibody Ab001.

Briefly, binding affinity of the α4β7 binding α4β7 binding antibody Ab001 to α4β7 was determined by Kinetic Exclusion Assay (KinExA) and flow cytometry, and functional activity was determined by inhibition of MAdCAM-mediated cellular adhesion.

The α4β7 binding α4β7 binding antibody Ab001 binds specifically to α4β7 and not to the related integrins α4β1 and αEβ7. In cellular assays, Ab001 binds α4β7 on the surface of RPMI-8866 cells with an affinity matching that of a comparator antibody, but it does not bind to Ramos cells expressing α4β1. The α4β7 binding α4β7 binding antibody Ab001 demonstrates high affinity for α4β7, with a K_D of 836 μM determined by KinExA. Functionally, the α4β7 binding α4β7 binding antibody Ab001 potently inhibits MAdCAM-1-mediated cellular adhesion with an IC_{50} comparable to that of a comparator antibody but has no inhibitory activity for VCAM-1-mediated cell adhesion.

Example 5: Simulation of Serum Concentrations of the α4β7 Binding Antibody

A simulation was performed using induction scenarios as follows: (1) with an α4β7 binding α4β7 binding antibody described herein such as Ab001 at subcutaneous doses of 600 mg at week 0 (W0) and 300 mg at week 2 (W2) to determine the average serum concentration during the induction period of 12 weeks; (2) with the comparator antibody at intravenous dose of 300 mg at week 0, week 2 and week 6; and (3) with the comparator antibody at intravenous dose of 300 mg at week 0 and week 2, and a subcutaneous dose of 108 mg Q2W starting at week 6. FIG. **7**A shows that $C_{avgW0\text{-}12}$ of Ab001 is greater than $C_{avgW0\text{-}12}$ of comparator antibody. FIG. **7**C shows that $C_{avgW0\text{-}12}$ of Ab001 is greater than $C_{avgW0\text{-}12}$ of comparator antibody. Simulated patients with Q12W SC achieve a C_{trough}≥36 μg/mL which was higher than C_{trough} of Simulated patients with IV administration of the comparator antibody and which was higher than C_{trough} of simulated patients with IV/SC administration of the comparator antibody. In addition, simulated patients with Q12W SC achieve maintain the $C_{trough} \geq 36$ μg/mL during the induction period.

FIG. **7**B depicts serum concentrations based on 300 mg Q8W IV dosing of comparator antibody and 300 mg Q12W SC dosing of the α4β7 binding antibody described herein such as Ab001 based on the half-life of the antibodies. FIG. **7**B shows that Q12W SC dosing of the α4β7 binding antibody described herein such as Ab001 results in a maintenance target $C_{trough} \geq 6$ μg/mL.

FIG. **7**D and FIG. **7**E show that simulated patients with Q12W SC achieve an effective minimum average serum or plasma concentration of the α4β7 binding antibody described herein such as Ab001.

FIG. **7**F depicts serum concentrations based on dosing of comparator antibody at intravenous dose of 300 mg at week 0 and week 2, and a subcutaneous dose of 108 mg Q2W starting at week 6 (26 injections per year for maintenance), and based on dosing of the α4β7 binding antibody described herein such as Ab001 at subcutaneous doses of 300 mg Q12W (4-6 injections per year for maintenance), based on the half-life of the antibodies. FIG. **7**F shows that the α4β7 binding antibody described herein such as Ab001 has a superior steady state profile for patients.

Example 6: PK Simulation of Serum Concentrations of the α4β7 Binding Antibody

FIGS. **8**A-**8**C are graphs showing human PK simulation (Simulated Using 75 mg/kg NHP—SC Dose (n=6, M/F)). FIG. **8**A is a graph showing the α4β7 binding antibody described herein such as Ab001 simulated human profiles at 600 mg. FIG. **8**B is a graph showing α4β7 binding antibody described herein such as Ab001 simulated human profiles 600 mg Q26W. FIG. **8**C is a graph showing α4β7 binding antibody described herein such as Ab001 simulated human profiles 400 mg, 200 mg, Q12W.

FIG. **8**A shows that a single 600 mg dose remains above the induction threshold concentration for 12 weeks and drops below the maintenance concentration at 32 weeks.

FIG. **8**B shows that simulation dosing of 600 mg α4β7 binding antibody described herein such as Ab001 Q26W (Q6M) maintains induction and maintenance concentration.

FIG. **8**C shows that simulation dosing of 400 mg, 200 mg α4β7 binding antibody described herein such as Ab001 Q12W maintains induction and maintenance concentration.

Example 7: Summary of Key Binding Characteristics, Preclinical In Vitro and In Vivo Studies for the α4β7 Binding Antibody Ab001

TABLE 5

Study Summary

Study Name	Study Description and Methods	Key Results and Conclusions
Functional Characterization of the α4β7 binding antibody Ab001 on α4β7 & α4β1	Test system: α4β7-positive RPMI-8866 cells, α4β1-positive Ramos cells α4β7 and α4β1 double positive HuT-78 cells; fluorescence-activated cell sorting (FACS) Concentration range: 5 pM-100 nM 3 cell lines; 6 concentrations; three replicates	the α4β7 binding antibody Ab001 potency, α4β7: EC_{50} = 0.15 nM the α4β7 binding antibody Ab001 potency, α4β1: EC_{50} = NB the α4β7 binding antibody Ab001 potency, α4β7, α4β1: EC_{50} = 0.25 nM Comparator antibody potency, α4β7: EC_{50} = 0.14 nM Comparator antibody potency, α4β1: EC_{50} = NB Comparator antibody potency, α4β1, α4β7: EC_{50} = 0.24 nM The α4β7 binding antibody Ab001 exhibits functional, high-potency binding to α4β7 but not to α4β1. Natalizumab was included in this study as a positive control for functional α4β1 binding.
Binding Potencies and Receptor Occupancy of the α4β7 binding antibody Ab001	Test system: Receptor Occupancy (IC_{50}) on CD4+, CD20+, and CD45RO+ Cells from Human Donors PBMCs and Monkey Donors PBMCs; titration of unlabeled Ab001 into samples equilibrated with EC50 concentrations of fluorescently labeled Ab001	the α4β7 binding antibody Ab001; Human whole blood: IC_{50} for all cell types = 0.53 nM the α4β7 binding antibody Ab001; Human PBMCs: IC_{50} for all cell types = 0.38 nM the α4β7 binding antibody Ab001; Monkey PBMCs: IC_{50} for all cell types = 0.50 nM Comparator antibody; Human whole blood: IC_{50} for all cell types = 0.44 nM Comparator antibody; Human PBMCs: IC_{50} for all cell types = 0.38 nM Comparator antibody; Monkey PBMCs: IC_{50} for all cell types = 0.56 nM
Determination of K_D against α4β7, α4β1, and αEβ7 by SPR or KinExA	Test system: KinExA and surface plasmon resonance Concentration range: Varied by target n = 5 concentrations	The α4β7 binding antibody Ab001 K_D v. α4β7 = 836 pM The α4β7 binding antibody Ab001 K_D v. α4β1 = NB The α4β7 binding antibody Ab001 K_D v. αEβ7 = NB Comparator antibody K_D v. α4β7 = 562 pM Comparator antibody K_D v. α4β1 = NB Comparator antibody K_D v. αEβ7 = NB Natalizumab was included in this study as a positive control for functional α4β1 binding.

TABLE 5-continued

Study Summary		
Study Name	Study Description and Methods	Key Results and Conclusions
Species Cross-reactivity and Binding Affinity Determination	Test system: SPR with multiple cycle kinetics	the α4β7 binding antibody Ab001 Cyno-α4β7 K_D = 2.72 nM the α4β7 binding antibody Ab001 Dog-α4β7 K_D = 4.81 nM the α4β7 binding antibody Ab001 Rabbit-α4β7 K_D = 6.26 nM the α4β7 binding antibody Ab001 Rat-α4β7 K_D = NB the α4β7 binding antibody Ab001 Mouse-α4β7 K_D = NB the α4β7 binding antibody Ab001 Human-α4β7 K_D = 4.65 nM Comparator antibody Cyno-α4β7 K_D = 1.2 nM Comparator antibody Dog-α4β7 K_D = 1.77 nM Comparator antibody Rabbit-α4β7 K_D = 5.34 nM Comparator antibody Rat-α4β7 K_D = NB Comparator antibody Mouse-α4β7 K_D = NB Comparator antibody Human-α4β7 K_D = 4.16 nM
Binding of Ab001 to Cells Isolated from Human, Rabbit, Mouse, and Rat PBMCs	Test system: binding to $CD4^+$ T cells, $CD4^+CD45RO+$ (human only), and either $CD20^+$ B cells or IgG^+ cells from cryopreserved PBMCs from human, mouse, rat, and rabbit, using flow cytometry	In human PBMCs, a dose-dependent increase in Ab001 binding was observed in $CD4^+$ T cells, $CD4^+$ $CD45RO^+$ memory T cells, and $CD20^+$ B cells. In rabbits, a dose-dependent increase in Ab001 binding was observed in a subset of PBMCs. No binding above background was observed in cells isolated from either mice or rats. The results were comparable to those of comparator antibody.
Binding of Ab001 to Human Monocytes, B Cells, NK Cells, and T cell Subtypes	Test system: binding to primary human monocytes, B cells, T cells (naïve, TEMRA, effector memory cells (EM), central memory cells (CM) and a human B-lymphoid cell line expressing α4β7 and α4β1 (RPMI-8226) was evaluated using fluorescence-activated cell sorting (FACS) and fluorescently labeled antibody	Ab001 preferentially bound to α4β7-expressing B cells, NK cells, and T cells (naïve, TEMRA, effector memory, and central memory) similar to comparator antibody
Functional Characterization of the α4β7 binding antibody Ab001 on MAdCAM-1 and VCAM-1 Mediated Adhesion	Test system: α4β1 and α4β7 expressing HuT-78 cells with coating of recombinant MAdCAM-1 or VCAM-1 Concentration range: 5 pM-100 nM n = 6 replicates	MAdCAM-1 Inhibition IC50: the α4β7 binding antibody Ab001 = 0.086 nM Comparator antibody = 0.083 nM VCAM-1 Inhibition IC50: the α4β7 binding antibody Ab001 = None Comparator antibody = None Natalizumab = 0.23 nM Conclusion: the α4β7 binding antibody Ab001 and a comparator antibody exhibit functional activities that are distinct from those of natalizumab.
Pharmacokinetic study of the α4β7 binding antibody Ab001 in cynomolgus monkeys	Doses: 10 mg/kg Route of Administration: IV N = 6 per group	$T_{1/2}$ = 17 days (experiment 1) $T_{1/2}$ = 22 days (experiment 2) Conclusion: The half-life of the α4β7 binding antibody Ab001 in male cynomolgus monkeys is 70% longer than that of a comparator antibody.
Mouse Tg276 pharmacokinetic study	Dose: 1, 5, and 10 mg/kg Route of administration: IV N = 5 males per group	Conclusion: the α4β7 binding antibody Ab001 exhibited extended pharmacokinetic half-life in Tg276 mice (>200% longer than a comparator antibody).

K_D: equilibrium dissociation constant between the antibody and its antigen

IC_{50}: half-maximal inhibitory concentration

EC_{50}: half maximal effective concentration

MAdCAM-1: Mucosal vascular addressin cell adhesion molecule 1

VCAM-1: Vascular cell adhesion protein 1

Example 8: Formulations

α4β7 binding antibody Ab001 drug substance has a deglycosylated molecular weight of 146,698.3 daltons. It is soluble to greater than 200 mg/mL as a colorless to brown, clear to opalescent liquid.

α4β7 binding antibody Ab001 drug product is a sterile liquid for injection supplied in 2 mL glass vials. Each single-use drug product vial contains a nominal volume of 2 mL of 150 mg/mL Ab001 in aqueous solution with histidine, arginine, ethylenedinitrotetraacetic acid disodium salt dihydrate (EDTA), and polysorbate 80.

Example 9: Toxicology and Toxicokinetics of the α4β7 Binding Antibody Ab001

Toxicokinetic assessment after single dose IV and SC dosing of α4β7 binding antibody Ab001 at 30, 75, and 150 mg/kg was conducted in male and female cynomolgus monkeys over 27 days. No PK outliers were observed, thus all animals were considered in the statistical analysis. α4β7 binding antibody Ab001 serum concentration-time profiles in male monkeys dosed intravenously (IV) displayed a biphasic decline with a mean terminal phase $t_{1/2}$ of 6.19±1.72 days, mean AUC_{last} of 27200 4880 day·μg/mL and a mean Vss of 53.0±4.30 mL/kg. A sharp increase in CL was noted after 14 days, consistent with the development of ADAs. When male and female monkeys were dosed SC, T_{max} values ranged between 1 and 7 days. Ab001 exhibited linear dose proportionality when considering C_{max} and AUC_{last}. Mean terminal $t_{1/2}$ values ranged from 10.2 to 24.2 days. There were no statistical differences (t test, p=0.05) in exposure at the 75 mg/kg dose between male and female monkeys when considering C_{max} and AUC.

α4β7 binding antibody Ab001 serum concentration-time profiles in male monkeys dosed IV (150 mg/kg) displayed a biphasic decline with a mean (+SD) terminal phase $t_{1/2}$ of 6.19±1.72 days, mean (+SD) AUC_{last} of 27200±3200 day·μg/mL and a mean (+SD) Vss of 53.0 4.30 mL/kg. A sharp increase in CL was noted after 14 days in this cohort, indicating the probable formation of ADAs.

T_{max} values for male and female monkeys dosed SC ranged between 1 and 7 days. α4β7 binding antibody Ab001 exhibited approximately dose-linear pharmacokinetics when considering C_{max} and AUC_{last}. Mean terminal $t_{1/2}$ values ranged from 10.2 to 24.2 days. There were no statistical differences (t-test, p=0.05) in exposure at 75 mg/kg between male and female monkeys when considering C_{max} and AUC_{last}.

Overall, serum exposures of α4β7 binding antibody Ab001 after SC dosing were proportional to dose when considering C_{max} (maximum mean concentration) and AUC_{last} (area under the concentration/time curve from 0 to 648 hr) (Table 15). Mean (±SD) C_{max} values ranged from 297±46.9 to 1780±315 μg/mL, and AUC_{last} ranged from 4590±600 to 30500±6460 day·μg/mL. Exposures (AUC_{last} and C_{max}) were similar in male and female animals after a 75 mg/kg SC dose. Ab001 IV dosing at 150 mg/kg resulted in a mean AUC_{last} exposure of 27200±3200 day·μg/mL and a mean C_{max} of 3670±433 μg/mL.

α4β7 binding antibody Ab001 was then administered to male and female cynomolgus monkeys (n=3/sex/group) every two weeks (Q2W) (doses on Day 1, Day 15, and Day 29) for 1 month at 0 (SC and IV; vehicle control), 25 (SC), 80 (SC), 160 (SC), and 160 mg/kg (IV). All animals survived until their scheduled necropsy on Study Day 30. Bioanalytical samples were collected, processed, and analyzed using a validated GyroLab® bioanalytical method, and toxicokinetic parameters were determined using non-compartmental analysis. $AUC_{0-28\ d}$ and C_{max} increased in a slightly more than dose-proportional manner after Q2W SC doses of 25, 80, and 160 mg/kg. No sex differences in exposure were observed. Following a single SC administration, median T_{max} values were observed between 3.00 to 7.00 days post dose across dose levels, while T_{max} values after the Day 15 dose were between 1.00 and 7.00 days. Mean (±SD) $AUC_{0-28\ d}$ exposures (combined sex) after biweekly SC dosing were 6180±2950, 28200 2260, and 71500±17000 day·μg/mL, respectively. The C_{max} (±SD) after the SC doses on Day 15 was 430±291, 1380±151, and 4040±1380 μg/mL, respectively. The mean $AUC_{0-28\ d}$ (±SD) exposure (combined sex) after biweekly IV dosing at 160 mg/kg was 73200±12500 day·μg/mL, while the C_{max} (±SD) was 4980±905 μg/mL. The mean accumulation ratio after a second biweekly dose was <1.4 (AUC) for all doses.

Anti-α4β7 binding antibody Ab001 binding antibodies were associated with decreased exposure in two females and one male animal at the 25 mg/kg SC dose, although these animals were still included in the TK calculations for this dose. The formation of ADAs had no impact on exposure at either the 80 mg/kg or the 160 mg/kg dose (NOAEL). Throughout the evaluation of the study, serum α4β7 binding antibody Ab001 concentrations (at C_{min} for 25 mg/kg dose) were a minimum of 3300× above the EC50 of Ab001 for saturable binding. Exposures (C_{max} and $AUC_{0-28\ d}$) were slightly more than dose-proportional, and no sex-related differences were observed.

Mean (±SD) $AUC_{0-28\ d}$ exposures (combined sex) after biweekly SC dosing were 6180 2950, 28200±2260, and 71500±17000 day·μg/mL at 25, 80, and 160 mg/kg, respectively. Mean (±SD) C_{max} after SC doses on Day 15 were 430±291, 1380±151, and 4040±1380 μg/mL at 25, 80, and 160 mg/kg, respectively. The mean (±SD) $AUC_{0-28\ d}$ exposure (combined sex) after biweekly IV dosing at 160 mg/kg was 73200±12500 day·μg/mL, while the mean (±SD) C_{max} was 4980±905 μg/mL. The mean accumulation ratio after a second biweekly dose was ≤1.40 (AUC) for all doses.

It was not clear whether α4β7 binding antibody Ab001 would demonstrate toxicity because of its half-life extension. The results showed that, despite this prior uncertainty, there were no macroscopic (e.g., organs) or microscopic (e.g., cells/tissues) findings at any dose.

Example 10: Human PK

Serum α4β7 binding antibody Ab001 concentration-time profiles in humans were predicted for single and multiple SC and IV injections of α4β7 binding antibody Ab001 over the dose range of 100 to 1000 mg. Steady-state α4β7 binding antibody Ab001 serum exposures were predicted using the following two resources: (1) α4β7 binding antibody Ab001 PK parameters in cynomolgus monkeys with interspecies extrapolation of CL parameters using published allometric exponents for half-life engineered monoclonal antibodies; and (2) literature values of PK parameters reported for vedolizumab, an approved therapeutic human IgG1 monoclonal antibody targeting α4β7 (Rosario et al. 2016, Clinical Drug Investigation 36 (11): 913-23).

α4β7 binding antibody assumptions were made: (i) Ab001 follows a biphasic decline after administration; (ii) pharmacokinetic parameters including CL and Vss do not change upon repeated dosing and do not change over the dose range of 100 to 1000 mg in a 70 kg human (as observed with vedolizumab); (iii) disease state does not have a significant influence on PK; (iv) target-mediated drug disposition (TMDD) has a small influence on PK; (v), there will be no anti-α4β7 binding antibody Ab001 antibody development or, if present, anti α4β7 binding antibody Ab001 antibodies do not have an impact on α4β7 binding antibody Ab001 PK.

The predicted terminal phase $t_{1/2}$ value of α4β7 binding antibody Ab001 in humans based on the above assumptions is approximately 64 days and is consistent with the expected extended half-life resulting from incorporation of the YTE mutation into the Fc domain. Vedolizumab has a reported human half-life of 25.5 days (Rosario et al. 2015, Alimentary Pharmacology & Therapeutics 42 (2): 188-202; Rosario et al. 2016). The predicted steady-state C_{max} and AUC values for α4β7 binding antibody Ab001 in humans for single and multiple IV and SC dosing are shown in Table 6.

TABLE 6

Predicted Exposure in Humans After Single and Multiple SC and IV Administration

Clinical		Single Dose			Two Doses Given Q2W		
Dose (mg/70 kg)	Route	C_{max} μg/mL	$AUC_{0\text{-}28d}$ day • μg/mL	$AUC_{0\text{-}inf}$ day • μg/mL	C_{max} μg/mL	$AUC_{0\text{-}28d}$ day • μg/mL	$AUC_{0\text{-}inf}$ day • μg/mL
100	SC	12.0	281	568	22.3	436	1450
300		37.2	900	2530	69.6	1360	5830
600		75.0	1830	5880	141	2760	12800
900		113	2760	9370	212	4150	19800
1000		125	3070	10500	236	4620	22100
100	IV	33.3	432	874	47.6	695	2150
300		100	1360	3640	145	2140	8170
600		200	2740	8230	291	4320	17500
900		300	4130	12900	437	6490	26800
1000		333	4590	14500	486	7220	29900

Example 11: PK Induction Simulation of Serum Concentrations of the α4β7 Binding Antibody

A simulation was performed using the following induction scenarios (1) with an α4β7 binding antibody described herein such as Ab001 at subcutaneous doses of 600 mg at week 0 (W0) and 300 mg at week 2 (W2) to determine the average serum concentration during the induction period of 12 weeks, (2) with an α4β7 binding antibody described herein such as Ab001 at subcutaneous doses of 600 mg at week 0 (W0), 300 mg at week 2 (W2) and 300 mg at week 6 (W6) to determine the average serum concentration during the induction period of 12 weeks, (3) with the comparator antibody at intravenous dose of 300 mg at week 0 (W0), week 2 (W2) and week 6 (W6) to determine the average serum concentration during the induction period of 12 weeks.

An analysis of FIG. **10**A indicates that $C_{maxW0\text{-}12}$, $C_{minW0\text{-}12}$, and $C_{avgW0\text{-}12}$ of Ab001 at the subcutaneous induction dose of 600 mg at week 0 (W0) and 300 mg at week 2 (W2) was calculated to be 89.7 μg/mL, 37.4 μg/mL, and 60.7 μg/mL, respectively. An analysis of FIG. **10**B indicates that $C_{maxW0\text{-}12}$, $C_{minW0\text{-}12}$, and $C_{avgW0\text{-}12}$ of Ab001 at the subcutaneous induction dose of 600 mg at week 0 (W0), 300 mg at week 2 (W2) and 300 mg at week 6 (W6) was calculated to be 94.7 μg/mL, 57.6 μg/mL, and 73.9 μg/mL, respectively. An analysis of FIG. **10**C indicates that $C_{maxW0\text{-}12}$, $C_{minW0\text{-}12}$, and $C_{avgW0\text{-}12}$ of the comparator antibody at the intravenous dose of 300 mg at week 0 (W0), week 2 (W2) and week 6 (W6) was calculated to be 164 μg/mL, 22.9 μg/mL, and 65.1 μg/mL, respectively.

Example 12: PK Maintenance Simulation of Serum Concentrations of the α4β7 Binding Antibody

A simulation was performed using maintenance scenarios (1) with an α4β7 binding antibody described herein such as Ab001 at subcutaneous doses of 600 mg every 26 weeks to determine the average serum concentration during the maintenance period from week 30 through week 60, (2) with an α4β7 binding antibody described herein such as Ab001 at subcutaneous doses of 300 mg every 12 weeks to determine the average serum concentration during the maintenance period from week 30 through week 60, (3) with the comparator antibody at intravenous dose of 300 mg every 8 weeks to determine the average serum concentration during the maintenance period from week 30 through week 60, and (4) with the comparator antibody at subcutaneous dose of 108 mg every 2 weeks to determine the average serum concentration during the maintenance period from week 30 through week 60.

An analysis of FIG. **11**A indicates that $C_{maxW30\text{-}60}$, $C_{minW30\text{-}60}$, and $C_{avgW30\text{-}60}$ of Ab001 at the subcutaneous maintenance dose of 600 mg every 26 weeks was calculated to be 74.8 μg/mL, 7.13 μg/mL, and 31.2 μg/mL, respectively. An analysis of FIG. **11**B indicates that $C_{maxW30\text{-}60}$, $C_{minW30\text{-}60}$, and $C_{avgW30\text{-}60}$ of Ab001 at the subcutaneous maintenance dose of 300 mg every 12 weeks was calculated to be 50.6 μg/mL, 17.8 μg/mL, and 31.3 μg/mL, respectively. An analysis of FIG. **11**C indicates that $C_{maxW30\text{-}60}$, $C_{minW30\text{-}60}$, and $C_{avgW30\text{-}60}$ of the comparator antibody at the intravenous dose of 300 mg every 8 weeks was calculated to be 136 μg/mL, 10.1 μg/mL, and 37.6 μg/mL, respectively. An analysis of FIG. **11**D indicates that $C_{maxW30\text{-}60}$, $C_{minW30\text{-}60}$, and $C_{avgW30\text{-}60}$ of the comparator antibody at the subcutaneous dose of 108 mg every 2 weeks was calculated to be 33.7 μg/mL, 27.2 μg/mL, and 31.1 μg/mL, respectively.

Example 13: Single Ascending Dose Simulation of α4β7 Binding Antibody

A single ascending dose simulation was performed for two scenarios: (1) with an α4β7 binding antibody described herein such as Ab001 at 600 mg to determine serum concentration of Ab001 during the period of 84 days from the administration, and (2) with an α4β7 binding antibody described herein such as Ab001 at 300 mg to determine serum concentration of Ab001 during the period of 84 days from the administration.

FIG. **12** shows a comparison of simulated serum concentration of Ab001 at 600 mg and 300 mg dose of Ab001. An analysis of FIG. 12 indicates that serum concentration of Ab001 in both scenarios remain within concentration range of 10 μg/mL and 40 μg/mL after 84 days of administration.

Example 14: Multiple Ascending Dose Simulation of α4β7 Binding Antibody

A multiple ascending dose simulation was performed for two scenarios: (1) with an α4β7 binding antibody described herein such as Ab001 at subcutaneous doses of 600 mg on Day 0 (D0) followed by 600 mg subcutaneous dose on Day 14 (D14) to determine serum concentration of Ab001 during the period of 56 days from the administration, and (2) with an α4β7 binding antibody described herein such as Ab001 at subcutaneous doses of 600 mg on Day 0 (D0) followed by 600 mg subcutaneous dose on Day 14 (D14) to determine serum concentration of Ab001 during the period of 56 days from the administration.

FIG. 13 shows a comparison of simulated serum concentration of Ab001 at 600 mg and 300 mg dose of Ab001. An analysis of FIG. 13 indicates that serum concentration of Ab001 in both scenarios remain above 40 μg/mL after 56 days of administration.

Example 15: Clinical Efficacy of α4β7 Binding Antibody

Vedolizumab has demonstrated some efficacy in treatment of ulcerative colitis (UC) and Crohn's disease (CD) (ENTYVIO Product Monograph). This antibody has a serum half-life of 26 days in human subjects (25 days according to the product insert). Various factors increase clearance of vedolizumab, including albumin, body weight, fecal calprotectin, prior treatment with TNF antagonist drugs, and presence of anti-vedolizumab antibody. In patients administered 300 mg vedolizumab as a 30-minute intravenous infusion on Weeks 0 and 2, median serum trough concentration at Week 6 was 25.6 mcg/mL (range 0.9 to 140.0) in ulcerative colitis and 24.5 mcg/mL (range 1.1 to 177.0) in Crohn's Disease. Median steady state serum trough concentrations were 9.8 mcg/mL (range 2.4 to 42.8) and 11.2 mcg/mL (0.4 to 54.5), respectively, in patients with ulcerative colitis and Crohn's disease, when 300 mg vedolizumab was administered intravenously every eight weeks starting at week 6.

Clinical trials evaluated efficacy and safety of intravenous (for induction) and intravenous or subcutaneous vedolizumab for the maintenance treatment of adult patients with moderately to severely active ulcerative colitis (Mayo score 6 to 12 with endoscopic sub score ≥2). The primary endpoint of the induction phase was proportion of patients with clinical response at Week 6. The primary endpoint for the subcutaneous study was the proportion of patients in clinical remission (complete Mayo score of ≤2 points and no individual subscore >1 point) at Week 52. Secondary endpoints were mucosal healing (Mayo endoscopic subscore of ≤1 point) at Week 52; durable clinical response (clinical response at Weeks 6 and 52); durable clinical remission (clinical remission at Weeks 6 and 52); and corticosteroid-free clinical remission (patients using oral corticosteroids at baseline who had discontinued corticosteroids and were in clinical remission) at Week 52.

The efficacy and safety of intravenous (for induction) and subcutaneous (for maintenance) vedolizumab for the treatment of adult patients with moderately to severely active Crohn's disease (CDAI score of 220 to 450) was evaluated in a randomized, double-blind, placebo-controlled study evaluating efficacy endpoints at Week 6 (induction) and Week 52 (maintenance). The primary endpoint was the proportion of patients with clinical remission (CDAI score ≤150). The secondary endpoints were enhanced clinical response; corticosteroid free remission; and clinical remission in TNFα antagonist naïve patients, at Week 52.

α4β7 binding antibody Ab001 is evaluated in clinical trials in patients with ulcerative colitis and Crohn's disease with various primary and secondary endpoints, such as clinical remission in both ulcerative colitis and Crohn's disease, endoscopic improvement (mucosal healing) in ulcerative colitis, durable clinical response in ulcerative colitis, durable clinical remission in ulcerative colitis, enhanced clinical response in Crohn's disease, clinical remission in advanced therapy (such as a TNFα antagonist) naïve patients in Crohn's disease, and corticosteroid-free clinical remission in both ulcerative colitis and Crohn's disease. Biomarker tests are conducted at Week 6, including C-reactive protein (CRP), fecal calprotectin, and symptoms.

In a clinical study, the characteristics of the α4β7 binding antibody Ab001 provide for greater proportion of patients achieving clinical remission in both ulcerative colitis and Crohn's disease than previously observed with vedolizumab.

In a clinical study, the characteristics of the α4β7 binding antibody Ab001 provide for greater endoscopic improvement in ulcerative colitis than previously observed for vedolizumab.

In a clinical study, the characteristics of the α4β7 binding antibody Ab001 provide for greater mucosal healing in ulcerative colitis than previously observed for vedolizumab.

In a clinical study, the characteristics of the α4β7 binding antibody Ab001 provide for greater durable clinical response in ulcerative colitis than previously observed for vedolizumab.

In a clinical study, the characteristics of the α4β7 binding antibody Ab001 provide for greater durable clinical remission in ulcerative colitis than previously observed for vedolizumab.

In a clinical study, the characteristics of the α4β7 binding antibody Ab001 provide for greater enhanced clinical response in Crohn's disease than previously observed for vedolizumab.

In a clinical study, the characteristics of the α4β7 binding antibody Ab001 provide for greater clinical remission in TNFα antagonist naïve patients in Crohn's disease than previously observed for vedolizumab.

In a clinical study, the characteristics of the α4β7 binding antibody Ab001 provide for greater corticosteroid-free clinical remission in both ulcerative colitis and Crohn's disease than previously observed for vedolizumab.

In a clinical study, the characteristics of the α4β7 binding antibody Ab001 provide for improvements in biomarkers, including CRP, fecal calprotectin, and symptoms at Week 6 than previously observed for vedolizumab.

Example 16: Two-Compartment PK Model

To inform first-in-human dosing of Ab001, a two-compartment pharmacokinetic (PK) model was built and calibrated to data from cynomolgus monkeys (N=34). The model included mechanisms for intercompartmental transport and elimination of drug, and it addressed target-mediated drug disposition (TMDD) through an empirical Michaelis-Menten elimination function. The model parameters were allometrically scaled to humans using scaling exponents specific to YTE-containing mAbs. Single and biweekly administration of drug via IV and SC were simulated. PK metrics, including the maximum drug concentration (C_{max}), area under the PK curve (AUC) and trough concentration (C_{min}), were calculated for each dosing scenario and are reported here.

The two-compartment PK model was calibrated to preclinical PK datasets for cynomolgus monkeys for both drug molecules, Ab001 and the comparator antibody. The parameters Thalf, Vmax, and Km were fitted to the mean measurements of the IV-administered data set for Ab001. For the SC administered dataset, the parameters $T_{1/2}$, V_{max}, and K_m were taken from IV fits, and optimized bioavailability (F) came out to be 100%, and absorption half-life ($T_{1/2}$, SC) to be 1.08 days. The fitted parameters were used to simulate cynomolgus monkey preclinical data for both IV and SC routes.

For comparator antibody preclinical fits, 10 and 50 mg/kg datasets were available for both IV and SC-administered routes. The data in the 10 mg/kg IV dose group ≥20 days were not included in calibration, as these points showed an anomalous increase in serum drug levels. The parameters $T_{1/2}$, V_{max}, K_m, P_{dist12}, and V_c were fitted to capture the IV-administered dataset. A lower value of optimized central volume was found for the comparator antibody, which was required to capture the C_{max} for the IV dataset. For the SC administered dataset, the parameters $T_{1/2}$, V_{max}, K_m, P_{dist12}, and V_c were taken from the IV calibrations and the model simulations were overlaid to SC preclinical data. The bioavailability (F) and absorption half-life ($T_{1/2}$, SC) were set to those from Ab001 SC dose calibrated values.

The Ab001 half-life in cynomolgus monkeys was estimated to be 17.1 days, and for the comparator antibody it was estimated to be 9.74 days. Based on the respective value for each antibody and applying the allometric scaling, the half-life in humans was estimated in this study to be 63.7 days, and the critical concentration above which TMDD has a negligible effect is estimated to be 8 μg/mL. In a prior analysis the estimated half-life of Ab001 in humans was 54 days using the FcRn scaling factors. In this study the human serum half-life of the comparator antibody was estimated to be 38.5 days.

INCORPORATION BY REFERENCE

All publications and patents cited throughout the text of this specification (including all patents, patent applications, scientific publications, manufacturer's specifications, instructions, etc.), whether supra or infra, are hereby incorporated by reference in their entirety for all purposes. To the extent the material incorporated by reference contradicts or is inconsistent with this specification, the specification will supersede any such material.

EQUIVALENTS

The disclosure may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments are therefore to be considered in all respects illustrative rather than limiting the disclosure described herein. Scope of the disclosure is thus indicated by the appended claims rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.

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```

What is claimed is:

1. A method of treating a gastrointestinal inflammatory disease in a subject in need thereof, the method comprising:
- administering to the subject in need thereof two or more doses of a citrate-free pharmaceutical composition, wherein the citrate-free pharmaceutical composition comprises from about 150 mg/ml to about 200 mg/ml of a long acting α4β7 targeting molecule, wherein the two or more doses are administered at least eight weeks apart, and
  - wherein the long acting α4β7 targeting molecule comprises:
- (i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1; and
- (ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

2. The method of claim **1**, wherein the two or more doses comprise from about 300 mg to about 450 mg of a long acting α4β7 targeting molecule.

3. The method of claim **1**, wherein the two or more doses are administered at least 12 weeks apart.

4. The method of claim **1**, wherein the two or more doses are administered subcutaneously.

5. The method of claim **1**, further comprising administering one or more additional doses at least two weeks before administration of the two or more doses of the citrate-free pharmaceutical composition.

6. The method of claim **5**, wherein the one or more additional doses comprise a citrate-free pharmaceutical composition comprising from about 150 mg/ml to about 200 mg/ml of a long acting α4β7 targeting molecule.

7. The method of claim **5**, wherein the one or more additional doses comprise from about 500 mg to about 1200 mg of the long acting α4β7 targeting molecule.

8. The method of claim **5**, wherein the one or more additional doses are administered intravenously.

9. The method of claim **5**, wherein the one or more additional doses are administered subcutaneously.

10. The method of claim **1**, wherein the gastrointestinal inflammatory disease is inflammatory bowel disease.

11. The method of claim **10**, wherein the inflammatory bowel disease is Crohn's disease.

12. The method of claim **10**, wherein the inflammatory bowel disease is ulcerative colitis.

13. A method of treating a gastrointestinal inflammatory disease in a subject in need thereof, the method comprising:
- administering subcutaneously to the subject in need thereof a citrate-free pharmaceutical composition comprising at least about 180 mg/ml of a long acting α4β7 targeting molecule that comprises:
- (i) a heavy chain consisting of an amino acid sequence according to SEQ ID NO: 1, and
- (ii) a light chain consisting of an amino acid sequence according to SEQ ID NO: 3.

14. The method of claim **13**, wherein the citrate-free formulation is administered as two or more doses at least twelve weeks apart.

15. The method of claim **14**, wherein the two or more doses comprise from about 300 mg to about 450 mg of the long acting α4β7 targeting molecule.

16. The method of claim **14**, further comprising administering one or more additional doses at least two weeks before subcutaneous administration of the two or more doses of the citrate-free pharmaceutical composition.

17. The method of claim **16**, wherein the one or more additional doses comprise from about 500 mg to about 1200 mg of the long acting α4β7 targeting molecule.

18. The method of claim **13**, wherein the gastrointestinal inflammatory disease is inflammatory bowel disease.

19. The method of claim **18**, wherein the inflammatory bowel disease is Crohn's disease.

20. The method of claim **18**, wherein the inflammatory bowel disease is ulcerative colitis.

* * * * *